



Understanding the impacts of dark pools on price discovery

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ARTICLE INFO

MSC:

D40

G14

G12

G18

Keywords:

Market fragmentation

Crossing network

Price discovery

Market quality

ABSTRACT

I study how crossing networks, a type of dark pool, affect price discovery and market liquidity in the presence of noisy and heterogeneous trader signals. I identify a buffer function of crossing networks that helps mitigate traders' losses from false signals. Additionally, I uncover an amplification effect. That is, when signal precision is high, crossing networks enhance price discovery. By contrast, when signal precision is low, crossing networks impair price discovery. These insights reconcile conflicting empirical evidence, yielding novel predictions and regulatory recommendations for equity and emerging markets.

1. Introduction

Over the years, financial systems across the world have experienced a widening of equity trading venues, among which dark pools have rapidly grown in popularity. In contrast with a traditional stock exchange, dark pools do not publicize information about their orders and price quotations before the trade. In this paper, I focus on a specific type of dark pools, known as crossing networks. Unlike a stock exchange in which public prices are formed to clear the buy and sell orders, a crossing network does not form such prices. Instead, they execute orders using prices derived from the stock exchanges. Crossing networks do not contribute to information aggregation in the exchange so they do not offer price discovery. Price discovery (i.e., the process and efficiency of prices aggregating information about assets' values) is essential to achieving the confidence of a broad community of market participants and ensuring the efficiency of the capital markets. Therefore, the question of whether crossing networks, and dark pools in general, will harm price discovery has become a rising concern and matter of debate for regulators and industry practitioners. Academic research, for its part, yields conflicting results. Ye (2011) predicts that, in theoretical studies, the addition of a crossing network strictly harms price discovery. By contrast, Zhu (2014) predicts that crossing networks strictly improve price discovery. Empirically, there are findings that support each of the different predictions.

I investigate the question of whether crossing network trading has a negative impact on price discovery and market liquidity. I first consider exogenous signals and then endogenize the information acquisition. In my model, there are informed speculators and uninformed liquidity traders. More specifically, informed traders possess heterogeneous private signals, with the distribution of these signals determined by the level of information precision. Uninformed liquidity traders have heterogeneous liquidity demands. Both types of traders have three options: (a) trading on an exchange, (b) trading in a crossing network, or (c) not trading (delaying their trades). The exchange is modeled as a platform where market makers post bid-ask prices and ensure execution, whereas the crossing network is modeled as a crossing-mechanism that utilizes the average of bid and ask (mid-price) from the exchange to execute orders. If there is an imbalance in buy and sell orders, probabilistic execution takes place, resulting in some buy or sell

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<https://doi.org/10.1016/j.finmar.2023.100882>

Received 20 May 2022; Received in revised form 14 December 2023; Accepted 25 December 2023

Available online 4 January 2024

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orders remaining unexecuted. I first derive the results when information precision is considered exogenous. I then demonstrate that the effects of crossing networks remain consistent when the model is extended to incorporate endogenous information acquisition.

I find a novel amplification effect of crossing networks on price discovery: price discovery in the exchange will be enhanced when it is high and will be impaired when it is low.¹ Furthermore, the presence of endogenous information acquisition does not alter the observed amplification effect. In an economy that encourages information acquisition (indicated by a low relative size of informed traders or low information acquisition costs), crossing networks improves price discovery. Conversely, in an economy that discourages information acquisition (i.e., a high relative size of informed traders or high information acquisition costs), crossing networks impair price discovery. These findings help reconcile some seemingly contradictory empirical results regarding the impact of dark pools and offer new empirical predictions concerning the information content of dark pool trades, dark pool market share, and their relationship with exchange spreads. I identify information precision as one of the key factors in determining the informational efficiency (price discovery) when markets are fragmented by dark pools (particularly in this paper, crossing networks). The implications of my results extend to policy considerations aimed at enhancing price discovery in equity markets and optimizing dark pool usage, particularly in emerging economies. Additionally, I discuss potential measures for assessing the information precision of markets.

The underlying rationale behind the amplification effect is as follows: First, in equilibrium, a notable sorting effect is observed. Specifically, among informed traders, those with strong signals trade in the exchange, those with modest signals trade in the crossing network, and those with weak signals do not trade. The sorting effect is a direct result of the trade-off of crossing network trading: crossing networks provide better prices than exchanges, together a higher non-execution probability to offsets this. Therefore, Informed traders with strong signals opt for exchanges because they possess high confidence in their profit-making abilities and prioritize guaranteed execution over securing a better price. Conversely, informed traders with moderate signals prefer crossing networks as they have less confidence in their profit-making potential and prioritize obtaining a better price over execution certainty. Lastly, informed traders with weak signals choose not to trade due to their lack of confidence in generating profits. A similar reasoning can be applied to liquidity traders, as their decision-making is influenced by the same trade-off considerations.

Second, I demonstrate that the sorting effect gives rise to an amplification effect. The varying information precision levels result in different signal strength distributions and lead to diverse venue choices among the traders. This, in turn, generates distinct impacts of crossing networks on price discovery. When information precision is high, a significant portion of informed traders receive strong signals, leading them to prefer trading on exchanges. Consequently, the addition of a crossing network attracts only a small fraction of informed traders, in comparison to liquidity traders. As a result, the exchange maintains a higher ratio of informed-to-uninformed traders, enhancing price discovery within the exchange. Conversely, when information precision is low, the majority of informed traders receive more modest signals and thus favor crossing networks. In this scenario, the crossing network attracts a higher fraction of informed traders relative to the liquidity traders, leading to a reduced informed-to-uninformed trader ratio within the exchange. As a consequence, price discovery within the exchange is impaired. In summary, higher information precision results in crossing networks primarily attracting liquidity traders, thereby improving price discovery in the exchange. Conversely, lower information precision leads to crossing networks drawing a larger proportion of informed traders, resulting in a diminished informed-to-uninformed trader ratio and a subsequent decline in price discovery within the exchange.

This paper highlights a crucial yet unexplored aspect of crossing networks in the literature: their ability to assist traders in mitigating information risk, which refers to the potential loss resulting from incorrect or unreliable information. This risk reduction is achieved through lower adverse selection costs associated with trading in crossing networks. When traders receive relatively weak signals, they face a heightened risk of incurring trading losses. Crossing networks serve as a valuable buffer zone for these traders, enticing those who previously refrained from trading or exclusively engaged in exchanges to participate in trading within these crossing networks. By including crossing networks (or say, crossing networks) in the trading landscape, these traders now have the option to participate in crossing network trading. This function of crossing networks becomes particularly significant when traders operate within a noisy information structure.

To the best of my knowledge, this paper is the first to introduce a noisy information structure in a fragmented market to study dark pools and their market impacts. The inclusion of information noise is not only more realistic than assuming perfect information but also provides valuable insights into the dynamics of price discovery and the underlying motivations driving traders' choices. As a result, my predictions exhibit greater robustness as the sorting and amplification effects hold across all equilibria. By contrast, It is often assumed in the theoretical literature that all informed traders possess perfectly precise information, which not only obscures the true motivations behind trading decisions but also introduces instability into the results.²

My research has direct implications for the ongoing debate surrounding crossing network usage. Cross-sectional variation in crossing network activity and its impacts highlights the need for a nuanced evaluation of these pools in different economic contexts. Firstly, regulators should identify firms with characteristics that may be negatively impacted by crossing network trading, such as high R&D, young, small, and less analyzed firms. Secondly, efforts should be made to facilitate information transmission and processing. This can be achieved through improvements in accounting and reporting disclosure systems. Additionally, enhancing the efficiency of judicial systems and strengthening law enforcement against insider trading would contribute to a more transparent and fair trading environment. Lastly, in emerging markets with poor legal systems and lower information precision, caution should

¹ Hendershott and Mendelson (2000) and Zhu (2014) also identify a dichotomous price efficiency. However, their findings generally require specific restrictions and do not demonstrate the presence of the amplification effect I observed in my study.

² For instance, Zhu (2014) examines specific equilibria where crossing networks enhance price discovery. However, it is important to note that other equilibria may exist in his model where crossing networks have adverse effects on price discovery. Unfortunately, these alternative equilibria are not discussed.

be exercised when considering crossing network trading. Regulators should carefully evaluate the potential risks associated with crossing networks in these markets. Further details are in Section 7.3.

I also produce testable predictions and help to reconcile some of the seemingly contradictory results in the previous empirical studies. One of the predictions that could motivate empirical and regulatory concerns is how much information is hidden in crossing networks. I predict an inverted U-shaped relationship between the information content of crossing network trades and liquidity levels (as measured by exchange spread). Specifically, assets with moderate exchange liquidity exhibit the highest information content in their crossing network trades, while the most liquid and illiquid assets show lower information content. This prediction aligns with previous theoretical studies. Further details are in Section 7.2.

In the realm of theoretical studies, a considerable body of research focuses on non-fragmented markets. Notably, two pioneering works on price discovery by [Glosten and Milgrom \(1985\)](#), and [Kyle \(1985\)](#) lay the foundation for understanding market dynamics. Additionally, there are studies that specifically investigate fragmented lit markets, such as [Pagano \(1989\)](#), [Chowdhry and Nanda \(1991\)](#), [Hasbrouck \(1995\)](#), [Viswanathan and Wang \(2002\)](#), and [Bloomfield et al. \(2015\)](#). [Antill and Duffie \(2021\)](#) provide a comprehensive analysis of size-discovery sessions in lit markets. There are a handful of papers that study information asymmetry in a market fragmented by lit and dark venues (see, e.g., [Hendershott and Mendelson, 2000](#); [Degryse et al., 2009](#); [Buti et al., 2017](#)). Yet, these models make assumptions regarding non-freedom of choice for traders or rely on exogenous or sticky prices. My paper is closely related to [Zhu \(2014\)](#) as I adopt and maintain the trading protocols and timeline established in their study. However, I depart from [Zhu \(2014\)](#) by introducing a noisy information structure and endogenizing (numerically) the process of information acquisition. Distinct equilibrium outcomes are predicted that provide valuable insights, complementing the statics of [Zhu \(2014\)](#). My paper is also related but divergent from [Ye \(2011\)](#), [Brolley \(2019\)](#), and [Bayona et al. \(2023\)](#). While these studies touch upon similar topics, there are notable differences in my approach. Specifically, my model allows for the free selection of traders, whereas [Ye \(2011\)](#) assumes that uninformed traders do not have the freedom to choose between different venues. [Brolley \(2019\)](#) and [Bayona et al. \(2023\)](#) consider a sequential trading game that contains a dark pool and a limit order market. Unlike my model, they assume either an exogenous execution risk in dark pools or an exogenous price impact, or a non-rationing mechanism in dark pools.

Empirical studies present conflicting findings regarding the impact of dark pool trading on price discovery, which aligns with my predictions. Some studies, such as [Jiang et al. \(2012\)](#), [Fleming and Nguyen \(2013\)](#), and [Buti et al. \(2022\)](#) support the notion that dark trading enhances price discovery. On the other hand, [Weaver \(2014\)](#), [Balakrishnan and Taori \(2017\)](#), and [Hatheway et al. \(2017\)](#), observe a decrease or delay in price informativeness associated with dark pool trading. Furthermore, [Hendershott and Jones \(2005\)](#) identify a negative impact on price discovery, while [Comerton-Forde and Putniņš \(2015\)](#) find that the effect of dark pool trading on price discovery varies based on the proportion of non-block dark trades. Specifically, they observe an improvement in price discovery when the proportion of non-block dark trades is low (below 10%), suggesting a low informational content in dark pools, whereas a high proportion of non-block dark trades has a detrimental effect on price discovery.

I extend my predictions to various measures of market quality, making it relevant to empirical studies that examine dark pool operations and other market quality indicators. Some authors analyze the information content of dark pool trades. For example, [Peretti and Tapiero \(2014\)](#) find that dark trades can predict price movement. Others analyze the trade-offs associated with dark trading, including [Conrad et al. \(2003\)](#), [Gresse \(2006\)](#), [Næs and Ødegaard \(2006\)](#), and [Ye \(2010\)](#). Another category of studies investigates the relationship between dark trading and the exchange spread. I predict a similar inverted U-shape pattern as observed in the works of [Ray \(2010\)](#) and [Preece \(2012\)](#). Additionally, my study highlights the cross-sectional variation and offers insights into the contradictory results reported in other papers. For instance, [ASIC \(2013\)](#), [Weaver \(2014\)](#), [Brugler \(2015\)](#), [Comerton-Forde and Putniņš \(2015\)](#), [Degryse et al. \(2015\)](#), [Hatheway et al. \(2017\)](#), find a positive association while [O'Hara and Ye \(2011\)](#), [Ready \(2014\)](#), and [Foley and Putniņš \(2016\)](#) identify a negative association between dark pool market share and exchange spread. Furthermore, other studies uncover cross-sectional differences (see, e.g., [Nimalendran and Ray, 2014](#); [Brolley, 2019](#); [Buti et al., 2022](#)). A more comprehensive discussion on the relationship between my predictions and the empirical literature is provided in Section 7.2.

Section 2 outlines key facts of informed trading within dark pools. I introduce the model in Section 3 and explore the equilibrium of the model with exogenous signals in Section 4. The effects of crossing networks on the economy are analyzed in Section 5. Section 6 presents results considering endogenous information acquisition. Section 7 delves into discussions regarding empirical and policy implications, while Section 8 explores different types of dark pools. The paper concludes in Section 9.

2. Informed trading in dark pools

In the early years, dark pools were primarily intended for large, uninformed traders to execute block transactions with a reduced price impact. This was possible because dark pools were not subject to NMS fair access requirements, allowing them to restrict or limit access to their services (see Reg ATS Rule 301(b)(5)). However, there have been significant changes in recent years. According to an industry insider at Rosenblatt Securities Inc., “it can be assumed that most pools are open to most investors connecting to the pool, provided the investors do not violate any codes of conduct”. This change is reflected in the trading sizes of dark pools. [Fig. 1](#) illustrates the average trading size in the United States, showing a convergence between dark pools and exchanges (NYSE and NASDAQ) since 2011. This suggests that the participant composition in these venues has become more similar, indicating a weakening of the exclusivity of dark pools to informed traders. Consequently, there is growing concern about the potential impact of dark pools on the economy. As more informed traders gain access to dark pools, their migration may undermine the information aggregation process on exchanges. It should be noted that I assume full access for informed traders, aligning with previous papers.

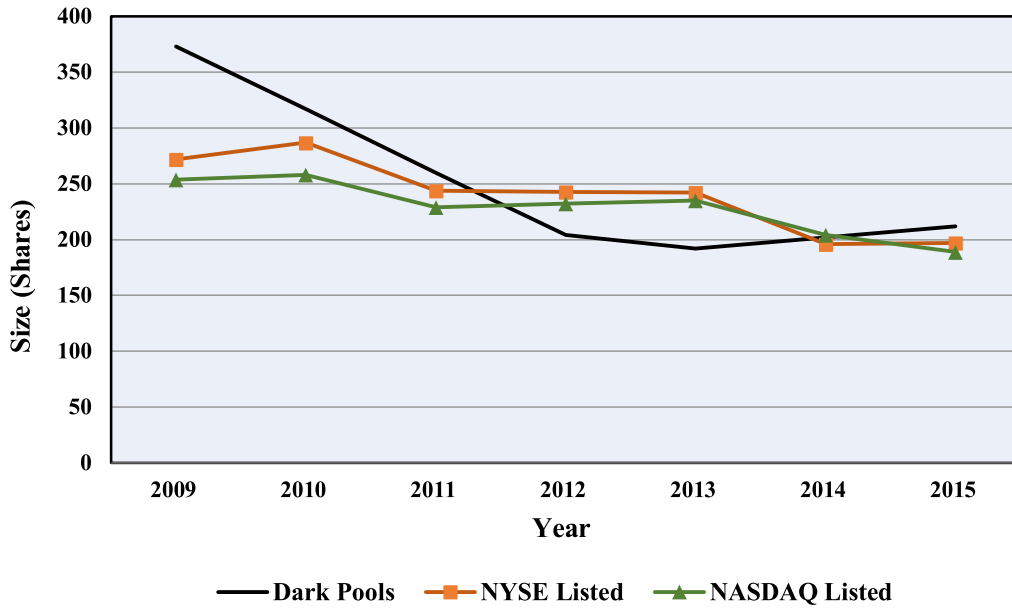


Fig. 1. Average trade size. The plot shows the annual average trade size of U.S. dark pools, NYSE and NASDAQ exchanges, from 2009 to 2015. Data source: Rosenblatt Securities.

3. The model

Now I introduce the basic set-up of the model. Suppose there is one risky asset that has an uncertain fundamental value:

$$\tilde{v} = \begin{cases} -\sigma_v, & \text{with probability } \frac{1}{2}, \\ \sigma_v, & \text{with probability } \frac{1}{2}. \end{cases}$$

That is to say, the risky asset has an unconditional mean zero and standard deviation σ_v . $\tilde{v}$ is realized in period 0, but this information is not revealed to the public.

There are two types of traders in the model: informed (speculators) and uninformed (liquidity traders). Both types are assumed to be risk-neutral. The population of informed speculators is represented by a continuum with measure μ , while the population of uninformed liquidity buyers (sellers) is also represented by a continuum with measure Z^+ (Z^-). The distributions of Z^+ and Z^- are identical and continuously on $[0, +\infty)$, with mean of $\frac{1}{2}\mu_z$. Z^+ and Z^- are also realized at period 0, implying that liquidity buyers and sellers arrive at the market simultaneously. The actual values of Z^+ and Z^- are not observable by any market participants.

My analysis spans multiple periods ($-1, 0, 1$, and 2). In period -1 , informed speculators determine their optimal information precision by weighing acquisition cost and benefits. In period 0, each informed speculator receives their own private signal regarding the value of the risky asset, $s_i = \tilde{v} + e_i$, where i is the index of informed traders and e_i represents the noise of the signal.³ In a symmetric equilibrium, informed traders choose the same optimal information precision level in period -1 . That is, e_i are identically independently distributed normal random variables, with mean 0 and standard deviation σ_e for all i .

Liquidity buyers and sellers enter the market without private information and seek to trade one unit of the risky asset to meet their liquidity needs. A delay cost d_j represents each liquidity trader's liquidity demand. If a liquidity trader's order cannot be executed in period 1, they incur a delay cost equal to $\sigma_v d_j$ per unit. The delay costs, denoted as $d_j s$, are i.i.d. random variables with a cumulative distribution function $G(x): [0, \bar{d}] \rightarrow [0, 1]$, where $G(x) \in C^2$, $1 \leq \bar{d} < \infty$, and $G'(x) + xG''(x) \geq 0, \forall x \in [0, 1]$.⁴ The values of $d_j s$ are determined in period 0.

There are two trading venues: an exchange (the lit market) and a crossing network (CN). The execution mechanism of the two venues is discussed in Sections 3.1 and 3.2.

In the timeline depicted in Fig. 2, informed speculators determine information precision in period -1 . In period 0, asset value $\tilde{v}$ is realized, and speculators receive private signals $s_i = \tilde{v} + e_i$. Liquidity buyers and sellers, Z^+ and Z^- , are also realized, along with traders' per unit delay cost d_j . Market makers announce bid-ask prices in the exchange. In period 1, traders submit orders to

³ According to Gyntelberg et al. (2010), stock market investors hold various types of private information about a firm's value, including knowledge of products, innovation prospects, management quality, and competitor strategies. Private information may also include passively collected information about macro-variables and other fundamentals, which may be dispersed among customers. Equity market order flow primarily reflects transactions by informed investors actively collecting private information. See Section 7.3 for more details.

⁴ This additional assumption is for the uniqueness of the equilibrium. It is satisfied by many commonly used distributions (e.g., a uniform distribution).

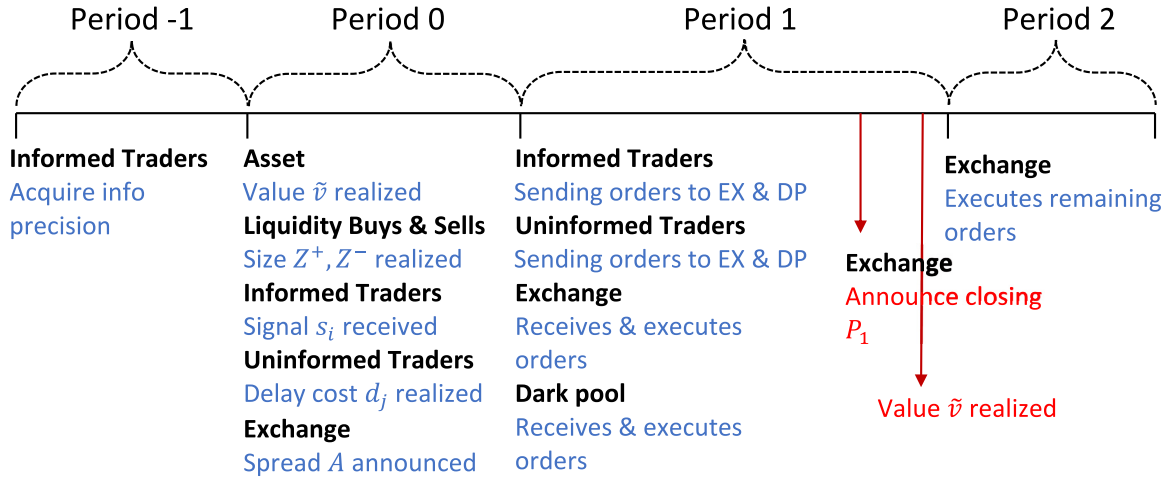


Fig. 2. Trading timeline.

the crossing network and the exchange, which are executed accordingly. At the end of period 1, the exchange announces a closing price P_1 , and $\tilde{v}$ is revealed. In period 2, the public becomes aware of the asset value, and any unexecuted crossing network orders are executed at the public price in the exchange. The distributions of $\tilde{v}$, Z^+ , Z^- , e_i , and d_j are publicly known information.

3.1. To trade in the exchange (lit market)

A lit market resembles an exchange framework inspired by [Glosten and Milgrom \(1985\)](#). It involves a risk-neutral market maker responsible for facilitating transactions. At the start of period 0, the market maker publicly discloses and commits to bid and ask prices for the risky asset, utilizing solely the available public information.

Given the symmetry of $\tilde{v}$, the midquote is naturally zero. Consequently, the ask price is denoted as $A(> 0)$, while the bid price is denoted as $-A$. Hence, I refer to $\frac{A}{\sigma_v}$ as the normalized (half) spread. Without loss of generality, I assume that the realized asset value is σ_v . Let $\bar{\gamma}_e$ and $\underline{\gamma}_e$ represent the respective fractions of informed speculators placing buy and sell orders on the exchange in period 1. Additionally, let α_e denote the fraction of uninformed liquidity traders participating in exchange trading. For now, I assume that traders do not split orders across venues. Therefore, the market maker's ex-post payoff is as follows:

$$\text{MM payoff} = \sigma_v[(\bar{\gamma}_e\mu - \underline{\gamma}_e\mu) + (\alpha_e Z^- - \alpha_e Z^+)] + A[\bar{\gamma}_e\mu + \underline{\gamma}_e\mu + \alpha_e Z^+ + \alpha_e Z^-].$$

The market maker's profit on the asset, which includes gains from both informed and uninformed traders, is represented by the first term. Additionally, the second term accounts for the gains obtained from transaction fees (spread) per exchange order. It is important to note that if the realized asset value is $-\sigma_v$, the market maker's payoff remains the same due to symmetry. Consequently, I refer to $\bar{\gamma}_e$ as the fraction of informed who make money (trade in the right direction), and $\underline{\gamma}_e$ the fraction of informed who lose money (trade in the wrong direction).

A market maker's objective is to break even on average.⁵ That is,

$$0 = \mathbb{E} \left\{ \sigma_v[(\bar{\gamma}_e\mu - \underline{\gamma}_e\mu) + (\alpha_e Z^- - \alpha_e Z^+)] + A[\bar{\gamma}_e\mu + \underline{\gamma}_e\mu + \alpha_e Z^+ + \alpha_e Z^-] \right\}.$$

Since $\mathbb{E}Z^+ = \mathbb{E}Z^- = \frac{1}{2}\mu_z$, the market maker's objective becomes:

$$0 = \sigma_v(-\bar{\gamma}_e + \underline{\gamma}_e)\mu + A[(\bar{\gamma}_e + \underline{\gamma}_e)\mu + \alpha_e\mu_z].$$

It implies that,

$$A = \frac{\bar{\gamma}_e - \underline{\gamma}_e}{\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu}} \sigma_v. \quad (1)$$

If $\bar{\gamma}_e \geq \underline{\gamma}_e (\geq 0)$ and $\alpha_e \geq 0$, then the normalized spread $0 \leq \frac{A}{\sigma_v} \leq 1$. In the next sections, I show that, in equilibrium, $\bar{\gamma}_e > \underline{\gamma}_e$. In other words, informed traders, as a group in general, trade in the right direction in the exchange.

At the end of period 1, a closing price $P_1 = \mathbb{E}[\tilde{v}|V_b, V_s]$ is announce based on the observed volumes V_b, V_s for buy and sell.

In period 2, the market maker's payoff is zero because the realization of $\tilde{v}$ has already been revealed and all trades will be made at the price that is equal to that realization.

⁵ One can think of this as a result of the competition among market makers. For simplicity, I assume that one market maker is operating.

The exchange is modeled as a market maker instead of using other trading protocols like limit order books, following a similar rationale as in [Zhu \(2014\)](#). First, combining dark pools and limit order books in a noisy information structure presents significant analytical challenges and is left for future research.⁶ Second, limit order books and dark pools present different trade-offs for informed traders. While the trade-off of price improvement and execution risk can be considered in limit order books, they are generally transparent, allowing depth to reveal information. On the other hand, dark pools typically do not provide depth-related information. This distinction results in a different trade-off for informed traders between dark pools and open limit order books. For further discussion on this matter, refer to Subsection 4.2 of [Brolley \(2019\)](#).

3.2. To trade in the crossing network

Dark pools employ two key trading protocols: a pricing mechanism, which determines the execution price of orders, and an execution mechanism, which governs the matching of buy and sell orders. In this study, I focus on crossing networks (a type of dark pools) that adopt the midpoint pricing mechanism so that crossing network orders are executed at the midpoint of the bid–ask spread in the exchange (which is 0 in my model). This implies that crossing networks provide price improvement compared to the exchange. While traders incur a transaction cost A when trading on the exchange, this cost is reduced to 0 in the crossing network. I then consider a rationing mechanism for the execution. Orders on the shorter side of the market are executed with a probability of one, while orders on the longer side are executed probabilistically to maintain market balance. For instance, if the realized value of $\tilde{v}$ is σ_v (the case when $\tilde{v} = -\sigma_v$ is symmetric), the expected execution rates for trading in the wrong direction and right direction are, respectively,

$$\bar{R} = \mathbb{E} \left[\min \left\{ 1, \frac{\bar{\gamma}_d \mu + \alpha_d Z^+}{\bar{\gamma}_d \mu + \alpha_d Z^-} \right\} \right], \quad (2)$$

$$\underline{R} = \mathbb{E} \left[\min \left\{ 1, \frac{\gamma_d \mu + \alpha_d Z^-}{\gamma_d \mu + \alpha_d Z^+} \right\} \right], \quad (3)$$

in which $\bar{\gamma}_d, \underline{\gamma}_d$ are the fractions of informed speculators who trade in the right direction and wrong direction respectively, and α_d is the fraction of uninformed liquidity traders who trade in the crossing network.

Therefore, $\underline{R}, \bar{R} \in [0, 1]$, which implies an execution risk in the crossing network, resulting a disadvantage of crossing networks compared to certain executions provided by the exchange. Furthermore, in the next section, I demonstrate that $\bar{\gamma}_d > \underline{\gamma}_d$, indicating the presence of information asymmetry in the crossing network. Consequently, $\underline{R} \leq \bar{R}$. As a result, the crossing network adverse selection cost can be defined as $(\bar{R} - \underline{R})\sigma_v$. The term $\bar{R} - \underline{R}$ is referred to as the normalized crossing network adverse selection cost.

Without loss of generality, I assume that the crossing network operates only in period 1 as the advantage diminishes once the realization of $\tilde{v}$ is revealed in period 2. Unexecuted orders from period 1 are subsequently routed to the exchange in period 2.

3.3. The informed speculators' choice

Upon receiving a signal, informed speculators update their beliefs about the asset's fundamental value using Bayes' rule. Denoting $B(s)$ as the probability of a high realization (σ_v) conditional on signal s , the updated belief is obtained by applying Bayes' rule:

$$B(s) = \Pr(\tilde{v} = \sigma_v | s) = \frac{\phi(\frac{s - \sigma_v}{\sigma_e})}{\phi(\frac{s - \sigma_v}{\sigma_e}) + \phi(\frac{s + \sigma_v}{\sigma_e})}, \quad (4)$$

where $\phi(x)$ is the pdf of a standard normal distribution function. $B(s) \in (0, 1)$ and $B(s)$ are strictly increasing in s .

Lemma 1 in the [Appendix](#) proves that an informed trader never splits orders among venues. And if he decides to trade, he buys if the signal is positive and sells if the signal is negative.⁷

By **Lemma 1**, the payoffs of trading in each venue and no trade are, respectively,

$$\text{Exchange (lit): } B(|s|)\sigma_v - (1 - B(|s|))\sigma_v - A, \quad (5)$$

$$\text{Crossing network: } B(|s|)\underline{R}\sigma_v - (1 - B(|s|))\bar{R}\sigma_v, \quad (6)$$

$$\text{Not trade: } 0. \quad (7)$$

⁶ Departing from my model, there is some direct effort that tries to model limit order book and dark pools with informed trading (see, e.g., [Brolley, 2019](#); [Bayona et al., 2023](#)). However, these efforts often assumed exogenous dark pool execution risk or price impact, or a non-rationing mechanism in the dark pool. There are models of LOB with private values but no asymmetric information (see, e.g., [Parlour, 1998](#); [Foucault, 1999](#); [Foucault et al., 2005](#); [Goettler et al., 2005](#); [Rosu, 2009](#); [Werner et al., 2023](#)). Studies that allow informed traders include [Glosten \(1994\)](#), [Chakravarty and Holden \(1995\)](#), [Seppi \(1997\)](#), [Biais et al. \(2000\)](#), [Kaniel and Liu \(2006\)](#), and [Goettler et al. \(2009\)](#), but adding a dark pool into their models and preserving a noisy information structure is highly challenging.

⁷ It is reasonable to assume that each informed trader only trade for one unit, giving that they are infinitesimal and hence non-strategic. It would be a completely different model if they were being strategic.

The problem for an informed agent is to select the optimal venue (either one of the two venues or no trade) that maximizes their payoff, given their confidence level $B(|s|)$. Therefore, the strategy of an informed speculator who receives a signal s is defined by a mapping

$$h_I(s) : (-\infty, \infty) \rightarrow \{\text{buy, sell}\} \times \{\text{exchange (lit), crossing network, do not trade}\}.$$

3.4. The uninformed liquidity traders' choice

If a type d uninformed agent fails to have their order executed in period 1, they will bear a (per unit) cost of $\sigma_v d$. A type d uninformed liquidity buyer's (seller's) per unit payoffs of trading in the exchange, in the crossing network, or delaying trade are, respectively,

$$\text{Exchange (lit):} \quad -A, \quad (8)$$

$$\text{Crossing network:} \quad -\frac{(\bar{R} - R)}{2} \sigma_v - (1 - \frac{(\bar{R} + R)}{2}) \sigma_v d, \quad (9)$$

$$\text{Delay trade:} \quad -\sigma_v d. \quad (10)$$

Similarly, Lemma 2 in the Appendix proves that liquidity traders' choices never split orders among venues. The strategy of a type d uninformed liquidity trader is defined by a mapping in the following:

$$h_{U,\iota}(d) : [0, \bar{d}] \rightarrow \{\text{exchange (lit), crossing network, delay trade}\},$$

where $\iota \in \{\text{buyer, seller}\}$.

4. The equilibrium with exogenous signals

In this section, I examine equilibrium conditions when informed traders' signals are exogenous. Two models are utilized: the multi-venue model, described in Section 3, and the single-venue model, where only the exchange (the lit market) is active.

4.1. Benchmark model: without a crossing network

In the benchmark model, everything remains the same except that the only available trading venue for traders is the exchange (the lit market). In this model, Lemmas 1 and 2 continue to hold, indicating that traders do not split their orders. This model is denoted as the single venue model, represented by the superscript "S". The equilibrium in this model is defined as follows:

Definition 1 (Benchmark: Without a Crossing Network). An equilibrium in the single-venue model consists of strategies for informed speculators $h^S_I(s)$, strategies for uninformed liquidity traders $h^S_{U,\iota}(d)$ (where $\iota \in \{\text{buyer, seller}\}$), an exchange spread A^S , and participation fractions $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S . The equilibrium satisfies the following conditions:

- (i) Given A^S , $h^S_I(s)$ and $h^S_{U,\iota}(d)$ are optimal for informed speculators with signal s and uninformed liquidity traders with per unit delay cost d , respectively.
- (ii) Given $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S , the exchange spread A^S ensures that the market maker in the exchange breaks even on average.
- (iii) $\bar{\gamma}_e^S$ and $\underline{\gamma}_e^S$ represent the fractions of informed traders trading in the right and wrong direction in the exchange, while α_e^S represents the fraction of uninformed traders participating in the exchange in period 1.

The existence and uniqueness of a cut-off equilibrium are characterized by Theorem 1.⁸

Theorem 1 (Existence and Uniqueness, Benchmark Model, No Crossing Network). For any $\sigma_e, \sigma_v \geq 0$, there exists an equilibrium where traders employ cut-off strategies. That is, the optimal strategies of the informed speculators and uninformed liquidity traders follow a cut-off strategy, i.e.,

$$h^S_I(s) = \begin{cases} (\text{buy, exchange (lit)}) & \text{if } s \geq \hat{s}, \\ (\text{sell, exchange (lit)}) & \text{if } s < -\hat{s}, \\ \text{do not trade} & \text{others.} \end{cases} \quad (11)$$

$$h^S_{U,\iota}(d) = \begin{cases} (\text{buy, exchange (lit)}) & \text{if } d \geq 2B(\hat{s}) - 1, \iota = \text{buyer} \\ (\text{sell, exchange (lit)}) & \text{if } d \geq 2B(\hat{s}) - 1, \iota = \text{seller} \\ \text{delay trade} & \text{others.} \end{cases} \quad (12)$$

The cut-off $\hat{s}$, the exchange spread A^S , and the participation fractions $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S are determined endogenously. Moreover, every equilibrium is a cut-off equilibrium, and the equilibrium is unique if $\sigma_e, \sigma_v > 0$, $G'(x) + xG''(x) \geq 0, \forall x \in [0, 1]$.

⁸ The cut-off characteristic is similar to the approaches taken by Kaniel and Liu (2006), Menkveld et al. (2017), and Brolley (2019).

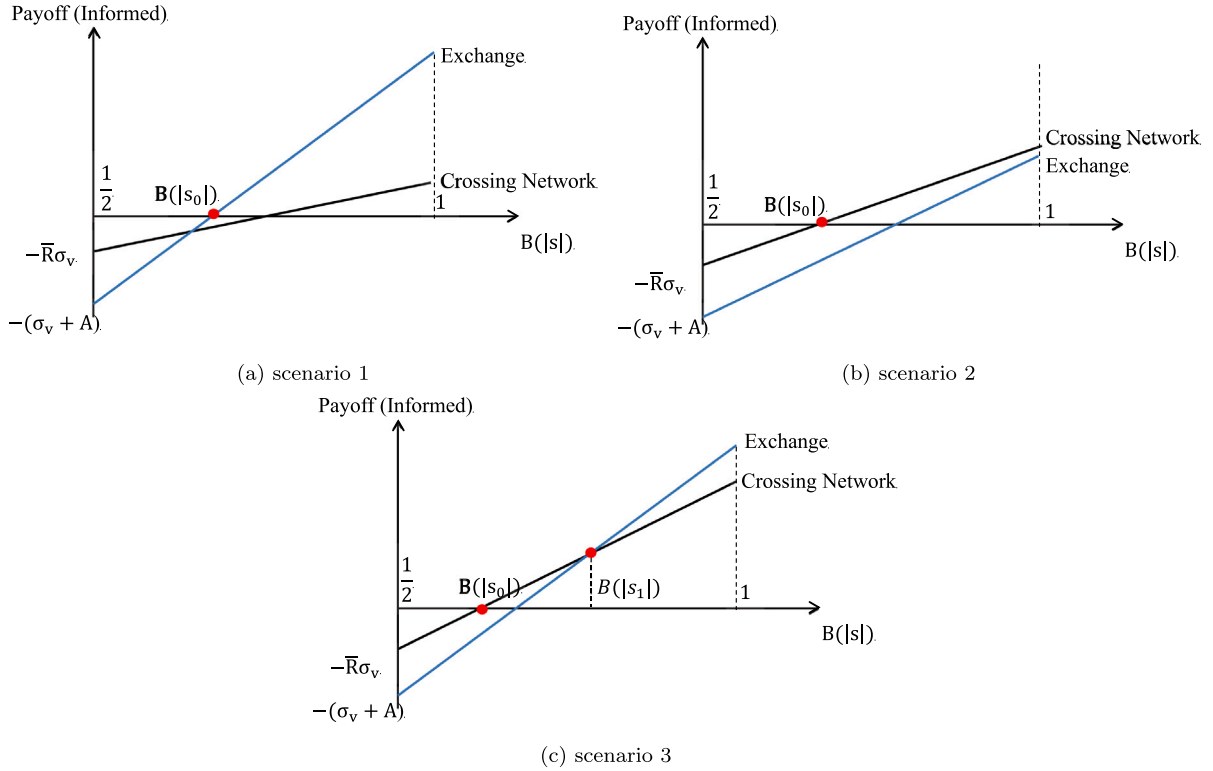


Fig. 3. Possible payoffs for informed traders, multi-venue. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Multi-venue model: with a crossing network

In the multi-venue model, there are two trading venues: an exchange (lit) and a crossing network (CN). To distinguish it from the single-venue model, the superscript $\mathbb{S}$ is omitted in the multi-venue model. The equilibrium of the multi-venue model is defined as follows:

Definition 2 (Multi-Venue with a Crossing Network). An equilibrium is a strategy for the informed speculators, denoted by $h_I(s)$, a strategy for the uninformed liquidity traders, denoted by $h_{U,i}(d)$, an exchange spread, A , two expected execution rate in the crossing network $\bar{R}$, $\underline{R}$, and a set of participation fractions $\bar{\gamma}_e, \underline{\gamma}_e, \bar{\gamma}_d, \underline{\gamma}_d, \alpha_e$, and α_d , s.t.

- (i) $h_I(s)$ is optimal for informed speculators with signal s , whereas $h_{U,i}(d)$ is optimal for uninformed liquidity traders with (per unit) delay cost d , given A , $\bar{R}$, and $\underline{R}$.
- (ii) the exchange spread A makes a market maker in the exchange break-even on average, given $\bar{\gamma}_e, \underline{\gamma}_e, \bar{\gamma}_d, \underline{\gamma}_d, \alpha_e$, and α_d ;
- (iii) the crossing network operates using a mid-pricing and a rationing execution mechanism. $\underline{R}$ and $\bar{R}$ are the respective expected execution probability for orders that are in the right and in the wrong directions;
- (iv) $\bar{\gamma}_e$ and $\underline{\gamma}_e$ measure the respective fractions of informed traders in the exchange who trade in the right and wrong directions. $\bar{\gamma}_d$ and $\underline{\gamma}_d$ measure the respective fractions of informed traders in the crossing network who trade in the right and wrong directions. α_e and α_d measure the respective fraction of uninformed traders who trade in the exchange (lit) and in the crossing network in period 1.

Next, consider an informed speculator with signal “ s ”. Suppose $1 \geq \bar{R} \geq \underline{R} > 0$ and $\sigma_v \geq A \geq 0$. The payoffs of the informed trading in different venues can be characterized by Eqs. (5), (6), (7), as well as Fig. 3(a), 3(b), and 3(c). These representations illustrate three possible scenarios. In scenarios 1 and 2, informed traders strictly prefer trading in one venue over not trading at all. Conversely, in scenario 3, informed traders participate in both venues and are sorted according to the strengths of their signals ($B(|s|)$). If either scenario 1 or 2 occurs in an equilibrium, it is referred to as a trivial equilibrium because all trades happen in a single venue. On the other hand, if scenario 3 occurs in an equilibrium, it is termed a non-trivial equilibrium.

Theorem 2 rigorously demonstrates that, in equilibrium, only scenario 3 can occur, ruling out the possibility of a trivial equilibrium.

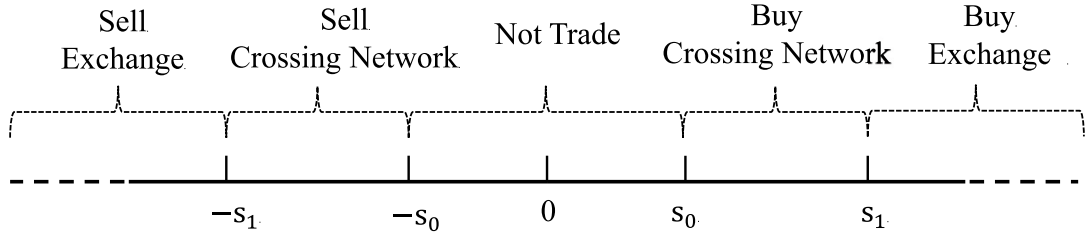


Fig. 4. Strategy of informed traders.

Let $s_0 > 0$ be the signal that renders the trader indifferent between trading in a crossing network and not trading, and let $s_1 > 0$ be the signal that renders the trader indifferent between trading in a crossing network and trading on the exchange. The values of s_0 and s_1 satisfy the following equations derived from Eqs. (5), (6), and (7):

$$B(s_0)(\bar{R} + \underline{R}) = \bar{R} \quad (13)$$

$$B(s_1) \left[(1 - \bar{R}) + (1 - \underline{R}) \right] \sigma_v = A + (1 - \bar{R})\sigma_v. \quad (14)$$

At this point, the existence and relationship between s_0 and s_1 have not been established. For now, (s_0, s_1) are assumed to exist and that $s_0 < s_1 < +\infty$. In such a case, the optimal strategy for an informed trader with signal s would be determined as follows:

$$h_I(s) = \begin{cases} (\text{buy, exchange (lit)}) & \text{if } s \geq s_1, \\ (\text{buy, crossing network}) & \text{if } s_0 \leq s < s_1, \\ (\text{sell, crossing network}) & \text{if } -s_1 \leq s < -s_0, \\ (\text{sell, exchange (lit)}) & \text{if } s < -s_1, \\ \text{do not trade} & \text{others.} \end{cases} \quad (15)$$

This is illustrated in Fig. 4. That is, it is strictly optimal that informed traders with strong signals will trade in the exchange, informed traders with modest signals will trade in the crossing network, and informed traders with weak signals will not trade.

If all informed traders follow such strategy, the exchange fraction of informed traders who trade in the right and wrong directions are, respectively,

$$\bar{\gamma}_e = \Pr(s \geq s_1 | \tilde{v} = \sigma_v) = \Pr(s \leq -s_1 | \tilde{v} = -\sigma_v) = 1 - \Phi\left(\frac{s_1 - \sigma_v}{\sigma_e}\right), \quad (16)$$

$$\underline{\gamma}_e = \Pr(s < -s_1 | \tilde{v} = \sigma_v) = \Pr(s > s_1 | \tilde{v} = -\sigma_v) = 1 - \Phi\left(\frac{s_1 + \sigma_v}{\sigma_e}\right). \quad (17)$$

And the crossing network fraction of informed who trade in the right and wrong directions are, respectively,

$$\begin{aligned} \bar{\gamma}_d &= \Pr(s_0 \leq s < s_1 | \tilde{v} = \sigma_v) = \Pr(-s_1 \leq s < -s_0 | \tilde{v} = -\sigma_v) \\ &= \Phi\left(\frac{s_1 - \sigma_v}{\sigma_e}\right) - \Phi\left(\frac{s_0 - \sigma_v}{\sigma_e}\right), \end{aligned} \quad (18)$$

$$\begin{aligned} \underline{\gamma}_d &= \Pr(-s_1 \leq s < -s_0 | \tilde{v} = \sigma_v) = \Pr(s_0 \leq s < s_1 | \tilde{v} = -\sigma_v) \\ &= \Phi\left(\frac{s_1 + \sigma_v}{\sigma_e}\right) - \Phi\left(\frac{s_0 + \sigma_v}{\sigma_e}\right). \end{aligned} \quad (19)$$

Similarly, the payoffs for uninformed traders in the exchange, crossing network, and delaying trade are provided respectively in Eqs. (8), (9), and (10). Let d_0 and d_1 represent the types of liquidity traders who are indifferent between delaying trade and trading in a crossing network, and those who are indifferent between trading in a crossing network and trading on the exchange, respectively. Then, by Eqs. (8), (9), and (10),

$$\begin{aligned} -\frac{(\bar{R} - \underline{R})}{2} \sigma_v - \left(1 - \frac{(\bar{R} + \underline{R})}{2}\right) \sigma_v d_0 &= -\sigma_v d_0, \\ -\frac{(\bar{R} - \underline{R})}{2} \sigma_v - \left(1 - \frac{(\bar{R} + \underline{R})}{2}\right) \sigma_v d_1 &= -A. \end{aligned}$$

Combining this with Eqs. (13) and (14), one has:

$$d_0 = 2B(s_0) - 1, \quad (20)$$

$$d_1 = 2B(s_1) - 1. \quad (21)$$

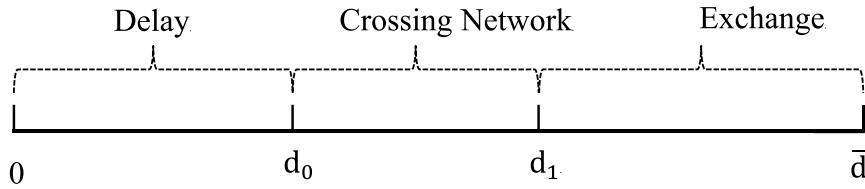


Fig. 5. Strategy of uninformed traders.

Thus, the optimal strategy for an uninformed trader is also a cut-off strategy:

$$h_{U,i}(d) = \begin{cases} (\text{buy, exchange (lit)}) & \text{if } d \geq 2B(s_1) - 1, i = \text{buyer} \\ (\text{sell, exchange (lit)}) & \text{if } d \geq 2B(s_1) - 1, i = \text{seller} \\ (\text{buy, crossing network}) & \text{if } 2B(s_0) - 1 \leq d < 2B(s_1) - 1, i = \text{buyer} \\ (\text{sell, crossing network}) & \text{if } 2B(s_0) - 1 \leq d < 2B(s_1) - 1, i = \text{seller} \\ \text{delay trade} & \text{otherwise.} \end{cases} \quad (22)$$

This is described in Fig. 5. The exchange fraction, α_e , and crossing network fraction, α_d , of uninformed liquidity traders, are, respectively,

$$\alpha_e = 1 - G(2B(s_1) - 1), \quad (23)$$

$$\alpha_d = G(2B(s_1) - 1) - G(2B(s_0) - 1). \quad (24)$$

In equilibrium, the cut-offs for uninformed traders are determined by the cut-offs for informed traders according to Eqs. (20) and (21). This interdependence implies that uninformed and informed traders move together, making it impossible for uninformed traders to shift between venues without influencing the informed traders and leading to a different equilibrium.

Given $\bar{\gamma}_e, \underline{\gamma}_e$, and α_e , the exchange spread A captured in Eq. (1) makes the market maker break even. Also, given $\bar{\gamma}_d, \underline{\gamma}_d$, and α_d , and given the distribution of Z^+ and Z^- , the expected execution rates in the crossing network, $\bar{R}$ and $\underline{R}$, are respectively determined by Eqs. (2) and (3).

Theorem 2 establishes the existence of a non-trivial cut-off equilibrium by demonstrating the existence of s_0 and s_1 . It also proves the impossibility of a trivial equilibrium. While an analytical proof of uniqueness encounters technical challenges, the numerical analysis strongly indicates a high probability of a unique equilibrium.

Theorem 2 (Existence of Equilibrium in the Multi-Venue Model with a Crossing Network). *Given any $\sigma_v, \sigma_e > 0$, there exists an equilibrium where traders adopt cut-off strategies. In this equilibrium, the optimal strategies for informed and uninformed traders, denoted as $h_I(s)$ and $h_U^U(d)$ respectively, are determined by Eqs. (15) and (22). The cut-off values (s_0 and s_1) satisfy Eqs. (13) and (14), with $0 < s_0 < s_1$. Importantly, every equilibrium, if multiple exist, is a cut-off equilibrium where both informed and uninformed traders actively participate in both venues, making it non-trivial. The exchange spread, A , the expected execution rates, $\bar{R}, \underline{R}$, are determined, respectively, by Eqs. (1), (2), and (3). The set of participation fractions, $\{\bar{\gamma}_e, \underline{\gamma}_e, \bar{\gamma}_d, \underline{\gamma}_d, \alpha_e, \alpha_d\}$, are determined by Eqs. (16), (17), (18), (19), (23), and (24).*

In the equilibrium described in Theorem 2, several notable features emerge. Firstly, there is a sorting effect of market fragmentation, where informed traders with different signal strengths prefer the exchange, crossing network, or not trading. Similarly, uninformed traders with varying impatience levels choose the exchange, crossing network, or delay trade.^{9,10} Secondly, all equilibria are non-trivial, as captured in Corollary 1. Informed and uninformed traders actively participate in both venues, creating liquidity and validating the “liquidity begets liquidity” concept. Furthermore, crossing networks serve to mitigate traders’ information risk by acting as a buffer zone for less well-informed traders, offering value in the presence of noisy information. This aspect, though overlooked in the literature, adds to the trade-offs and reinforces the importance of crossing networks, particularly in the face of increased information noise or trader risk aversion.

Corollary 1 (Liquidity Begets Liquidity). *The condition $\alpha_d > 0$ holds if and only if $\bar{\gamma}_d - \underline{\gamma}_d > 0$. This means that in the crossing network, the presence of informed traders attracts uninformed traders to participate, and vice versa.*

⁹ The absence of a sorting effect can lead to unstable predictions. For instance, in Zhu (2014), uninformed traders can collectively shift from the crossing network to the exchange without affecting the behavior of informed traders, potentially increasing adverse selection costs in the crossing network. This can result in decreased price discovery, which contradicts their previous conclusions. These equilibria, however, have not been extensively discussed. In my model, the sorting effect ensures that traders always move together, providing robust predictions across all equilibria.

¹⁰ The sorting effect observed in my model is conceptually related to the studies by Menkveld et al. (2017) and Brolley and Malinova (2021). Menkveld explores pecking orders in the absence of information asymmetry, while Brolley focus on limit order markets in which investors are sorted based on their valuations of a security’s fundamental value.

Corollary 2 highlights that crossing networks significantly reduce traders' participation in the exchange. Additionally, **Corollary 3** reveals the presence of adverse selection in both the exchange and the crossing network.

Corollary 2. Given any $\sigma_e, \sigma_v > 0$, $s_1 > \hat{s}$. In correspondent, $d_1 > \hat{d}$.

Corollary 3 (Adverse Selection). $\forall \sigma_e, \sigma_v > 0$, $0 < \underline{\gamma}_e < \overline{\gamma}_e$, $0 < \underline{\gamma}_d < \overline{\gamma}_d$, and $\bar{R} - \underline{R} > 0$.

5. Are crossing networks good or bad? It depends

This section addresses key questions: 5.1 examines crossing networks' impact on market liquidity and transaction costs, 5.2 evaluates their influence on price discovery, 5.3 explores factors determining their detrimental likelihood, 5.4 analyzes market responses to information precision variations, and 5.5 discusses the role of crossing networks in the trade off of utility and price discovery.

In this section, the information precision level, $\frac{1}{\sigma_e}$, is assumed to be an exogenous parameter. In Section 6, I will discuss the case when σ_e is endogenous.

5.1. Crossing network impacts on liquidity and transaction costs

Proposition 1 presents a comparative analysis between the single-venue and multi-venue models, examining how crossing networks affect market participation and exchange spread.

Proposition 1. Given any $\sigma_v, \sigma_e > 0$, and when information precision is sufficiently high (i.e., σ_e is small enough), the introduction of a crossing network alongside an exchange has the following effects: (a) it reduces participation in the exchange for both informed and uninformed traders, while increasing overall market participation; and (b) it widens the exchange spread. To be more specific, if the condition $\frac{\mu_-}{\mu} \geq \frac{R}{1-R} \frac{1}{1-G(\hat{k})}$ is satisfied, where $R = \mathbb{E} \left[\min \left\{ 1, \frac{Z^+}{Z^-} \right\} \right]$ and $\hat{k}$ is uniquely determined by $\hat{k} = \frac{1}{1+[1-G(\hat{k})]\frac{\mu_-}{\mu}}$, then

- (i)(Participation) $(\overline{\gamma}_e^S - \underline{\gamma}_e^S) \geq (\overline{\gamma}_e - \underline{\gamma}_e)$, $\alpha_e^S \geq \alpha_e$, and if σ_e is sufficiently small or large, $\alpha_e^S \leq \alpha_e + \alpha_d$. And,
- (ii)(Exchange spread) $\frac{A^S}{\sigma_v} \leq \frac{A}{\sigma_v}$ if σ_e is small.

Proposition 1 shows that adding a crossing network alongside an exchange decreases exchange participation but increases total market participation. This suggests that crossing networks create additional liquidity by attracting traders from both “Not trade” and “Trade in the exchange” categories to “Trade in the crossing network”. This is captured in Fig. 6 in which $\alpha_e \leq \alpha_e^S \leq \alpha_e + \alpha_d$.

The impact of a crossing network on the exchange spread is contingent on various factors, including the level of information asymmetry and the intensity of informed and uninformed trading. The addition of a crossing network leads to the migration of both informed and uninformed traders, and the outcome depends on the dominance of one group over the other. When informed traders possess high information precision (low σ_e), a significant portion of them prefer to remain in the exchange, resulting in an increase in information asymmetry and widening of the exchange spread ($\frac{A}{\sigma_v}$). Conversely, when informed traders have low information precision (high σ_e), a substantial fraction migrates to the crossing network, leading to a reduction in their relative proportion in the exchange. In this scenario, the impact on the exchange spread varies depending on the intensity of migration. Further details are provided in Appendix B.6.¹¹

5.2. Crossing network impacts on price discovery

Price discovery is measured by the informativeness of P_1 . At the end of period 1, the market maker announces a closing price $P_1 = \mathbb{E}[\tilde{v}|V_b, V_s]$, in which V_b and V_s are the buy and sell volume respectively. P_1 is perceived as a proxy for the fundamental value of the asset, because $\mathbb{E}[\tilde{v}|P_1, V_b, V_s] = \mathbb{E}[\mathbb{E}[\tilde{v}|P_1, V_b, V_s]|P_1] = P_1$.

Price discovery is assessed through the informativeness of P_1 , which represents the closing price announced by the market maker at the end of period 1. The value of P_1 is determined based on the buy volume V_b and sell volume V_s , and is perceived as an approximation of the asset's fundamental value. This is because, due to the law of iterated expectations, $\mathbb{E}[\tilde{v}|P_1, V_b, V_s] = \mathbb{E}[\mathbb{E}[\tilde{v}|P_1, V_b, V_s]|P_1] = P_1$. Thus, P_1 is considered a proxy for the fundamental value of the asset.

To facilitate comparison with previous studies, I adopt similar measures as Zhu (2014). Without loss of generality, I assume that the true value $\tilde{v}$ is equal to $+\sigma_v$. The likelihood ratio is defined as follows:

¹¹ Note that $\frac{A}{\sigma_e}$ depends on both $\frac{\overline{\gamma}_e - \underline{\gamma}_e}{\overline{\gamma}_e + \underline{\gamma}_e}$ and $\frac{\overline{\gamma}_e - \underline{\gamma}_e}{\alpha_e}$, when σ_e is large, $\frac{\overline{\gamma}_e - \underline{\gamma}_e}{\alpha_e}$ decreases when adding a crossing network but not necessarily $\frac{\overline{\gamma}_e - \underline{\gamma}_e}{\overline{\gamma}_e + \underline{\gamma}_e}$. The overall effect on $\frac{A}{\sigma_e}$ is uncertain.

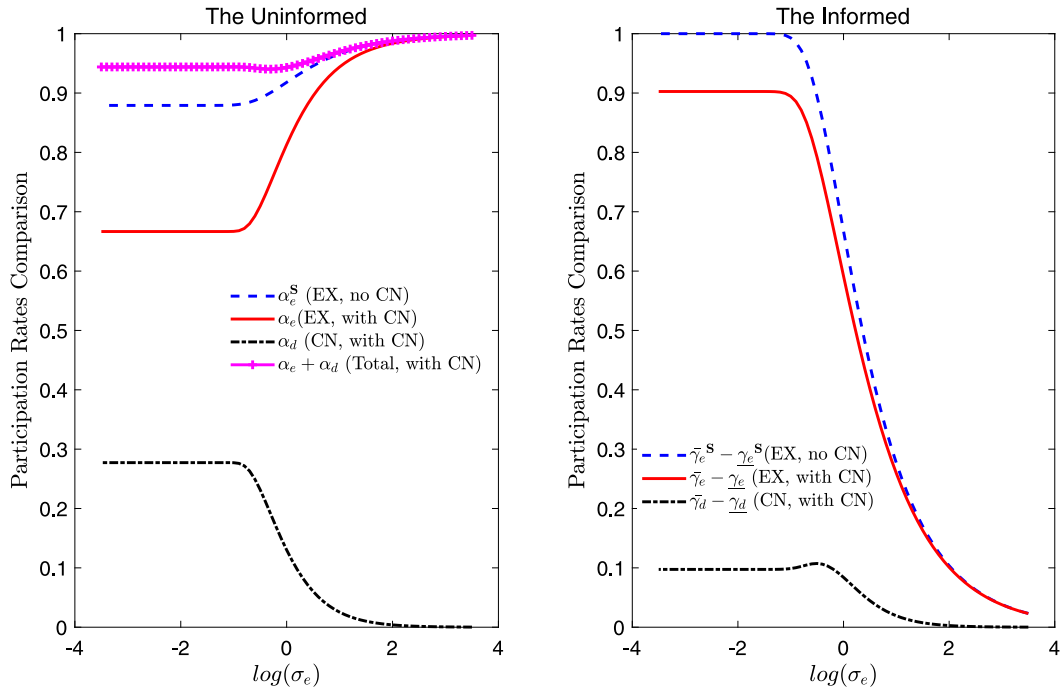


Fig. 6. The participation rates. The left figure shows the uninformed traders' exchange participation rate without a crossing network (α_e^S) and the participation rates in the exchange (α_e), crossing network (α_d), and total participation ($\alpha_e + \alpha_d$) with a crossing network, as a function of $\log(\sigma_e)$. The right figure illustrates the informed traders' right minus wrong participation in the exchange without a crossing network ($\tilde{\gamma}_e^S - \underline{\gamma}_e^S$), as well as their participation in the exchange ($\tilde{\gamma}_e - \underline{\gamma}_e$) and the crossing network ($\tilde{\gamma}_d - \underline{\gamma}_d$) when a crossing network is present. In both plots, I assume $\log(\sigma_v) = 0$, $\mu_z = 60$, and $\mu = 30$.

$$\begin{aligned}
 r &= \log \frac{\Pr(\tilde{v} = +\sigma_v | V_b, V_s)}{\Pr(\tilde{v} = -\sigma_v | V_b, V_s)} \\
 &= \log \frac{\phi_z(Z^+ = \frac{1}{\alpha_e} [V_b - \bar{\gamma}_e \mu]) \cdot \phi_z(Z^- = \frac{1}{\alpha_e} [V_s - \underline{\gamma}_e \mu])}{\phi_z(Z^- = \frac{1}{\alpha_e} [V_b - \bar{\gamma}_e \mu]) \cdot \phi_z(Z^+ = \frac{1}{\alpha_e} [V_s - \underline{\gamma}_e \mu])}.
 \end{aligned}$$

And because

$$\begin{aligned}
 P_1 &= \sigma_v \Pr(\tilde{v} = +\sigma_v | V_b, V_s) + (-\sigma_v) \Pr(\tilde{v} = -\sigma_v | V_b, V_s) \\
 &= \frac{\Pr(\tilde{v} = +\sigma_v | V_b, V_s) - \Pr(\tilde{v} = -\sigma_v | V_b, V_s)}{\Pr(\tilde{v} = +\sigma_v | V_b, V_s) + \Pr(\tilde{v} = -\sigma_v | V_b, V_s)} \sigma_v,
 \end{aligned}$$

one has:

$$P_1 = \frac{e^r - 1}{e^r + 1} \sigma_v.$$

Clearly, a higher value of r indicates that P_1 is closer to the true value σ_v . In the extreme case where $r = +\infty$, P_1 equals σ_v , indicating complete informativeness. Hence, r can be viewed as a measure of informativeness.

Another commonly used measure of informativeness in previous studies is the scaled root-mean-squared error (RMSE). It is defined as follows:

$$\text{RMSE} = \frac{[\mathbb{E}[(\tilde{v} - P_1)^2 | \tilde{v} = \sigma_v]]^{.5}}{\sigma_v} = \mathbb{E} \left[\frac{4}{(e^r + 1)^2} | \tilde{v} = \sigma_v \right].$$

It is scaled by σ_v . Since $r \in (0, 1)$, the scaled pricing error (RMSE) is between 0 and 1. If RMSE is higher, there are more pricing errors, and there is less price discovery.

Since V_b and V_s are random variables, r is also a random variable. When μ_z, σ_z^2 are large enough, the density of $\phi_z(\cdot)$ can be approximated by a normal distribution $\mathcal{N}(.5\mu_z, .5\sigma_z^2)$.¹² Substituting the density functions, an approximate r is given by:

$$r^{\text{Approx}} = \frac{2(\bar{\gamma}_e - \underline{\gamma}_e)\mu}{\alpha_e^2 \sigma_z^2} (V_b - V_s).$$

¹² When μ_z and σ_z^2 are large enough, Z^+ is approximately normal.

Given that $\bar{v} = \sigma_v$ and that $V_b - V_s$ has a distribution of $\mathcal{N}\left((\bar{\gamma}_e - \underline{\gamma}_e)\mu, \alpha_e^2 \sigma_z^2\right)$, r^{Approx} has a distribution of:

$$\mathcal{N}\left(2I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)^2, 4I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)^2\right),$$

where:

$$I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e) = \frac{(\bar{\gamma}_e - \underline{\gamma}_e)\mu}{\alpha_e \sigma_z}.$$

Thus, the magnitude of $I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)$ serves as a measure of price discovery in the exchange. In line with the previous studies, I also refer it as the signal-to-noise ratio. I then consider two measures of price discovery: the signal-to-noise ratio, $I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)$, and the scaled RMSE under the normal approximation. Fig. 7 illustrates a numerical example.¹³

Now I introduce a measure of information advantage for the informed traders:

$$\sigma = \frac{\sigma_v}{\sigma_e}.$$

An informed speculator's information advantage is defined as the product of the asset's fundamental uncertainty σ_v and the precision of its signals, $\frac{1}{\sigma_e}$. A higher σ_v indicates greater undisclosed information and potentially higher profitability for the informed speculators. Similarly, a lower σ_e signifies higher precision in their private information, leading to greater informational profit. Proposition 2 elucidates the influence of σ on the impact of crossing networks on price discovery.

Proposition 2. Price discovery in the exchange, as measured by the informativeness of P_1 , increases with the information advantage of informed traders, σ . Moreover, there exists a threshold value, $\bar{\sigma} > 0$, such that: (a) when $\sigma < \bar{\sigma}$, the addition of a crossing network diminishes price discovery, and (b) when σ is large, the addition of a crossing network enhances price discovery. Specifically, suppose $\hat{k} \leq \frac{\mu_z}{\mu} < +\infty$, where $\hat{k}$ is uniquely determined by $\hat{k} = \frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}}$, then $I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)$ and $I(\bar{\gamma}_e^S, \underline{\gamma}_e^S, \alpha_e^S)$ increase in σ , while $RMSE$ and $RMSE^S$ decrease in σ . And, when $\sigma_e > 0$ is large enough, $\exists \bar{\sigma} > 0$ such that:

- (i) if $\sigma \in (0, \bar{\sigma})$, adding a crossing network will strictly decrease the informativeness of the price in the exchange, that is, $I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e) < I(\bar{\gamma}_e^S, \underline{\gamma}_e^S, \alpha_e^S)$, and $RMSE > RMSE^S$;
- (ii) if σ is sufficiently large, adding a crossing network will increase the informativeness of the price in the exchange, that is, $I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e) \geq I(\bar{\gamma}_e^S, \underline{\gamma}_e^S, \alpha_e^S)$, and $RMSE \leq RMSE^S$.

The intuition is as follows. The impact of adding a crossing network alongside an exchange on price discovery depends on the ratio of informed traders to uninformed traders in the exchange. When the information advantage of informed traders, σ , is high, a larger proportion of informed traders choose to remain in the exchange, resulting in a relatively small migration to the crossing network compared to the liquidity traders. As a result, the addition of a crossing network increases the signal-to-noise ratio in the exchange, enhancing the informativeness of P_1 . Conversely, when σ is low, a higher proportion of informed traders prefer to migrate to the crossing network, reducing the proportion of informed traders in the exchange. This leads to a decrease in the signal-to-noise ratio and a decline in price discovery.

In Fig. 7, the right plots show the signal-to-noise ratio as a function of $\sigma = \frac{\sigma_v}{\sigma_e}$. As σ increases, indicating a higher informational advantage for informed traders, the signal-to-noise ratio also increases, indicating greater price discovery. Adding a crossing network alongside an exchange has a different impact on price discovery depending on the level of σ . When σ is low (i.e., low σ_v or high σ_e), the introduction of a crossing network decreases price discovery. However, when σ is high (i.e., high σ_v or low σ_e), the crossing network enhances price discovery. The left plot provides a visual representation of the crossing network's impact on price discovery in a two-dimensional context, considering both σ_v and σ_e .

The results reveal an amplification effect of crossing networks on price discovery. Crossing networks enhance price discovery when it is already high, but impair price discovery when it is low. Therefore, caution should be exercised when introducing dark pools (CNs, particularly) in economies with already low levels of price discovery. A more comprehensive discussion of such economies is in Section 7.3.

My findings suggest that the prediction of Zhu (2014) (crossing networks strictly increasing price discovery) can be viewed as an extreme case in my model when information precision is very high (i.e., $\sigma_e \rightarrow 0$). However, real-world information structures exhibit cross-sectional differences, challenging the applicability of this extreme case. My study shows that tailored policies are needed to address this issue (discussed in Sections 7.1 and 7.3).

5.3. What determines the likelihood of a harmful crossing network?

To address this question, I introduce a proxy termed the “crossing network's impact on price discovery likelihood”. It establishes an upper bound on σ_v as the following, beyond which crossing networks have a detrimental effect on price discovery.

$$\bar{\sigma}_v = \sup_{x>0} \left\{ x | \forall \sigma_v \in (0, x), I(\bar{\gamma}_e^S, \underline{\gamma}_e^S, \alpha_e^S) > I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e) \right\}.$$

¹³ By the same argument as in previous studies, under the normal approximation, a higher signal-to-noise ratio $I(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)$ always corresponds to a lower scaled RMSE. That is, they are in nature the same measure. Therefore, I only plot the signal-to-noise ratio.

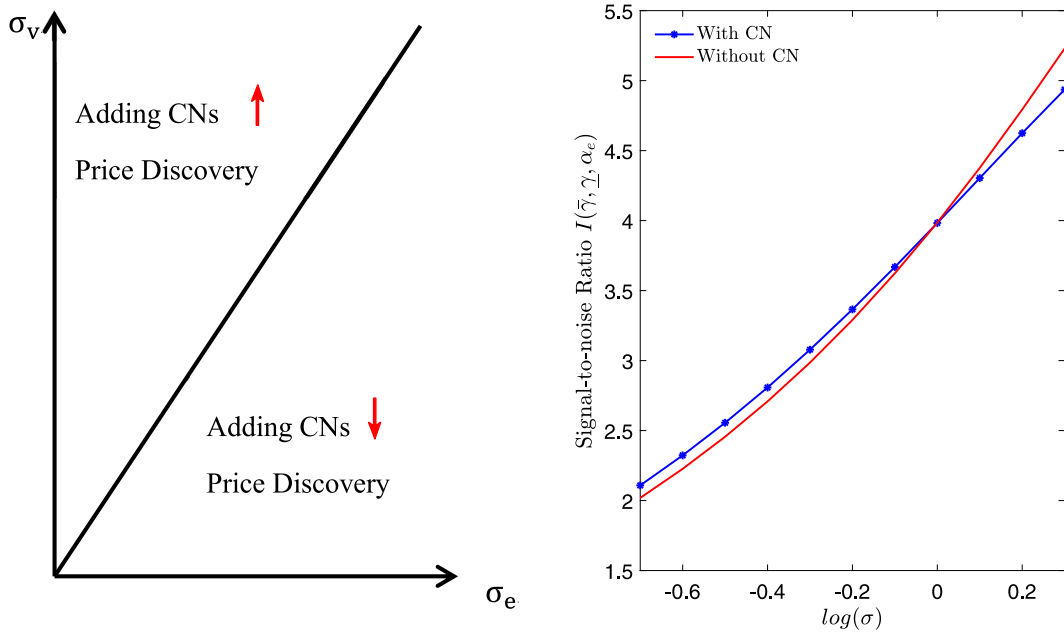


Fig. 7. Price discovery. The left figure shows the crossing network impact on price discovery with 2-dimensions: σ_v and σ_e . The right figures shows the signal-to-noise ratio $I(\tilde{\gamma}, \underline{\gamma}, \alpha_e)$ as a function of $\sigma = \frac{\sigma_v}{\sigma_e}$.

By Proposition 2, such $\bar{\sigma}_v$ must exist. A higher $\bar{\sigma}_v$ reflects a higher fraction of assets whose price discovery will be harmed by adding a crossing network.

I analyze two factors and their associations with $\bar{\sigma}_v$: (i) the level of information precision, represented by the inverse of σ_e , and (ii) the relative proportion of informed traders, denoted as $\frac{\mu}{\mu_z}$. Proposition 3 provides a concise summary of my findings.

Proposition 3. *Crossing networks are prone to impair price discovery when there is lower information precision (higher σ_e) and a higher proportion of informed traders (higher $\frac{\mu}{\mu_z}$). That is, suppose $\hat{k} \leq \frac{\mu_z}{\mu} < +\infty$, where $\hat{k}$ is uniquely determined by $\hat{k} = \frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}}$, then*

- (i) $\bar{\sigma}_v$ increases in σ_e . As $\sigma_e \rightarrow 0^+$, $\bar{\sigma}_v \rightarrow 0$, and as $\sigma_e \rightarrow +\infty$, $\bar{\sigma}_v \rightarrow +\infty$. And,
- (ii) for any sequence of $\{(\frac{\mu}{\mu_z})_n\}$, there exists a subsequence $\{(\frac{\mu}{\mu_z})_n\}$ such that as $(\frac{\mu}{\mu_z})_n$ increases, $\bar{\sigma}_v$ increases, also, as $(\frac{\mu}{\mu_z})_n \rightarrow 0^+$, $\bar{\sigma}_v \rightarrow 0$.¹⁴

Proposition 3 reveals that the impact of crossing networks on price discovery depends on the information environment of the economy. Fig. 8 provides a numerical example illustrating this relationship. Crossing networks are more likely to enhance price discovery in an economy with a favorable information environment, characterized by higher information precision ($\frac{1}{\sigma_e}$) and a lower relative size of informed traders ($\frac{\mu}{\mu_z}$). On the other hand, crossing networks are more likely to impair price discovery in an economy with a poor information environment, characterized by lower information precision and a higher relative size of informed traders. These findings align with the research conducted by Balakrishnan and Taori (2017), which suggests that crossing network trading may harm price discovery. Regulators can use these insights when making decisions regarding the operation of crossing networks. More comprehensive analysis and evidence can be found in Section 7.3.

5.4. How market respond when information precision varies?

In the previous sections, I examined the impact of adding a crossing network alongside an exchange. In this section, I shift the focus to understanding how the market responds to changes in information precision within a given economy, whether it be a single or multi-venue setup.

Crossing networks offer a trade-off by reducing the informational risk for less well-informed traders. As information precision decreases (or increases), the demand for such risk mitigation increases (or decreases) accordingly, leading to the migration of traders between venues. Propositions 4 and 5 capture this behavior. Fig. 6 provides a numerical example illustrating these findings.

¹⁴ I cannot directly show that $\bar{\sigma}_v$ increases in $\frac{\mu}{\mu_z}$, but I can show an upper bound of $\bar{\sigma}_v$ that is increasing in $\frac{\mu}{\mu_z}$.

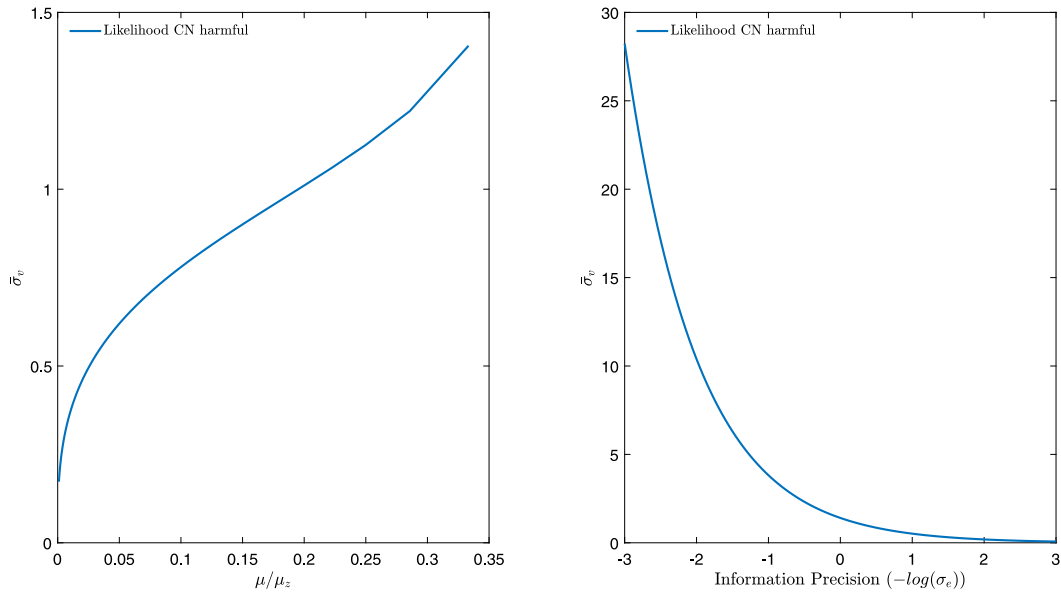


Fig. 8. The likelihood CNs harm price discovery $\bar{\sigma}_v$. The left-hand figure shows the threshold $\bar{\sigma}_v$ as a function of the relative size of informed traders, $\frac{\mu}{\mu_z}$. The right-hand figure shows the threshold $\bar{\sigma}_v$ as a function of the information precision, $-\log(\sigma_e)$. On the left, $\log(\sigma_e) = 0$. On the right, $\frac{\mu}{\mu_z} = 0.2$.

Proposition 4 (Transaction Costs). Suppose σ_e is sufficiently large. As information precision increases, both the exchange spreads and the adverse selection costs in the crossing network also increase. Specifically, as σ_e decreases:

- (Without a CN): The exchange spread $\frac{A^S}{\sigma_v}$ strictly increases.
- (With a CN): Both the exchange spread $\frac{A}{\sigma_v}$ and the adverse selection costs in the crossing network $\hat{R} - \underline{R}$ increase.

Proposition 5 (Participation Rates). Suppose σ_e is sufficiently large. As information precision increases, both informed traders' participation in the exchange and the crossing network increases. By contrast, for uninformed traders, as information precision increases, their participation in the exchange decreases while their participation in the crossing network increases, leading to a decrease in total uninformed participation. Specifically, as σ_e decreases:

- (Without a CN): For informed traders, their “meaningful” participation in the exchange, $\bar{\gamma}_e^S - \underline{\gamma}_e^S$, strictly increases. For uninformed traders, their exchange participation, α_e^S , strictly decreases.
- (With a CN): For informed traders, their “meaningful” participation in both the exchange and the crossing network, $\bar{\gamma}_e - \underline{\gamma}_e$ and $\bar{\gamma}_d - \underline{\gamma}_d$, increases. For uninformed traders, their exchange participation, α_e , decreases, while their crossing network participation, α_d , increases. The total uninformed participation, $\alpha_e + \alpha_d$, decreases.¹⁵

Remark 1. When σ_e is large (as in Propositions 4 and 5), crossing network participation for informed traders and crossing network adverse selection costs increase with information precision. However, when σ_e is small, they may decrease with information precision. While explicit comparative statics are unable to be provided when σ_e is small, I observe this inverted U-shape relationship in the numerical example in Fig. 6.¹⁶ Although I offer an explanation within the context, the technical proof is beyond the scope of this discussion.

When the information precision level is low (i.e., high σ_e), an increase in precision leads to a concentration of signals within the relatively modest range. This results in more informed traders transitioning from not trading to the crossing network. Consequently, there is a higher proportion of informed participation in the crossing network, leading to an increase in CN adverse selection. Conversely, when the information precision level is high (i.e., low σ_e), an increase in precision concentrates signals within the relatively strong range. This prompts more informed traders to migrate from the crossing network to the exchange, causing a lower proportion of informed trades in the crossing network. As a result, the crossing network adverse selection decreases.

¹⁵ $\bar{\gamma}_e - \underline{\gamma}_e$ and $\bar{\gamma}_d - \underline{\gamma}_d$ capture the “meaningful” participation of informed trades, representing the fractions of informed trades that trade in the right direction net the fractions that trade in the wrong direction.

¹⁶ In all the plots, parameters are set where $\mu_z = 60$, $\mu = 30$, Z^+ and Z^- follow gamma distributions with mean 30 and variance 30, and $G(d) = \frac{d}{3}$ for $\bar{d} \in [0, 3]$.

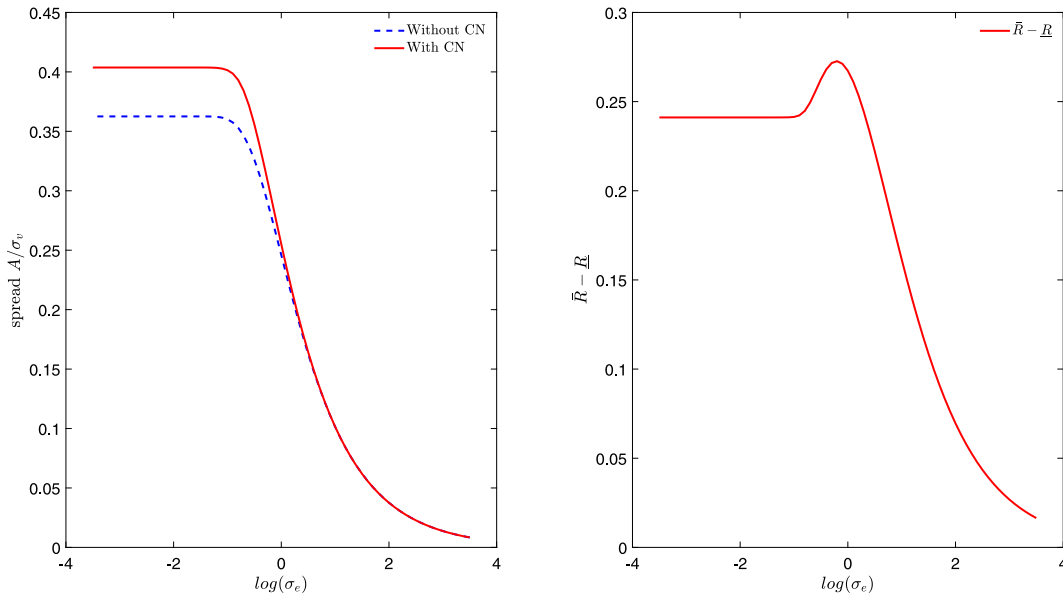


Fig. 9. Transaction costs. The left-hand figure shows the normalized spreads on the exchange, with and without a crossing network, and how they vary with $\log(\sigma_e)$; the right-hand figure shows the adverse selection cost in the crossing network and how it varies with $\log(\sigma_e)$. In both figures, $\log(\sigma_e) = 0$.

In the exchange, higher signal precision leads to increased participation of informed traders ($\bar{\gamma}_e - \gamma_e$) and wider exchange spreads (A), while uninformed trader participation (α_e) decreases. This occurs because higher signal precision strengthens informed traders' signals, prompting them to transition from not trading to trading in the crossing network and from trading in the crossing network to trading in the exchange. This increased information asymmetry within the exchange raises the exchange spread. Additionally, liquidity traders shift from trading in the exchange to trading in the crossing network, resulting in decreased uninformed participation.

In the crossing network, the participation of informed traders ($\bar{\gamma}_d - \gamma_d$) and the level of adverse selection ($\hat{R} - \underline{R}$) exhibit an inverted U-shaped relationship with signal precision, as shown in Fig. 9. When signal precision is low (high σ_e), an increase in precision concentrates signals in the relatively modest range, causing more informed traders to transition from not trading to the crossing network. This results in a higher proportion of informed participation and increased adverse selection in the crossing network. Conversely, when signal precision is high (low σ_e), increased precision concentrates signals in the relatively strong range, leading more informed traders to migrate from the crossing network to the exchange. Consequently, the proportion of informed trades in the crossing network decreases, reducing adverse selection.

My model offers distinctive predictions compared to previous studies, highlighting two major differences. First, regardless of the σ_v , my model predicts that both informed and uninformed traders migrate to crossing networks in a clustering fashion. This aligns with empirical evidence and challenges the notion that informed traders only trade in crossing networks for highly valuable assets (e.g., Zhu, 2014). As demonstrated in Fig. 1, there is no persistent gap between the average size of crossing networks and lit markets. This reinforces the notion that crossing networks serve as risk mitigators for information and remain lucrative for all types of traders, whether informed or uninformed.

Second, my model does not exhibit a squeeze-out effect in crossing networks. I predict that the participation of informed and liquidity traders in crossing networks can simultaneously increase, particularly when the information precision of informed traders is relatively low. By contrast, Zhu (2014) predicts that as informed traders' participation grows in crossing networks, it squeezes out liquidity traders, resulting in a decrease in α_d . This discrepancy arises from their extreme assumption of exact information for traders, leading to exceptionally high trading intensity by informed traders in the crossing network. In my model, however, this intensity is neutralized to some extent by speculators trading in the wrong direction.

5.5. Tradeoffs of utility and price discovery

In this subsection, I explore the relationship between traders' ex-ante expected utility and the level of price discovery in the economy. Fig. 10 provides a numerical illustration of this relationship in equilibrium.

Key observations can be made as follows. Firstly, in general, when information precision is high (high $-\log(\sigma_e)$), both informed and uninformed traders experience lower expected utility when a crossing network is introduced. Secondly, as information precision improves (i.e., $-\log(\sigma_e)$ increases), the expected utility of informed traders increases, while the expected utility of uninformed traders decreases, regardless of the presence of a crossing network. Thirdly, for informed traders, their expected utility may increase with

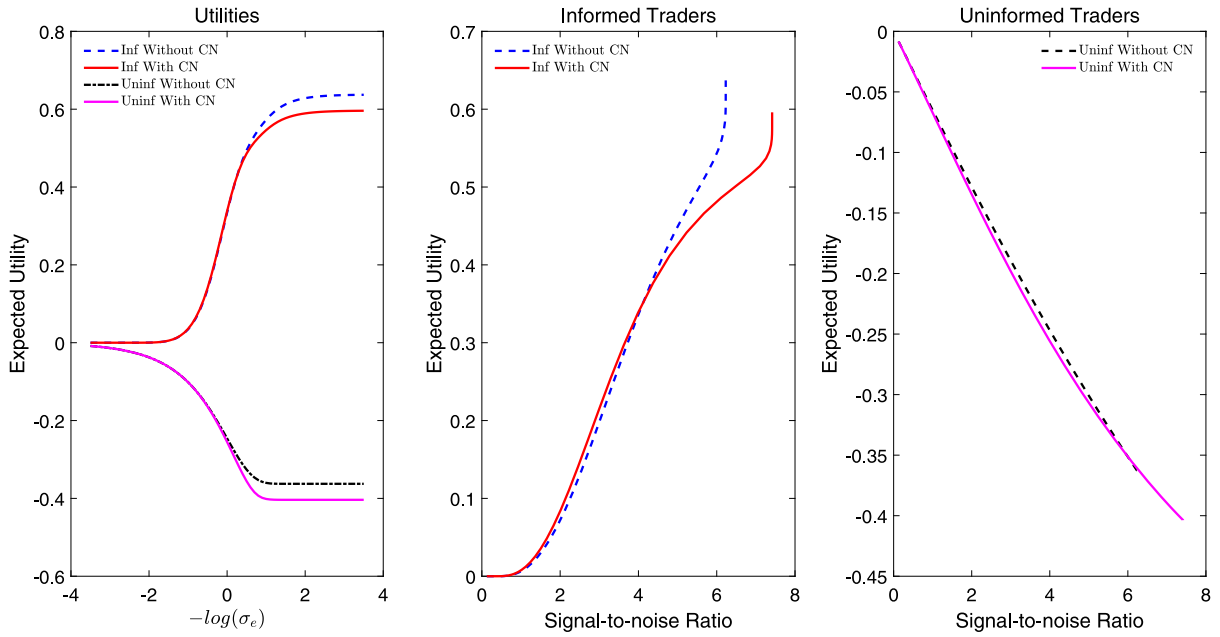


Fig. 10. The relationship of utility and price discovery. The left figure shows the expected utilities of the informed and uninformed traders as functions of $-\log(\sigma_e)$, with or without a crossing network. The middle and the right figures show the relationship between the traders' expected utility and the economy's signal-to-noise ratio, $I(\bar{\gamma}^*, \gamma^*, \alpha_e^*)$, for informed and uninformed traders, respectively.

the signal-to-noise ratio, indicating that improved information precision benefits both the level of price discovery and the expected utility of informed traders. However, for uninformed traders, this relationship is reversed, indicating a trade-off.

Furthermore, I find that the expected utility of informed traders (uninformed traders) generally exhibits faster growth (decrease) with the signal-to-noise ratio in an economy without crossing networks compared to an economy with a crossing network. However, it is worth noting that in the extreme case, the signal-to-noise ratio could be significantly higher when a crossing network is present.

Overall, these findings provide insight on the interplay between traders' expected utility and the economy's price discovery, highlighting the contrasting effects of information precision on informed and uninformed traders and the influence of crossing networks on the relationship.

6. Endogenous information acquisition

In previous sections, The equilibrium outcomes are examined based on an exogenous information precision level. In this section, I extend the analysis to an endogenous model where informed traders choose their optimal information precision level, σ_e^* , by weighing the associated benefits and costs. The “*” denotes the results of the endogenous model. The equilibrium in the endogenous model is formally defined in [Appendix C](#).

The crucial question is whether the preservation of the previous predictions holds under endogenous information acquisition. Through extensive numerical computations, I address this question and find that endogenous information acquisition does not alter the results. In the numerical computations, I assume a cost function of information precision, $C(\sigma_e) = .1(\frac{\sigma_e}{\sigma_0})^{-c}$, where $C(\sigma_e)$ decreases as σ_e increases. The coefficient c represents the difficulty of acquiring information. [Fig. 11](#) demonstrates that the amplification effect of crossing networks, which is observed in the exogenous model, remains present in the endogenous model. I specifically focus on two dimensions of information acquisition: the competition level, $\frac{\mu}{\mu_z}$, and the cost, c . The results indicate that when the economic environment encourages information acquisition (i.e., low $\frac{\mu}{\mu_z}$ or low information acquisition costs), crossing networks enhance price discovery. Conversely, when the environment discourages information acquisition (i.e., high $\frac{\mu}{\mu_z}$ or high information acquisition costs), crossing networks have a detrimental effect on price discovery. A comprehensive analysis of the optimal level of information precision and the impact of crossing networks on market liquidity and transaction costs is provided in [Appendix C](#).

7. Discussion: Empirical and regulatory implications

In the following analyses, I focus on the informed traders' information advantage, represented by $\sigma = \frac{\sigma_v}{\sigma_e}$, or the information imprecision, σ_e , when σ_v is fixed. I refer to a better information environment as one with more precise information and less informed traders. While I aim to attribute my findings to different information structures, I maintain a cautious interpretation of these predictions. It is important to note that crossing network activity and its effects exhibit substantial cross-sectional variation and should be assessed differently for various assets and economic conditions.

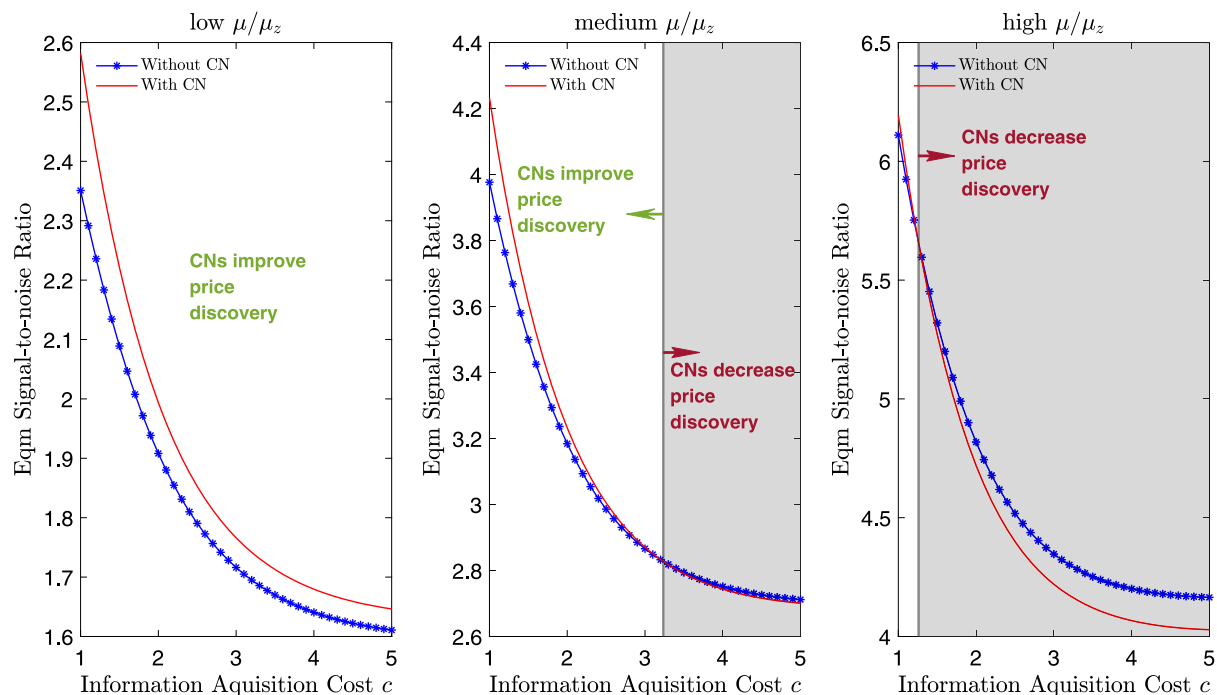


Fig. 11. The crossing network impact on price discovery in the endogenous model. The three figures show the equilibrium signal-to-noise ratio, $I(\tilde{y}^*, \gamma^*, \alpha^*)$, as a function of the information acquisition cost coefficient c . The three figures represent three different levels of $\frac{\mu}{\mu_z}$ (the size of the informed traders), $\frac{\mu}{\mu_z} = 0.23, 0.39$, and 0.61 , respectively.

7.1. Measurement for information precision

I now briefly discuss the measurement of information precision and the information environment in the literature. In this paper, two key factors are considered: the precision of private information and the proportion of informed traders. A better information environment is characterized by higher precision in traders' private information and a smaller number of informed traders.¹⁷ An examination of the measurement of information precision for individual stocks enables cross-sectional analysis. Moreover, an investigation of the information environment at the macroeconomic level informs regulatory considerations regarding crossing networks.

(i) Firm characteristics: Previous studies identify certain firm characteristics that are associated with lower levels of information precision in traders' predictions. These characteristics include greater growth volatility (e.g., high R&D firms, young firms), smaller size, and fewer analyst followers Lang and Lundholm (1993), Baginski and Hassell (1997), Li et al. (2012), Maffett (2012). Consequently, firm attributes such as age, size, number of analyst followers, R&D ratios, and measures of growth volatility can serve as proxies for the level of information precision in a firm's equity. Additionally, some researchers use variations and errors in analysts' forecasts as proxies for the level of information precision among traders (Gleason and Lee, 2003; Botosan et al., 2004).

(ii) Motivation of information acquisition The motivation for information acquisition can be influenced by various factors. One important factor is the profitability of trades and the costs associated with acquiring information. Frankel et al. (2006) find that the informativeness of analysts' reports depends on the trade's profitability and the costs of obtaining information. Additionally, the level of trader ability for information acquisition and processing tends to be higher in mature financial markets with increased competition, innovation in trading technologies, and years of trading experience (Clement, 1999; Chen et al., 2005; Louis et al., 2014). These factors can also serve as proxies for the information precision of individual stocks.

(iii) Information environment The measurement of the information environment at the macro-level considers factors such as the strength of legal institutions and law enforcement against insider trading, the effectiveness of the public disclosure system, and the availability and efficiency of media channels. Public disclosure and media channels generally contribute to improving the precision of informed traders' forecasts.¹⁸ Strong legal systems also play a significant role in reducing the presence of insiders.

¹⁷ Private information does not necessarily refer to insider information but includes publicly available information that is difficult for the majority of the public to collect, transmit, and process.

¹⁸ While the relationship between public and private information is subject to debate, researchers generally find a positive correlation between the precision of public and private information, suggesting that public disclosures can be processed into private information by informed investors. See Kim and Verrecchia (1994) and Botosan et al. (2004).

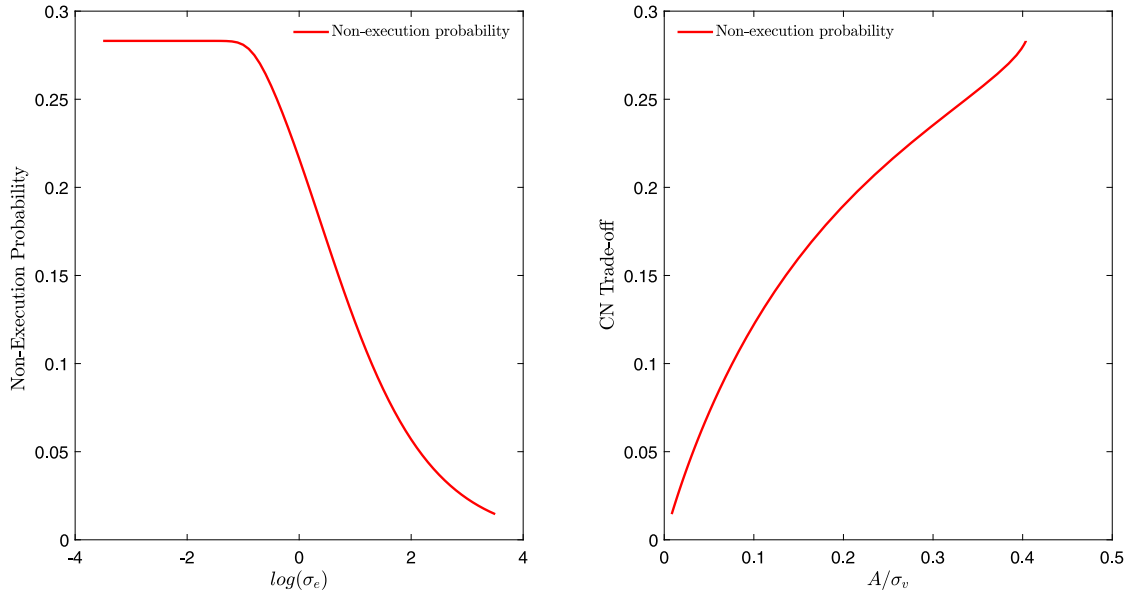


Fig. 12. Execution probability and trade-off of a crossing network. The left figure shows the non-execution probability as a function of $\log(\sigma_e)$. The right-hand figure shows the non-execution probability as a function of the exchange spread A/σ_v . In both figures, $\log(\sigma_v) = 0$.

7.2. A summary of empirical predictions

1. Trade-off of crossing networks: The model predicts that the price execution trade-off of crossing networks is more pronounced in an economy with a better information environment.¹⁹ Fig. 12 illustrates that as information precision (measured by $\frac{1}{\sigma_e}$) increases, the non-execution probability in crossing networks (i.e., $1 - \frac{\bar{R}+R}{2}$) also increases. Additionally, the exchange spread of an asset rises as its crossing network's non-execution probability (i.e., $1 - \frac{\bar{R}+R}{2}$) increases, as indicated by $\frac{A}{\sigma_v}$ in the figure.

This phenomenon can be explained by considering two factors that influence execution in crossing networks: the total participation of traders and the level of information asymmetry within the pool. Increased participation reduces the discrepancy between the two sides of the trade, resulting in higher execution rates. On the other hand, higher information asymmetry has the opposite effect, reducing execution rates. Consequently, when information precision improves, the exchange spread rises in tandem with the non-execution probability.

2. The relationship between crossing network market share and exchange spreads depends on the context. In an economy, industry, or asset with high (low) information precision, all else being equal, crossing network market share decreases (increases) as information precision and exchange spreads increase. Specifically, the relationship follows an inverted U-shape pattern for both information precision and exchange spreads. These predictions are based on Propositions 4, 5, and Remark 1.

To measure crossing network usage, I examine the volumes traded in each venue. Particularly, I consider the volumes in the crossing network, the exchange, and the total consolidated volume. After informed traders cancel their unexecuted orders and exit the market in period 2, the remaining orders are executed in the exchange. The expected trading volume in the crossing network, in the exchange, and total consolidated volume are, respectively:

$$V_d = (\bar{R}\underline{y}_d + \underline{R}\bar{y}_d)\mu + \frac{\bar{R}+R}{2}\alpha_d\mu_z, \quad (25)$$

$$V_e = (\underline{y}_e + \bar{y}_e)\mu + \alpha_e\mu_z + (1 - \alpha_e - \alpha_d)\mu_z + \left(1 - \frac{\bar{R}+R}{2}\right)\alpha_d\mu_z, \quad (26)$$

$$V = V_d + V_e. \quad (27)$$

I distinguish the components of dark volumes by dark uninformed volumes and dark informed volumes respectively as:

$$V_d^U = \left(1 - \frac{\bar{R}+R}{2}\right)\alpha_d\mu_z, \quad (28)$$

$$V_d^I = V_d - V_d^U. \quad (29)$$

¹⁹ Empirical studies consistently document this price execution trade-off, as highlighted by Conrad et al. (2003), Gresse (2006), Næs and Ødegaard (2006), Ye (2010), He and Lepone (2014), and Kwan et al. (2015).

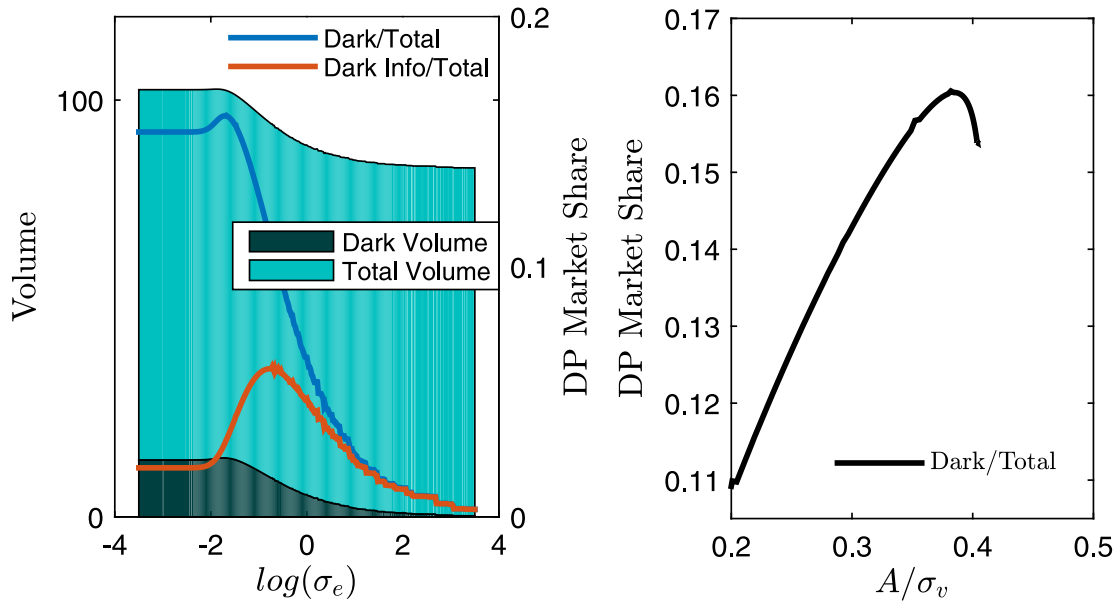


Fig. 13. Dark volumes and market share. The left figure shows the crossing network volume, total volume, and crossing network market share as a function of $\log(\sigma_e)$. The right-hand figure shows the crossing network market share as a function of the exchange spread A/σ_v . In both figures, $\log(\sigma_v) = 0$.

Fig. 13 illustrates the equilibrium behavior of crossing network market share and crossing network informed volume share.

This prediction highlights the significance of traders' information structure. It aligns with similar findings in Ray (2010) and Preece (2012), who also observe an inverted U-shaped relationship between dark pool usage and exchange spread. However, empirical studies using different datasets yield contradictory results. For instance, Weaver (2014), Brugler (2015), and Hatheway et al. (2017) find a positive association between dark trading and exchange spread, while O'Hara and Ye (2011), Ready (2014), and Foley and Putniņš (2016) report a negative association. Studies focusing on Australian dark trading by ASIC (2013) and Comerton-Forde and Putniņš (2015) find a positive relationship, while Degryse et al. (2015) find a positive relationship for European dark fragmentation. My model suggests that these seemingly contradictory findings can be attributed to the cross-sectional variation in the information structure of different datasets. This cross-sectional difference is reflected in studies such as O'Hara and Ye (2011), Nimalendran and Ray (2014), and Buti et al. (2022). More cross-sectional studies that consider firm characteristics and countries are needed to gain further insights.

3. Information content in crossing networks and its correlation with exchanges: The prediction is twofold. Firstly, it is expected that the information content of crossing network trades exhibits an inverted U-shaped relationship with information precision. Secondly, I anticipate an inverted U-shaped relationship between the information content of crossing network trades and the exchange spread, which serves as a proxy for liquidity. This prediction aims to establish a connection between typically unobservable measures and observable indicators such as exchange spreads. The basis for this prediction can be found in Propositions 4 and 5.

I employ two measures to examine the correlation between crossing network information content and exchanges. The first measure is the *CN Predictive Fraction*, which represents the fraction of crossing network volumes that exhibit predictive behavior, indicating higher information content. It is computed as follows:

$$CN \text{ Predictive Fraction} = \frac{R(\bar{\gamma}_d \mu + .5\alpha_d \mu_z)}{V_d}.$$

Additionally, I consider the normalized adverse selection costs, $\bar{R} - \bar{R}$, as another measure. Fig. 14 illustrates the inverted U-shape relationship between these measures and exchange liquidity, represented by normalized spreads. Surprisingly, those with intermediate exchange spreads may have higher levels of informative trades in crossing networks compared to those with smaller or larger spreads.²⁰ However, it is important to note that while execution rates can be estimated from traders' desks, the quote data in dark pools remains private.

4. Impacts of adding a crossing network. My predictions indicate that the addition of a crossing network alongside an exchange leads to two main effects: (i) a decrease in exchange volume and an increase in overall volume, and (ii) an amplification effect on price discovery. These predictions stem directly from Propositions 1, 2, and Remark 1.

²⁰ Limited research addresses this issue. Peretti and Tapiero (2014) concludes that dark pool trades can significantly forecast price movements. Nimalendran and Ray (2014) study trades in a large crossing network and find a positive association between information content in a dark pool and exchange spread. The study of Aquilina et al. (2016) partially documents this cross-sectional variation in dark pool information content.

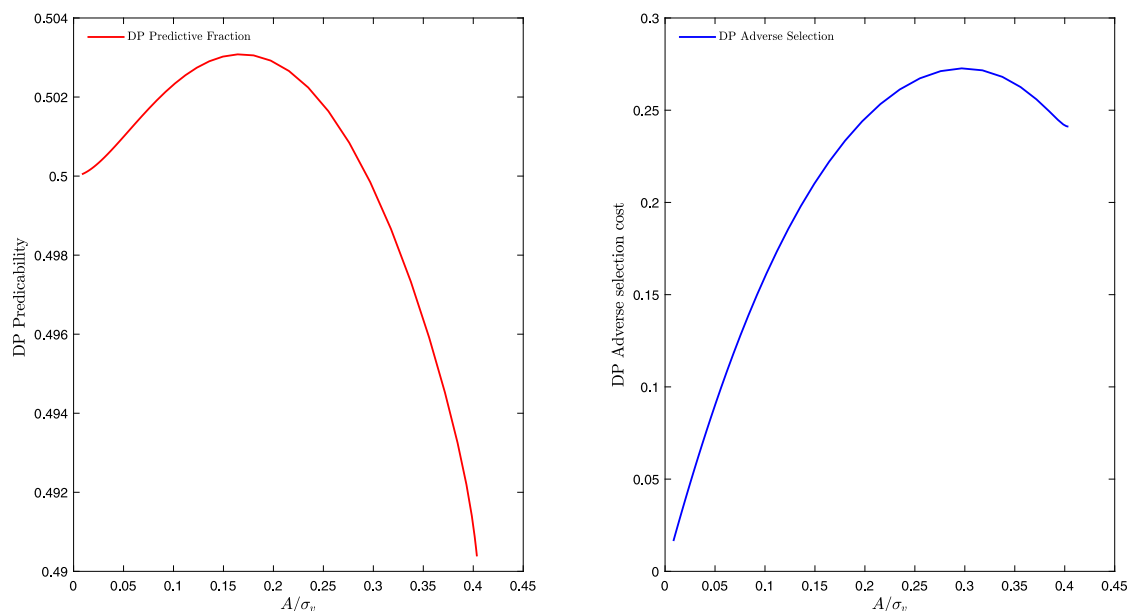


Fig. 14. Predictability of crossing network trades. The left figure shows the crossing network predictive fraction as a function of spread A/σ_v . The right-hand shows the crossing network adverse selection costs as a function of spread A/σ_v .

Limited research has directly examined the impact of introducing dark pool trading. For instance, studies such as [Hendershott and Mendelson \(2000\)](#) and [Hendershott and Jones \(2005\)](#) observed a reduction in price efficiency when Island ECN discontinued displaying its limit order book. Additionally, [Chlistalla and Lutat \(2011\)](#) find that the entrance of the dark pool Chi-X in the U.S. led to a decrease in spread.

Other studies explore the relationship between price discovery and the intensity of crossing network trading within a fragmented market structure. [O'Hara and Ye \(2011\)](#) and [Jiang et al. \(2012\)](#) reported a positive association between price discovery and crossing network trading, while [Weaver \(2014\)](#) and [Hatheway et al. \(2017\)](#) find the opposite. [Comerton-Forde and Putniņš \(2015\)](#) conduct a comprehensive cross-sectional study and show that when the fraction of non-block trades in crossing networks is high (above 10%, indicating a substantial presence of informational orders), dark trading hinders price discovery. Conversely, if crossing networks contain fewer non-block orders, dark trading improves price discovery. Their findings align with my prediction of an inverted U-shape relationship between crossing network information content (the ratio of informed orders) and information precision. Further research is still needed to explore the impact of crossing network activity on the price efficiency for different types of stocks across the market.

7.3. Regulatory considerations

This paper provides insight on the potential exacerbation of economic issues in fragmented markets due to the presence of crossing networks. Based on the analysis, I propose the following recommendations for a more balanced and efficient market environment that minimizes the adverse consequences associated with crossing network fragmentation.

First, in specific circumstances, regulators may contemplate the oversight of trading within crossing networks based on firm-specific attributes. Previous research reveals that companies characterized by high R&D intensity, youthful age, modest size, and limited analyst coverage generally exhibit diminished information precision. The introduction of crossing networks to such entities could potentially have an adverse impact on price discovery. In cases where these objectives hold significant importance, regulatory frameworks can take into account the distinctive characteristics of these firms.

Second, enhancing the information environment can mitigate the negative impacts of crossing networks. Measures such as a robust judicial system to combat insider trading and information leakage, an efficient public disclosure system encompassing improved accounting practices and financial reporting regulations, and more effective financial media channels can contribute to a healthier information environment. Supporting this recommendation, [Balakrishnan and Taori \(2017\)](#) provide direct evidence that the presence of crossing network trading, coupled with information leakage arising from broker-customer conflicts, can hamper or delay price discovery. [Boulton et al. \(2019\)](#) also document that disclosure opacity can affect the venue choice of traders.

Last, caution is advised when considering crossing network trading in emerging markets due to their inadequate legal systems, weak enforcement against insider trading, and poor quality of information disclosure. For example, [Bhattacharya and Daouk \(2002\)](#) find significantly lower enforcement of insider trading laws in 81 emerging markets compared to developed countries. Studies in mainland China by [Yu and Lu \(2009\)](#) and [Wang and Wu \(2011\)](#) reveal the low quality of financial information and

restatements of financial reports. In Taiwan, Tang et al. (2013) discovered a flawed corporate governance system exacerbating the information environment, and Budsaratragoon et al. (2012) identify severe informational asymmetry, lax enforcement, and low pricing efficiency in Thailand. Given the amplification effect of crossing networks on price discovery, introducing crossing network trading in these countries may exacerbate the situation. Therefore, careful consideration is necessary to address existing challenges before implementing crossing network trading in such markets.

8. Discussion: General types of dark pools

In this section, I briefly discuss different types of dark pools that do not primarily focus on mid-point execution. As dark pools have evolved, the trading mechanisms within them have expanded beyond mid-point execution, such as the crossing networks I examined. Many dark pools now incorporate a diverse combination of order types and execution mechanisms. They allow both peg orders and limit orders, with some enabling the formation of a certain equilibrium price within the National Best Bid and Offer (NBBO) range.²¹

The main difference between crossing networks and dark pools that allow limit pricing and market clearing within the NBBO is as follows: Crossing networks utilize a derived price to execute orders. As a result, only the execution rate in crossing networks can reflect the imbalance between buy and sell orders within the venue. On the other hand, dark pools that permit limit orders and facilitate market clearing establish an equilibrium price based on the demand and supply within the pool. It is important to note that this equilibrium price is bounded by the NBBO and may not fully reflect the overall demand and supply situation. In scenarios where orders are concentrated at the far touch of the NBBO, additional mechanisms are required for execution within these dark pools. Therefore, both the execution prices and execution rates within these dark pools collectively reflect the demand and supply imbalance. However, it is essential to highlight that the established equilibrium prices in these pools are not publicly displayed.

A crucial question arises regarding whether the main findings of my model can be extended to other types of dark pools that allow non-midpoint execution. While I exercise caution in drawing conclusions, I conjecture that the amplification effect of dark pools on price discovery would still hold in limit order dark pools. This expectation stems from the inherent trade-off between price improvement and execution risk, which underlies my results and remains relevant across different dark pool configurations, as long as they ensure executions within the NBBO. In these pools, where equilibrium prices are bound by the NBBO, orders on the heavier side will be partially rationed. This implies that execution risk in these pools remains higher than that in the lit exchange. With the anticipation of better prices but poorer execution rates in dark pools, traders, weighing the expected profit against execution rates, will make venue and order type choices. Considering the continuity of the model's setups, a new threshold of σ will determine different impacts of the dark pool, resulting in the amplification effect. Moreover, the participation of both informed and uninformed traders will increase as both informed and uninformed traders move together and limit orders are available in dark pools, the adverse selection costs will continue to be lower compared to the exchange.

9. Conclusion

In this paper, I examine the influence of crossing networks on price discovery and market quality within a noisy information framework. My findings demonstrate that the introduction of a crossing network alongside the traditional exchange has an amplification effect on price discovery. Specifically, it enhances price discovery in the presence of high-precision information and impairs price discovery when information precision is low. These effects persist even when information acquisition is endogenous. The results provide a reconciling perspective on the conflicting empirical findings in previous studies and highlight potential avenues for further research to unravel the intricate relationship between crossing network trading and market quality.

I highlight the crossing network's function as an informational risk mitigator. In equilibrium, information is sorted by market fragmentation. Traders with strong signals trade in the exchange, traders with modest signals trade in the crossing network, and traders with weak signals do not trade. When information precision is low, a large proportion of informed traders with modest signals crowd in the crossing network to reduce their information risk. Adding a crossing network, shifts a higher fraction of informed traders from the exchange, compared with liquidity traders, leaving a lower informed-to-uninformed trader ratio in the exchange and thus decreasing price discovery. By contrast, when information precision is high, a large proportion of informed traders with strong signals crowd in the exchange. Adding a crossing network shifts only a small fraction of informed traders from the exchange, compared to liquidity traders, increasing the informed-to-uninformed trader ratio in the exchange and increasing price discovery.

Several noteworthy observations complement the overall impact on market quality. Firstly, in situations of low information precision, the market may experience a decline in price discovery accompanied by an increased exchange spread. In such cases, crossing networks prove to be consistently detrimental to the exchange. Secondly, crossing networks consistently attract both informed and liquidity traders, who tend to cluster together. Therefore, I can expect to observe the presence of both informed and uninformed traders in all trading venues. Lastly, the ability of dark trades to forecast price movements follows an inverted U-shaped relationship with the exchange spread. Consequently, assets with moderate exchange liquidity exhibit a higher information content in their crossing network trades.

²¹ It is important to note that although these dark pools permit both pegged and limit orders, they still maintain some degree of adherence to the midpoint crossing principle. For example, CitiBLOC and MS RPool ATS6 exclusively allow midpoint executions, Luminex Trading & Analytics LLC permits only mid-peg orders, QUA treats limit orders as midpoint pegs with limits, and market orders as midpoint pegs. BIDS ATS, Citi-ONE ATS, MS RPool ATS4, and Sigma X2 select execution prices that are equal to or closest to the midpoint. Level ATS and CrossStream exclusively facilitate midpoint execution for certain order types. BlockCross and MS RPool ATS4 prioritize midpoint execution before considering other prices. Intelligent Cross LLC employs a midpoint discrete matching process to match orders, and so on. According to a 2022 report by Goldman Sachs SigmaX2, the second largest dark pool in the U.S., approximately 64% of its total executed shares involve orders pegged to the mid-price.

CRedit authorship contribution statement

Linlin Ye: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Acknowledgments

The work was partially supported by the Natural Science Foundation of China No. 71803168 and the Natural Science Foundation of Guangdong Province, China, No. 2018A030310052. The author thanks the editor and reviewer for their constructive suggestions. The author thanks Pierre-Oliver Weill, Avaniidhar Subrahmanyam, Tomasz Sadzik, Mark Garmaise, Haoxiang Zhu, Mao Ye, Joseph Ostroy, John Wiley, Obara Ichiro, Simon Board, Ivo Welch, Antonio Bernardo, Francis Longstaff, Bernard Herskovic, William Mann, Daniel Andrei, Shuyang Sheng, and Jernej Copic for valuable comments and discussions. The author also thanks Alex Kemmsies in Rosenblatt Rosenblatt Securities and Sayena Mostowfi in TABB Group for providing the industry data.

Appendix A. A summary of variables

Variable	Definition
$\tilde{v}$	The value of the risky asset, with mean 0 and std σ_v .
P_1	The period 1 closing price announced by the exchange.
s_i	Informed trader i 's signal.
σ_e	$\frac{1}{\sigma_e}$ measures the information precision.
e_i	The noise in trader i 's signal (mean 0 and std σ_e).
μ	The measure of the informed traders.
$Z^+(Z^-)$	The measure of the liquidity buyers (sellers), with mean $\frac{1}{2}\mu_z$.
d_j	(Uninformed) liquidity trader j 's delay cost, with cdf $G(\cdot)$.
A	The period 1 exchange trading spread.
$\bar{R}(\underline{R})$	The period 1 expected crossing network execution rates of trading in the wrong(right) direction.
$\alpha_e(\alpha_d)$	The fraction of liquidity traders in the exchange (crossing network).
$\bar{\gamma}_e(\underline{\gamma}_e)$	The fraction of informed traders trading in the right (wrong) direction in the exchange.
$\bar{\gamma}_d(\underline{\gamma}_d)$	The fraction of informed traders trading in the right (wrong) direction in the crossing network.
$\mathcal{I}(\bar{\gamma}_e, \underline{\gamma}_e, \alpha_e)$	Signal-to-noise ratio, a measure of price discovery in the exchange.

Appendix B. The proofs

B.1. Lemma 1 and the proof

Lemma 1 (Non-Split Orders and Trading Direction, Informed).²² *If an informed trader decides to trade, it is strictly optimal to buy if the informed trader's signal $s > 0$ and to sell if $s < 0$. Moreover, with probability one, an informed trader strictly prefers to send the entire order to one of the two venues, or do not trade at all.*

Proof. Since each trader is infinitesimal and orders are limited by the amount, their action has no impact on the market parameters (i.e., the exchange spread A and the crossing network execution probabilities $(\bar{R}, \underline{R})$). Therefore, splitting the order cannot affect the (per unit) profit in each venue. Without loss of generality, I focus on the case of a positive signal (the case for a negative signal is similar). Suppose that the informed traders have a signal $s > 0$. Then the informed trader has a belief $B(s) > \frac{1}{2}$. Because the profit of a buy order in each venue is strictly higher than the profit of a sell order, it is optimal to choose the buy direction. From their perspective, given the exchange spread A and the crossing network execution probabilities $(\bar{R}, \underline{R})$, the expected (per unit) profit for trading in the lit market, crossing network, and not trade depends on one's confidence level $B(|s|)$ and is determined by Eqs. (5), (6), and (7), respectively.

Because these payoffs are linear in $B(|s|)$, given any belief $B(|s|)$, there is always one venue that is no worse than any of the other venues. This relationship is shown in Fig. 3. When $s \neq \pm s_0$ or $\pm s_1$, the payoff of trading in one venue is strictly better than others, and it is optimal to send the entire order to that venue. When $s = \pm s_0$ or $\pm s_1$, there are two venues that yield the same payoff, and the trader can choose to split the order or not between these two venues. However, since the realization of the signal

²² A non-split order is strictly optimal for the traders.

among the informed traders is continuously distributed, the measure of informed traders who receive a particular signal is zero. That is, such traders who are indifferent to these two venues have a mass of zero in the market. Therefore, in probability one, all informed traders send the entire order to either the exchange or the crossing network, or do not trade at all.

The intuition is rather straightforward. Since there is a continuum of informed traders and each trader's signal is drawn from the same continuous distribution, signals are realized continuously by the law of large numbers. Therefore, beliefs are distributed continuously. Since no individual has an impact on the market, and the expected profit in each venue is linear in the agents' beliefs, it is with probability one that, for any informed trader with signal s , one venue (or not trade) is strictly better than others.

Furthermore, the magnitude of $B(|s|)$ reflects the probability that this trading direction is right. Thus, $|s|$ can be regarded as the strength of one's signal, and $B(|s|)$ can be regarded as the agent's confidence level in their information. A strong signal (i.e., a high $|s|$) represents a strong belief that the trading direction is right, whereas weak signals (i.e., low $|s|$) represents a weak belief in the trading direction. In the next section, I will show how much credit an informed trader gives to the trader's private information is crucial in determining the optimal strategies of venue selection. $\square$

B.2. Lemma 2 and the proof

Lemma 2 (No Split Orders, Uninformed). A liquidity trader (buyer or seller) strictly prefers to send the entire order to one of the venues or delay trade.

Proof. From a type d liquidity trader's perspective, the expected per unit payoff from trading in the lit market, crossing network, and completely deference are determined by Eqs. (8), (9), and (10), respectively. Since each individual has no impact to the market, given A , $\bar{R}$, $\underline{R}$, the per unit payoff in each venue is fixed. There is always one venue that is no worse than the others. In addition, the payoff is linear in the number of units transacted. Hence there is no need to split among different venues or different periods. $\square$

B.3. Proof of Theorem 1

Hereafter, some variables are normalized, dividing by σ_e , i.e., let $s = \frac{s}{\sigma_e}$, $s_0 = \frac{s_0}{\sigma_e}$, $s_1 = \frac{s_1}{\sigma_e}$, $\hat{s} = \frac{\hat{s}}{\sigma_e}$, $\sigma = \frac{\sigma_v}{\sigma_e}$. Then it is equivalent to prove that, given $\sigma \geq 0$, there is a unique cut-off $\hat{s}$ such that $h_I^S(s)$, $h_{U,i}^S(d)$, A^S , $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S consist an equilibrium, in which

$$h_I^S(s) = \begin{cases} (\text{buy, exchange (lit)}) & \text{if } s \geq \hat{s}, \\ (\text{sell, exchange (lit)}) & \text{if } s < -\hat{s}, \\ \text{do not trade} & \text{otherwise,} \end{cases} \quad (\text{B.1})$$

$$h_{U,i}^S(d) = \begin{cases} (\text{buy, exchange (lit)}) & \text{if } d \geq 2B(\hat{s}) - 1, i = \text{buyer,} \\ (\text{sell, exchange (lit)}) & \text{if } d \geq 2B(\hat{s}) - 1, i = \text{seller,} \\ \text{delay trade} & \text{otherwise,} \end{cases} \quad (\text{B.2})$$

$$\bar{\gamma}_e^S = 1 - \Phi(\hat{s} - \sigma), \quad (\text{B.3})$$

$$\underline{\gamma}_e^S = 1 - \Phi(\hat{s} + \sigma), \quad (\text{B.4})$$

$$\alpha_e^S = 1 - G(2B(\hat{s}) - 1), \quad (\text{B.5})$$

$$\frac{A^S}{\sigma_v} = \frac{\bar{\gamma}_e^S - \underline{\gamma}_e^S}{\bar{\gamma}_e^S + \underline{\gamma}_e^S + \alpha_e^S \frac{\mu_z}{\mu}}, \quad (\text{B.6})$$

where $\hat{s}$ is determined by:

$$2B(\hat{s}) - 1 = \frac{A^S}{\sigma_v}. \quad (\text{B.7})$$

The proof of the theorem in two steps. First, I show that if $\hat{s}$ is given, the other variables $h_I^S(s)$, $h_{U,i}^S(d)$, A^S , $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S solved from (B.1)–(B.6) form an equilibrium. Then I show that such $\hat{s}$ exists and is unique.

Suppose that $\hat{s}$ exists. By Eq. (B.7), an informed trader with signal $\hat{s}$ is indifferent between trading in the exchange or not trading. Since $B(s)$ is increasing in s , $h_I^S(s)$ is an optimal strategy for informed traders. Similarly, since a type $\hat{d} = 2B(\hat{s}) - 1$ uninformed liquidity trader is indifferent between trading on the exchange and deferring trade, $h_{U,i}^S(d)$ is an optimal strategy for uninformed traders. By the law of large numbers, given $h_I^S(s)$ and $h_{U,i}^S(d)$, the fraction of uninformed traders who trade in the exchange would be $\alpha_e^S = \Pr(d \geq \hat{d}) = 1 - G(2B(\hat{s}) - 1)$. Thus, the fraction of informed traders who trade in the right direction would be $\bar{\gamma}_e^S = \Pr(s \geq \hat{s}) = 1 - \Phi(\hat{s} - \sigma)$, and the fraction of informed traders who trade in the wrong direction would be $\underline{\gamma}_e^S = \Pr(s < \hat{s}) = 1 - \Phi(\hat{s} + \sigma)$. In addition, for a given $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S , A^S can be found from Eq. (B.6) because it would make the market maker on the exchange breaks even on average. Thus, $h_I^S(\hat{s})$, $h_{U,i}^S(\hat{d})$, A^S , $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S indeed form an equilibrium.

Then I prove that such $\hat{s}$ exists and is unique. After substituting the expressions of A^S , $\bar{\gamma}_e^S$, $\underline{\gamma}_e^S$, and α_e^S into equation (B.7), the following equation for $\hat{s}$ is obtained:

$$\frac{\Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma)}{2 - \Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma) + (1 - G(2B(\hat{s}) - 1)) \frac{\mu_z}{\mu}} = 2B(\hat{s}) - 1. \quad (\text{B.8})$$

Define

$$f(s) = (2B(s) - 1) \left[2 - \Phi(s + \sigma) - \Phi(s - \sigma) + (1 - G(2B(s) - 1)) \frac{\mu_z}{\mu} \right] - [\Phi(s + \sigma) - \Phi(s - \sigma)],$$

and the derivative of $f(s)$ is:

$$f'(s) = 2B'(s) \left[2 - \Phi(s + \sigma) - \Phi(s - \sigma) + (1 - G(2B(s) - 1)) \frac{\mu_z}{\mu} \right] - 2(2B(s) - 1)G'(2B(s) - 1)B'(s).$$

One can easily find that $f'(\frac{1}{2}) < 0$, $f(+\infty) > 0$, $f'(0) > 0$, $f'(+\infty) = 0$. Because $G'(x) + xG''(x) \geq 0$, $\forall x \in [0, 1]$, one has $f''(s) < 0$. Thus, there exists a unique $\hat{s}$ such that $f(\hat{s}) = 0$. $\square$

B.4. Proof of Theorem 2

Hereafter, some variables are normalized, dividing them by σ_e , i.e., $s = \frac{s}{\sigma_e}$, $s_0 = \frac{s_0}{\sigma_e}$, $s_1 = \frac{s_1}{\sigma_e}$, $\hat{s} = \frac{\hat{s}}{\sigma_e}$, $\sigma = \frac{\sigma}{\sigma_e}$. Then finding the equilibrium is equivalent to solving the following system of equations:

$$B(s_0)(\bar{R} + \underline{R}) = \bar{R}, \quad (\text{B.9})$$

$$B(s_1) \left[(1 - \bar{R}) + (1 - \underline{R}) \right] = \frac{A}{\sigma_v} + (1 - \bar{R}), \quad (\text{B.10})$$

$$\bar{R} = \mathbb{E} \left[\min \left\{ 1, \frac{\bar{\gamma}_d \mu + \alpha_d Z^+}{\gamma_d \mu + \alpha_d Z^-} \right\} \right], \quad (\text{B.11})$$

$$\underline{R} = \mathbb{E} \left[\min \left\{ 1, \frac{\gamma_d \mu + \alpha_d Z^-}{\bar{\gamma}_d \mu + \alpha_d Z^+} \right\} \right], \quad (\text{B.12})$$

$$\frac{A}{\sigma_v} = \frac{\bar{\gamma}_e - \underline{\gamma}_e}{(\bar{\gamma}_e + \underline{\gamma}_e) + \alpha_e \frac{\mu_z}{\mu}}, \quad (\text{B.13})$$

$$\bar{\gamma}_e = 1 - \Phi(s_1 - \sigma), \quad (\text{B.14})$$

$$\underline{\gamma}_e = 1 - \Phi(s_1 + \sigma), \quad (\text{B.15})$$

$$\bar{\gamma}_d = \Phi(s_1 - \sigma) - \Phi(s_0 - \sigma), \quad (\text{B.16})$$

$$\underline{\gamma}_d = \Phi(s_1 + \sigma) - \Phi(s_0 + \sigma), \quad (\text{B.17})$$

$$\alpha_e = 1 - G(2B(s_1) - 1), \quad (\text{B.18})$$

$$\alpha_d = G(2B(s_1) - 1) - G(2B(s_0) - 1), \quad (\text{B.19})$$

where:

$$B(s) = \frac{\phi(s - \sigma)}{\phi(s - \sigma) + \phi(s + \sigma)}. \quad (\text{B.20})$$

Before proving the existence of solutions to these equations, following lemma shall be introduced. This lemma can be used to define $\underline{R}$, $\bar{R}$, and $\frac{A}{\sigma_v}$, when $s_0 = s_1$.

Lemma 3. Let $s_0 \geq 0$ and $s_1 = s_0 + \epsilon$, then,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0^+} \underline{R} &= \mathbb{E} \left[\min \left\{ 1, \frac{\phi(s_0 + \sigma)\mu + 2G'(2B(s_0) - 1)B'(s_0)Z^-}{\phi(s_0 - \sigma)\mu + 2G'(2B(s_0) - 1)B'(s_0)Z^+} \right\} \right], \\ \lim_{\epsilon \rightarrow 0^+} \bar{R} &= \mathbb{E} \left[\min \left\{ 1, \frac{\phi(s_0 - \sigma)\mu + 2G'(2B(s_0) - 1)B'(s_0)Z^+}{\phi(s_0 + \sigma)\mu + 2G'(2B(s_0) - 1)B'(s_0)Z^-} \right\} \right], \\ \lim_{\epsilon \rightarrow 0^+} \frac{A}{\sigma_v} &= \frac{\Phi(s_0 + \sigma) - \Phi(s_0 - \sigma)}{2 - \Phi(s_0 + \sigma) - \Phi(s_0 - \sigma) + [1 - G(2B(s_0) - 1)] \frac{\mu_z}{\mu}}. \end{aligned}$$

Moreover, if $s_0 = 0$ or $\sigma = 0$, then $\lim_{\epsilon \rightarrow 0^+} \underline{R} = \lim_{\epsilon \rightarrow 0^+} \bar{R} = 1$.

Proof. Suppose that ϵ is sufficiently small. Because $s_0 \geq 0$ and $s_1 = s_0 + \epsilon$, by the Taylor expansion, $\bar{\gamma}_d = \phi(s_0 + \sigma)\epsilon + o(\epsilon)$, $\underline{\gamma}_d = \phi(s_0 - \sigma)\epsilon + o(\epsilon)$, and $\alpha_d = 2G'(2B(s_0) - 1)B'(s_0)\epsilon + o(\epsilon)$. Therefore:

$$\underline{R} = \mathbb{E} \left[\min \left\{ 1, \frac{\phi(s_0 + \sigma)\mu\epsilon + 2G'(2B(s_0) - 1)B'(s_0)Z^- \epsilon}{\phi(s_0 - \sigma)\mu\epsilon + 2G'(2B(s_0) - 1)B'(s_0)Z^+ \epsilon} + o(\epsilon) \right\} \right],$$

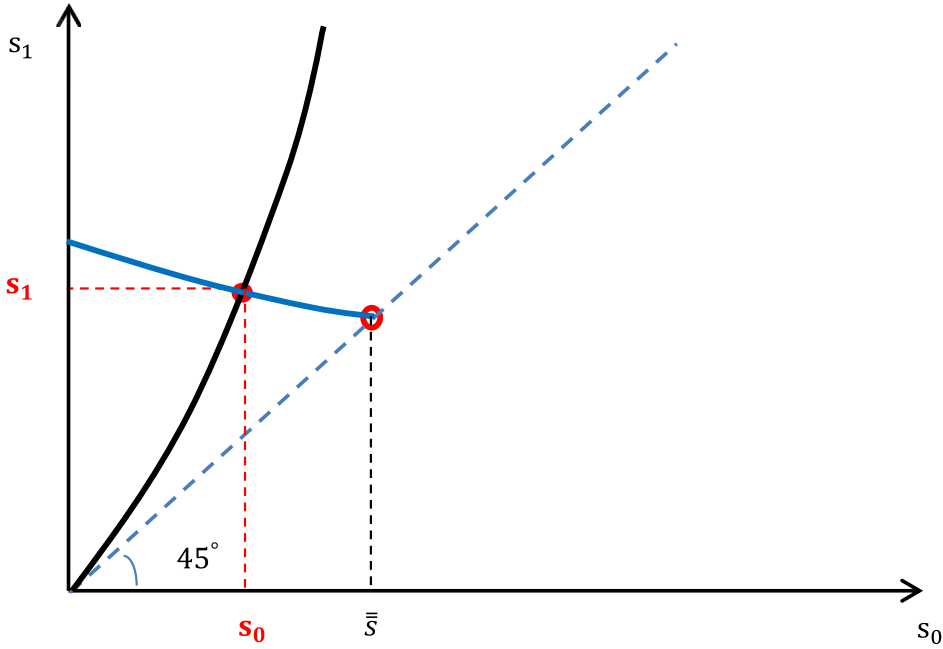


Fig. 15. Equilibrium existence. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$\bar{R} = E \left[\min \left\{ 1, \frac{\phi(s_0 - \sigma)\mu\epsilon + 2G'(2B(s_0) - 1)B'(s_0)Z^+\epsilon}{\phi(s_0 + \sigma)\mu\epsilon + 2G'(2B(s_0) - 1)B'(s_0)Z^-\epsilon} + o(\epsilon) \right\} \right].$$

Similarly, by the Taylor expansion, $\bar{\gamma}_e = 1 - \Phi(s_0 - \sigma) - \phi(s_0 - \sigma)\epsilon + o(\epsilon)$, $\underline{\gamma}_e = 1 - \Phi(s_0 + \sigma) - \phi(s_0 + \sigma)\epsilon + o(\epsilon)$, and $\alpha_e = [1 - G(2B(s_0) - 1)] - 2G'(\cdot)B'(s_0)\epsilon + o(\epsilon)$. Therefore:

$$\begin{aligned} \frac{A}{\sigma_v} = & \frac{\Phi(s_0 + \sigma) - \Phi(s_0 - \sigma) - [\phi(s_0 + \sigma) - \phi(s_0 - \sigma)]\epsilon}{2 - \Phi(s_0 + \sigma) - \Phi(s_0 - \sigma) + [1 - G(2B(s_0) - 1)]\frac{\mu_z}{\mu} - [\phi(s_0 + \sigma) + \phi(s_0 - \sigma) + 2G'(\cdot)B'(s_0)]\epsilon} \\ & + o(\epsilon). \end{aligned}$$

Let $\epsilon \rightarrow 0^+$, the lemma is proved. $\square$

The theorem can be proved in a similar way as in the proof of Theorem 1. First, I show that if s_0 and s_1 are given, then the other variables, $h_I(\cdot)$, $h_{U,i}(\cdot)$, A , $\bar{R}$, $\underline{R}$, $\bar{\gamma}_e$, $\underline{\gamma}_e$, $\bar{\gamma}_d$, $\underline{\gamma}_d$, α_d , and α_e , who are solved from Eqs. (B.11)–(B.19), will together form an equilibrium. Then I show that (s_0, s_1) exists and is unique.

Given A , $\bar{R}$, $\underline{R}$, $\bar{\gamma}_e$, $\underline{\gamma}_e$, $\bar{\gamma}_d$, $\underline{\gamma}_d$, α_d , and α_e , and given s_0 and s_1 who are determined by Eqs. (B.9), (B.10) and suppose $0 < s_0 < s_1$, it is optimal for informed speculators and uninformed liquidity buyers (and sellers) to following the strategy described respectively by $h_I(\cdot)$ and $h_{U,i}(\cdot)$, $i \in \{\text{buyer, seller}\}$.

Consider an informed speculator who receives a signal $s \geq 0$ (the case when $s \leq 0$ is symmetric with respect to the vertical axis, and hence the analysis is similar and skipped here). Suppose that $0 < s_0 < s_1$. From one perspective, the expected payoffs in the lit market, the crossing network, and no-trade are, respectively, $[B(s)\sigma_v - (1 - B(s))\sigma_v] - A$, $B(s)\underline{R}\sigma_v - (1 - B(s))\bar{R}\sigma_v$, and 0. Fig. 3(c) captures the payoff as a function of $B(s)$. As one can see in the graph, since the payoffs are linear with respect to $B(s)$, and $B(s)$ is strictly increasing with respect to s , the informed speculator with signal s should use the exchange (the lit market) to trade when the trader's signal $s \geq s_1$, and the crossing network when $s_0 \leq s < s_1$, and stay outside when $s < s_0$. This is marked as the red line in Fig. 3(c).

The fractions of each type of traders in each venue, $\bar{\gamma}_e$, $\underline{\gamma}_e$, $\bar{\gamma}_d$, $\underline{\gamma}_d$, α_e , and α_d , are determined by Eqs. (B.14), (B.15), (B.16), (B.17), (B.18), (B.19), respectively, and A , $\bar{R}$, and $\underline{R}$ are given by Eqs. (B.13), (B.11), (B.12). Thus properties (ii), (iii) and (iv) in Definition 2 are satisfied.

The next step is to show that such pair of cut-off (s_0, s_1) exists and satisfies $0 < s_0 < s_1$. In order to show this, consider Eqs. (B.9) and (B.10) and the intersection for the two lines represented by these two equations.

For Eq. (B.9), one can see that it behaves as the black line in Fig. 15 for the following reasons:

- (i) Suppose $s_0 = 0$, $s_1 = 0$, then $B(s_0) = \frac{1}{2}$, and by Lemma 3, $\bar{R} = \underline{R} = 1$. Therefore $(s_0, s_1) = (0, 0)$ satisfies Eq. (B.9).

- (ii) Now suppose $s_0 > 0$, then $\frac{1}{2} < B(s_0) < 1$. To satisfy (B.9), need that $\bar{R} < \bar{R} \leq 1$, thus $|\gamma_d| < |\bar{\gamma}_d|$. To obtain this, it must be true that $s_1 > s_0$ if such s_1 exists. This means that the line representing equation (B.9) must be above the 45 degree line. By continuity, such s_1 must exist for a small enough s_0 . (Note that if s_0 is too large, such s_1 may not exist.)
- (iii) Next, there exist some $\bar{s}$ such that $s_1 \rightarrow +\infty$ when $s_0 \rightarrow \bar{s}$. Eq. (B.9) can be rewritten as $B(s_0) = \frac{\bar{R}}{\bar{R} + \bar{R}}$. As $s_1 \rightarrow +\infty$, $\bar{\gamma}_d \rightarrow 1 - \Phi(s_0 - \sigma)$, $\gamma_d \rightarrow 1 - \Phi(s_0 + \sigma)$, $\alpha_d \rightarrow 1 - B(s_0)$. Hence $\bar{R} \rightarrow \mathbb{E} \left[\min \left\{ 1, \frac{1 - \Phi(s_0 - \sigma) + [1 - B(s_0)]Z^+}{1 - \Phi(s_0 + \sigma) + [1 - B(s_0)]Z^-} \right\} \right]$, and $\underline{R} \rightarrow \mathbb{E} \left[\min \left\{ 1, \frac{1 - \Phi(s_0 + \sigma) + [1 - B(s_0)]Z^-}{1 - \Phi(s_0 - \sigma) + [1 - B(s_0)]Z^+} \right\} \right]$. Therefore, for any $s_0 \in [0, \infty)$, there must exist $\bar{\gamma}_d > \gamma_d$, thus $\bar{R} > \underline{R}$. Then let $s_1 \rightarrow +\infty$, the left-hand side of the equation, $B(s_0)$, is equal to $\frac{1}{2}$ if $s_0 = 0$, and is equal to 1 if $s_0 \rightarrow +\infty$. However, the right-hand side of the equation, $\frac{\bar{R}}{\bar{R} + \underline{R}}$, is greater than $\frac{1}{2}$ if $s_0 = 0$, and equal to $\frac{1}{2}$ if $s_0 \rightarrow +\infty$. This is because $\lim_{s \rightarrow +\infty} \frac{1 - \Phi(s_0 - \sigma)}{1 - B(s_0)} = \lim_{s \rightarrow +\infty} \frac{1 - \Phi(s_0 + \sigma)}{1 - B(s_0)} = 0$, so $\lim_{s \rightarrow +\infty} \bar{R} = \lim_{s \rightarrow +\infty} \underline{R} = \mathbb{E} \left[\min \left\{ 1, \frac{Z^+}{Z^-} \right\} \right]$. By continuity, there must exist an $\bar{s} \in (0, +\infty)$ such that, as $s_0 \rightarrow \bar{s}$, $s_1 \rightarrow +\infty$, LHS = RHS. That is, Eq. (B.9) is satisfied.

For Eq. (B.10), it can be rewritten as:

$$B(s_1) = \frac{\frac{A}{\sigma_v}}{(1 - \bar{R}) + (1 - \underline{R})} + \frac{(1 - \bar{R})}{(1 - \bar{R}) + (1 - \underline{R})}. \quad (\text{B.21})$$

- (i) If $s_0 = 0$, then there must exist a $s_1 > 0$ to satisfy Eq. (B.10). Note that for any given $\sigma \in (0, +\infty)$, $A > 0$ is satisfied. If $s_1 = 0$, then $B(s_1) = \frac{1}{2}$ and $\bar{R} = \underline{R} = 1$. Plugging into Eq. (B.10) gives us $A = 0$, which contradicts the fact that $A > 0$. If $s_1 < 0$, then $B(s_1) < \frac{1}{2}$, $\bar{\gamma}_d < \gamma_d$, and $0 < \bar{R} < \underline{R} < 1$ (I do not consider any $\bar{R}, \underline{R} < 0$). Hence $\frac{(1 - \bar{R})}{(1 - \bar{R}) + (1 - \underline{R})} > \frac{1}{2} > B(s_1)$. In order for Eq. (B.10) to be satisfied, it must be $A < 0$, which contradicts with that fact that $A > 0$. Then the existence of s_1 can be shown using the continuity of Eq. (B.21). Its left-hand side $B(s_1)$ is increasing in s_1 and $B(0) = \frac{1}{2}$, $\lim_{s_1 \rightarrow \infty} B(s_1) = 1$. If $s_1 = 0$, the right-hand side equals $\frac{\frac{A}{\sigma_v}}{(1 - \bar{R}) + (1 - \underline{R})} + \frac{1}{2} > \frac{1}{2}$. However, when $s_1 \rightarrow \infty$, $A \rightarrow 0$ and $1 > \bar{R} > \underline{R}$, hence the right-hand side equals $0 + \frac{1 - \bar{R}}{(1 - \bar{R}) + (1 - \underline{R})} < \frac{1}{2}$. By the continuity of Eq. (B.21), there must exist a $s_1 \in (0, +\infty)$ such that the equation is satisfied.
- (ii) Next, there exists an $\bar{s} > 0$ and small enough $\epsilon > 0$ such that for $s_0 = \bar{s}$, $s_1 = \bar{s} + \epsilon$, Eq. (B.10) is satisfied as $\epsilon \rightarrow 0^+$. Consider any $s_0 = s$, $s_1 = s + \epsilon$, when $\epsilon > 0$ is sufficiently small. By Lemma 3, Eq. (B.10) is equivalent to:

$$B(s) = \frac{\frac{A}{\sigma_v}}{(1 - \bar{R}) + (1 - \underline{R})} + \frac{(1 - \bar{R})}{(1 - \bar{R}) + (1 - \underline{R})}, \quad (\text{B.22})$$

where:

$$\begin{aligned} \frac{A}{\sigma_v} &= \frac{\Phi(s + \sigma) - \Phi(s - \sigma)}{2 - \Phi(s + \sigma) - \Phi(s - \sigma) + [1 - G(2B(s) - 1)] \frac{\mu_z}{\mu}}, \\ \underline{R} &= \mathbb{E} \left[\min \left\{ 1, \frac{\phi(s - \sigma)\mu + 2G'(2B(s) - 1)B'(s)Z^+}{\phi(s + \sigma)\mu + 2G'(2B(s) - 1)B'(s)Z^-} \right\} \right], \\ \bar{R} &= \mathbb{E} \left[\min \left\{ 1, \frac{\phi(s + \sigma)\mu + 2G'(2B(s) - 1)B'(s)Z^-}{\phi(s - \sigma)\mu + 2G'(2B(s) - 1)B'(s)Z^+} \right\} \right]. \end{aligned}$$

Consider s on $[0, \infty)$. The left-hand side of Eq. (B.22) increases with respect to s , with $B(0) = \frac{1}{2}$, and $\lim_{s \rightarrow \infty} B(s) = 1$. Now consider the right-hand side of Eq. (B.22). By Lemma 3, if $s \rightarrow 0^+$, the limit of the right-hand side is $\frac{\frac{A}{\sigma_v}}{(1 - \bar{R}) + (1 - \underline{R})} + \frac{1}{2} > \frac{1}{2}$. If $s \rightarrow \infty$, $A \rightarrow 0$ and $1 > \bar{R} > \underline{R}$, hence the limit of the right-hand side is $0 + \frac{1 - \bar{R}}{(1 - \bar{R}) + (1 - \underline{R})} < \frac{1}{2}$. By continuity, there must exist a $\bar{s} \in (0, \infty)$ such that Eq. (B.22) is satisfied at $(\bar{s}, \bar{s})$ (i.e., $s_0 = s_1 = \bar{s}$).

The above argument is summarized in Fig. 15. Given $\sigma > 0$ fixed, the black curve represents the (s_0, s_1) pairs that satisfy Eq. (B.9). It goes through the point $(0, 0)$, is always above the line $s_1 = s_0$, and $s_1 \rightarrow +\infty$ when $s_0 \rightarrow \bar{s}$. The red curve represents the (s_0, s_1) pairs that satisfy Eq. (B.10). When $s_0 = 0$, $s_1 \in (0, \infty)$. And there exists some $\bar{s} \in (0, +\infty)$ such that $s_0 = s_1 = \bar{s}$, satisfies Eq. (B.10). Then because all functions are continuous, there must exist a pair (s_0, s_1) , $0 < s_0 < s_1 < +\infty$, such that both Eqs. (B.9) and (B.10) are satisfied; It is the intersection of the black curve and the red curve in Fig. 15. The existence is then established. $\square$

B.5. Proof of Corollary 3

Proof. If $\sigma \in (0, +\infty)$, by Theorem 2, $0 < s_0 < s_1$. Therefore by definition of (4), and (1), (2), (3), (16), (17), (18), (19), (24), it must be that $\frac{A}{\sigma_v}, \alpha_d, \alpha_e \in (0, 1)$ and $0 < \underline{\gamma}_e < \bar{\gamma}_e < 1, 0 < \underline{\gamma}_d < \bar{\gamma}_d < 1$. Therefore, $0 < \underline{R} < \bar{R} < 1$. $\square$

B.6. Proof of Proposition 1

To prove Proposition 1, following lemmas are needed.

Lemma 4. For any given $\sigma \in (0, +\infty)$, $\hat{s}(\sigma) < s_1(\sigma)$.

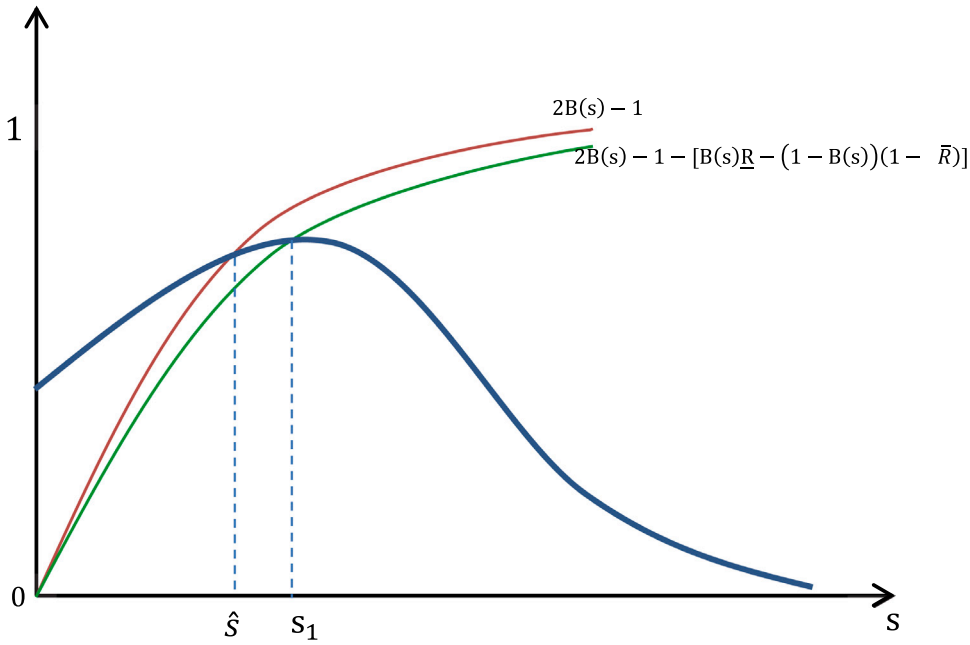


Fig. 16. $h(s)$ and $\hat{h}(s)$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Proof. Substitute the expressions of $\frac{A}{\sigma_v}$ into Eqs. (B.7) and (B.10), then $\hat{s}$, s_1 are respectively determined by the following two equations:

$$\begin{aligned} \frac{\Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma)}{2 - \Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma) + (1 - G(2B(\hat{s}) - 1))\frac{\mu_z}{\mu}} &= 2B(\hat{s}) - 1, \\ \frac{\Phi(s_1 + \sigma) - \Phi(s_1 - \sigma)}{2 - \Phi(s_1 + \sigma) - \Phi(s_1 - \sigma) + (1 - G(2B(s_1) - 1))\frac{\mu_z}{\mu}} &= 2B(s_1) - 1 \\ &\quad - [B(s_1)\underline{R} - (1 - B(s_1))\bar{R}]. \end{aligned}$$

Let $f(s) = \frac{\Phi(s+\sigma) - \Phi(s-\sigma)}{2 - \Phi(s+\sigma) - \Phi(s-\sigma) + (1 - G(2B(s) - 1))\frac{\mu_z}{\mu}}$, and its derivative is:

$$f'(s) = \frac{D_1(s) + D_2(s)}{\left[2 - \Phi(s + \sigma) - \Phi(s - \sigma) + (1 - G(2B(s) - 1))\frac{\mu_z}{\mu}\right]^2},$$

where:

$$\begin{aligned} D_1(s) &= (\phi(s + \sigma) - \phi(s - \sigma)) \left(2 - \Phi(s + \sigma) - \Phi(s - \sigma) + (1 - G(2B(s) - 1))\frac{\mu_z}{\mu} \right) < 0, \\ D_2(s) &= -(\Phi(s + \sigma) - \Phi(s - \sigma)) \left(-\phi(s + \sigma) - \phi(s - \sigma) - 2G'(2B(s) - 1)B'(s)\frac{\mu_z}{\mu} \right) > 0. \end{aligned}$$

and since $G'(s) + sG''(s) \geq 0$, one can check that $D'_1(s) < 0$, $D'_2(s) < 0$, so that $D'_1(s) + D'_2(s) < 0$. Also, $f'(0) > 0$. Therefore we can represent $f(s)$ as the blue curve in Fig. 16. Since $G'(s) + sG''(s) \geq 0$, one can represent $f(s)$ as the blue curve in Fig. 16.

Let $\hat{h}(s) = 2B(s) - 1$ and $h(s) = 2B(s) - 1 - [B(s)\underline{R} - (1 - B(s))\bar{R}]$. By Eq. (B.9), for any $s > s_0$, $B(s) > B(s_0) = \frac{\bar{R}}{\underline{R} + \bar{R}}$. That is, $[B(s)\underline{R} - (1 - B(s))\bar{R}] > 0$. Therefore, $\hat{h}(s) > h(s)$. In Fig. 16, $\hat{h}(s)$ is represented by the red curve, while $h(s)$ is represented by the green curve, which is below $\hat{h}(s)$. Obviously, the intersection point s_1 is larger than $\hat{s}$. The Lemma is proved. $\square$

Lemma 5. If $\sigma \rightarrow +\infty$, there exists a unique $\hat{k} \in (\frac{1}{2}, 1)$ such that $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e^S = 1$, $\lim_{\sigma \rightarrow +\infty} \underline{\gamma}_e^S = 0$, $\lim_{\sigma \rightarrow +\infty} \alpha_e^S = 1 - G(\hat{k})$, and $\lim_{\sigma \rightarrow +\infty} \frac{A^S}{\sigma_v} = \hat{k}$, where $\hat{k}$ is determined by:

$$\hat{k} = \frac{1}{1 + \left[1 - G(\hat{k})\right]\frac{\mu_z}{\mu}}. \quad (\text{B.23})$$

In addition, such $\hat{k}$ is smaller if $\frac{\mu_z}{\mu}$ is larger.

Proof. Suppose $\lim_{\sigma \rightarrow +\infty} \hat{s}\sigma = +\infty$. Then, when $\sigma \rightarrow +\infty$, $2B(\hat{s}) - 1 \rightarrow 1$. Thus equation (B.7) gives that $\frac{A^S}{\sigma_v} = 1$. However, $\hat{\alpha}_e = 1 - G(1) > 0$, which implies $\frac{A^S}{\sigma_v} < 1$. Therefore:

$$\lim_{\sigma \rightarrow +\infty} \hat{s}\sigma < +\infty.$$

Let $\lim_{\sigma \rightarrow +\infty} \hat{s}\sigma = \hat{C} \in [0, +\infty)$, where $\hat{C}$ will be determined later. Then $\lim_{\sigma \rightarrow +\infty} \hat{s} = 0$ and $\lim_{\sigma \rightarrow +\infty} (\hat{s} - \sigma) = -\infty$. Therefore, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e^S = 1$, $\lim_{\sigma \rightarrow +\infty} \gamma_e^S = 0$. Let $\hat{k} = \lim_{\sigma \rightarrow +\infty} 2B(\hat{s}) - 1 = \frac{1-e^{-2\hat{C}}}{1+e^{-2\hat{C}}}$, then $\lim_{\sigma \rightarrow +\infty} \alpha_e^S = 1 - G(\hat{k})$ and $\lim_{\sigma \rightarrow +\infty} \frac{A^S}{\sigma_v} = \frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}}$. However, $\hat{k}$ has to satisfy equation (B.23) such that Eq. (B.7) is satisfied.

Let $f(k) = k - \frac{1}{1+[1-G(k)]\frac{\mu_z}{\mu}}$, one can easily verify that $f(0) < 0$, and $f(1) > 0$. Therefore, there exists a $\hat{k} \in (0, 1)$ such that $f(\hat{k}) = 0$, and $\hat{C} = \frac{1}{2} \ln \frac{1+\hat{k}}{1-\hat{k}}$. $\square$

Lemma 6. Let $R = \mathbb{E} \left[\min \left\{ 1, \frac{Z^+}{Z^-} \right\} \right]$. Consider any equilibrium $s_0(\sigma), s_1(\sigma)$ for $\sigma \rightarrow +\infty$, then $\lim_{\sigma \rightarrow +\infty} s_0\sigma < +\infty$. In addition, the limits of variables can be determined in the following two statements:

- (i) If $\lim_{\sigma \rightarrow +\infty} s_1\sigma < +\infty$, then $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e = 1$, $\lim_{\sigma \rightarrow +\infty} \gamma_e = 0$, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_d = 0$, $\lim_{\sigma \rightarrow +\infty} \gamma_d = 0$, $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(k_1)$, $\lim_{\sigma \rightarrow +\infty} \alpha_d = G(k_1)$, $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = \frac{1}{1+[1-G(k_1)]\frac{\mu_z}{\mu}}$, $\lim_{\sigma \rightarrow +\infty} \bar{R} = R$, and $\lim_{\sigma \rightarrow +\infty} R = R$, where $k_1 \in (\frac{1}{2}, 1)$ is determined by:

$$(1 - R)k_1 = \frac{1}{1 + [1 - G(k_1)] \frac{\mu_z}{\mu}}. \quad (\text{B.24})$$

- (ii) If $\lim_{\sigma \rightarrow +\infty} s_0\sigma = +\infty$, then $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e = 1 - k_3$, $\lim_{\sigma \rightarrow +\infty} \gamma_e = 0$, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_d = k_3$, $\lim_{\sigma \rightarrow +\infty} \gamma_d = 0$, $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(1)$, $\lim_{\sigma \rightarrow +\infty} \alpha_d = G(1) - G(2k_2 - 1)$, $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = \frac{1 - k_3}{1 - k_3 + [1 - G(1)]\frac{\mu_z}{\mu}}$, $\lim_{\sigma \rightarrow +\infty} \bar{R} = \frac{k_2}{1 - k_2} \frac{[1 - G(1)]\frac{\mu_z}{\mu}}{1 - k_3 + [1 - G(1)]\frac{\mu_z}{\mu}}$, and $\lim_{\sigma \rightarrow +\infty} R = \frac{[1 - G(1)]\frac{\mu_z}{\mu}}{1 - k_3 + [1 - G(1)]\frac{\mu_z}{\mu}}$, where $k_2 \in [\frac{1}{2}, 1)$ and $k_3 \in [0, 1)$ are determined by:

$$\frac{[1 - G(1)]\frac{\mu_z}{\mu}}{1 - k_3 + [1 - G(1)]\frac{\mu_z}{\mu}} = \mathbb{E} \left[\min \left\{ 1, \frac{Z^-}{\frac{k_3}{G(1) - G(2k_2 - 1)} + Z^+} \right\} \right], \quad (\text{B.25})$$

$$k_2 = \frac{\mathbb{E} \left[\min \left\{ 1, \frac{\frac{k_3}{G(1) - G(2k_2 - 1)} + Z^+}{Z^-} \right\} \right]}{\mathbb{E} \left[\min \left\{ 1, \frac{\frac{k_3}{G(1) - G(2k_2 - 1)} + Z^+}{Z^-} \right\} \right] + \mathbb{E} \left[\min \left\{ 1, \frac{Z^-}{\frac{k_3}{G(1) - G(2k_2 - 1)} + Z^+} \right\} \right]}. \quad (\text{B.26})$$

Proof. Consider any continuously differentiable functions $s_0(\sigma), s_1(\sigma)$.

The first step is to show $\lim_{\sigma \rightarrow +\infty} s_0\sigma < +\infty$ by contradiction. Suppose that $\lim_{\sigma \rightarrow +\infty} s_0\sigma = +\infty$, then $B(s_0) = \frac{1}{1+e^{-2s_0\sigma}} \rightarrow 1$. Since $s_1 > s_0$, $\lim_{\sigma \rightarrow +\infty} s_1\sigma = +\infty$ and $B(s_1) = \frac{1}{1+e^{-2s_1\sigma}} \rightarrow 1$. In addition, Eq. (B.9) gives $\frac{\bar{R}}{R} = 1$, i.e., $\bar{R} = R$.

If $\lim_{\sigma \rightarrow +\infty} (s_0 - \sigma) < +\infty$, then $\bar{\gamma}_d > 0 = \gamma_d$, which is a contradiction to $\bar{R} = R$. If $\lim_{\sigma \rightarrow +\infty} (s_0 - \sigma) = +\infty$, then $\lim_{\sigma \rightarrow +\infty} s_1 - \sigma = +\infty$. Therefore, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e = \lim_{\sigma \rightarrow +\infty} \gamma_e = 0$, $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = 0$, and by Eq. (B.10), $\lim_{\sigma \rightarrow +\infty} B(s_1) = \lim_{\sigma \rightarrow +\infty} \frac{1 - \bar{R}}{1 - \bar{R} + 1 - R} = \frac{1}{2}$, which is a contradiction to $\lim_{\sigma \rightarrow +\infty} B(s_1) = 1$. Therefore:

$$\lim_{\sigma \rightarrow +\infty} s_0\sigma = C \in [0, +\infty).$$

The next step is to show the two statements;

- (i) Suppose that $\lim_{\sigma \rightarrow +\infty} s_0\sigma = C_0 \in [0, +\infty)$ and $\lim_{\sigma \rightarrow +\infty} s_1\sigma = C_1 \in [0, +\infty)$, then $\lim_{\sigma \rightarrow +\infty} (s_0 - \sigma) \rightarrow -\infty$ and $\lim_{\sigma \rightarrow +\infty} (s_1 - \sigma) \rightarrow -\infty$. Therefore, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e = 1$, $\lim_{\sigma \rightarrow +\infty} \gamma_e = \lim_{\sigma \rightarrow +\infty} \bar{\gamma}_d = \lim_{\sigma \rightarrow +\infty} \gamma_d = 0$.

To show that $\lim_{\sigma \rightarrow +\infty} \bar{R} = \lim_{\sigma \rightarrow +\infty} R = R$, if $C_0 = C_1$, Lemma 3 and $\lim_{\sigma \rightarrow +\infty} \frac{\phi(s_0 - \sigma)}{B'(s_0)} = \lim_{\sigma \rightarrow +\infty} \frac{\phi(s_0 + \sigma)}{B'(s_0)} = 0$ gives that $\lim_{\sigma \rightarrow +\infty} \bar{R} = \lim_{\sigma \rightarrow +\infty} R = R$. If $C_0 < C_1$, because $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_d = \lim_{\sigma \rightarrow +\infty} \gamma_d = 0$ and $\lim_{\sigma \rightarrow +\infty} \alpha_d > 0$, $\lim_{\sigma \rightarrow +\infty} \bar{R} = \lim_{\sigma \rightarrow +\infty} R = \mathbb{E} \left[\min \left\{ 1, \frac{Z^+}{Z^-} \right\} \right] = R$.

Then equation (B.9) gives that $\lim_{\sigma \rightarrow +\infty} B(s_0) = 1/2$ and $\lim_{\sigma \rightarrow +\infty} s_0\sigma = 0$. Let $k_1 = \lim_{\sigma \rightarrow +\infty} 2B(s_1) - 1$, then $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(k_1)$, $\lim_{\sigma \rightarrow +\infty} \alpha_d = G(k_1)$ and $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = \frac{1}{1+[1-G(k_1)]\frac{\mu_z}{\mu}}$. Eq. (B.10) can then be rewritten in the following form:

$$(2B(s_1) - 1)(1 - R) = \frac{1}{1 + [1 - G(2B(s_1) - 1)] \frac{\mu_z}{\mu}},$$

and k_1 has to satisfy equation (B.24).

Let $f(k) = (1-R)k - \frac{1}{1+[1-G(k)]\frac{\mu_z}{\mu}}$. one can verify that $f(0) < 0$ and $f(1) > 0$ if $1 + [1-G(1)]\frac{\mu_z}{\mu} > \frac{1}{1-R}$. There is a $k_1 \in (0, 1)$ such that $f(k_1) = 0$, and $C_1 = \frac{1}{2} \ln \frac{1+k_1}{1-k_1}$.

- (ii) Suppose that $\lim_{\sigma \rightarrow +\infty} s_0 \sigma = C_2 \in [0, +\infty)$ and $\lim_{\sigma \rightarrow +\infty} s_1 \sigma = +\infty$. Then one have $\lim_{\sigma \rightarrow +\infty} \gamma_e = 0$, $\lim_{\sigma \rightarrow +\infty} \gamma_d = 0$, and $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(1)$. Suppose that $\lim_{\sigma \rightarrow +\infty} (s_1 - \sigma) = C_3 \in [-\infty, +\infty]$. Let $k_2 = \lim_{\sigma \rightarrow +\infty} B(s_0) = \frac{1}{1+e^{-2C_2}} \in [\frac{1}{2}, 1)$ and $k_3 = \lim_{\sigma \rightarrow +\infty} \bar{\gamma}_d = \Phi(C_3) \in [0, 1]$. Then $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e = 1 - k_3$, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_d = k_3$, $\lim_{\sigma \rightarrow +\infty} \alpha_d = G(1) - G(2k_2 - 1)$, $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = \frac{1-k_3}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}}$. Combining Eqs. (B.9) and (B.10), one has $\lim_{\sigma \rightarrow +\infty} \bar{R} = \frac{k_2}{1-k_2} \frac{[1-G(1)]\frac{\mu_z}{\mu}}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}}$, $\lim_{\sigma \rightarrow +\infty} R = \frac{[1-G(1)]\frac{\mu_z}{\mu}}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}}$. In addition, by Eqs. (B.11) and (B.12), k_2 and k_3 have to satisfy Eqs. (B.25) and (B.26).

Suppose that $1 + [1-G(1)]\frac{\mu_z}{\mu} \leq \frac{1}{1-R}$. For Eq. (B.25), the left-hand side is increasing with respect to k_3 , while the right-hand side is decreasing with respect to k_3 . In addition, when $k_3 = 0$, $LHS - RHS = \frac{[1-G(1)]\frac{\mu_z}{\mu}}{1+[1-G(1)]\frac{\mu_z}{\mu}} - \mathbb{E} \left[\min \left\{ 1, \frac{Z^-}{Z^+} \right\} \right] = 1 - R - \frac{1}{1+[1-G(1)]\frac{\mu_z}{\mu}} \leq 0$, and when $k_3 = 1$, $LHS - RHS = 1 - \mathbb{E} \left[\min \left\{ 1, \frac{Z^-}{G(1)-G(2k_2-1)+Z^+} \right\} \right] > 0$. Thus, given any $k_2 \in [\frac{1}{2}, 1)$, there exists a unique $k_3(k_2) \in (0, 1)$ that solves Eq. (B.25). Furthermore, as k_2 increases, the right-hand side of Eq. (B.25) decreases, thus $k_3(k_2)$ is decreasing with respect to k_2 . When $k_2 \rightarrow 1$, $k_3(k_2) \rightarrow 0$. Thus $\bar{R} \rightarrow \frac{[1-G(1)]\frac{\mu_z}{\mu}}{1+[1-G(1)]\frac{\mu_z}{\mu}} \geq 0$.

For Eq. (B.26), substitute k_3 with the expression solved from (B.25), and it becomes a function of k_2 only. When $k_2 = 1/2$, $k_3 \in [0, 1)$. Then $LHS - RHS \leq 0$. While when $k_2 \rightarrow 1$, $LHS - RHS \geq 0$. Therefore, there exist $k_2 \in [1/2, 1)$ and $k_3 \in [0, 1)$ such that Eqs. (B.25) and (B.26) are satisfied. Additionally, $C_2 = \frac{1}{2} \ln \frac{k_2}{1-k_2}$, $C_3 = \Phi^{-1}(k_3)$. $\square$

Now I proceed to prove Proposition 1. From Lemma 4, $\hat{s} < s_1$ for all $\sigma \in (0, +\infty)$. Thus, $\bar{\gamma}_e^S - \underline{\gamma}_e^S = \Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma) > \Phi(s_1 + \sigma) - \Phi(s_1 - \sigma) = \bar{\gamma}_e - \underline{\gamma}_e$ and $\alpha_e^S = 1 - G(2B(\hat{s}) - 1) > 1 - G(2B(s_1) - 1) = \alpha_e$.

Let $\hat{k}$, k_1 , k_2 , and k_3 as in Eqs. (B.23), (B.24), (B.25), and (B.26), and suppose $1 - R > \frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}}$, then as $\sigma \rightarrow +\infty$, by Lemma 5:

$$\lim_{\sigma \rightarrow +\infty} \frac{A^S}{\sigma_v} = \frac{1}{1 + [1 - G(\hat{k})]\frac{\mu_z}{\mu}},$$

$$\lim_{\sigma \rightarrow +\infty} \alpha_e^S = 1 - G(\hat{k}).$$

By Lemma 6(i), $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = \frac{1}{1+[1-G(k_1)]\frac{\mu_z}{\mu}}$ and $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(k_1)$. One can verify that $\hat{k} < k_1$ from Eqs. (B.23) and (B.24). Therefore $\frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}} < \frac{1}{1+[1-G(k_1)]\frac{\mu_z}{\mu}}$ and $1 - G(\hat{k}) > 1 - G(k_1)$. That is, $\lim_{\sigma \rightarrow +\infty} \frac{A^S}{\sigma_v} < \lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v}$. One can easily verify that $\lim_{\sigma \rightarrow +\infty} \alpha_e^S < \lim_{\sigma \rightarrow +\infty} \alpha_e + \alpha_d$.

By Lemma 6(ii), $\lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v} = \frac{1-k_3}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}}$ and $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(1)$. Then by Eq. (B.25), $\frac{1-k_3}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}} = 1 - \mathbb{E} \left[\min \left\{ 1, \frac{Z^-}{G(1)-G(2k_2-1)+Z^+} \right\} \right] > 1 - \mathbb{E} \left[\min \left\{ 1, \frac{Z^-}{Z^+} \right\} \right] = 1 - R$. Since $1 - R > \frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}}$, one has $\frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}} < \frac{1-k_3}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}}$. That is, $\lim_{\sigma \rightarrow +\infty} \frac{A^S}{\sigma_v} < \lim_{\sigma \rightarrow +\infty} \frac{A}{\sigma_v}$.

Since $\hat{k} < k_1$, $k_3 > 0$, it is proved that $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e^S - \underline{\gamma}_e^S \leq \lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e - \underline{\gamma}_e$, $\lim_{\sigma \rightarrow +\infty} \alpha_e^S \geq \lim_{\sigma \rightarrow +\infty} \alpha_e$.

Next, consider the case when $\sigma \rightarrow 0^+$. Recall that when $\sigma = 0$, $\frac{A^S}{\sigma_v} = \frac{A}{\sigma_v} = 0$. So one needs to compare their derivatives at 0. From the proof of Lemma 9, $\lim_{\sigma \rightarrow 0^+} \hat{s} = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(\hat{s})}{2-2\phi(\hat{s})+\frac{\mu_z}{\mu}}$, and $\lim_{\sigma \rightarrow 0^+} s_0 = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(s_1)}{2-2\phi(s_1)+\frac{\mu_z}{\mu}}$. Since $\frac{2\phi(s)}{2-2\phi(s)+\frac{\mu_z}{\mu}}$ decreases in s , either $\lim_{\sigma \rightarrow 0^+} s_0 < \lim_{\sigma \rightarrow 0^+} \hat{s} < \lim_{\sigma \rightarrow 0^+} s_1$, or $\lim_{\sigma \rightarrow 0^+} s_0 = \lim_{\sigma \rightarrow 0^+} \hat{s} = \lim_{\sigma \rightarrow 0^+} s_1$. Therefore there are two cases to consider: (i) $\lim_{\sigma \rightarrow 0^+} s_0 < \lim_{\sigma \rightarrow 0^+} \hat{s} < \lim_{\sigma \rightarrow 0^+} s_1$. Since $\frac{d\frac{A}{\sigma_v}}{d\sigma}$ increases in s when $\sigma \rightarrow 0^+$, $\frac{A^S}{\sigma_v} < \frac{A}{\sigma_v}$, as $\sigma \rightarrow 0^+$. (ii) $\lim_{\sigma \rightarrow 0^+} s_0 = \lim_{\sigma \rightarrow 0^+} \hat{s} = \lim_{\sigma \rightarrow 0^+} s_1$. In this case, $\frac{d\frac{A^S}{\sigma_v}}{d\sigma} = \frac{d\frac{A}{\sigma_v}}{d\sigma}$, it is undetermined whether $\frac{A^S}{\sigma_v} < \frac{A}{\sigma_v}$ or $\frac{A^S}{\sigma_v} > \frac{A}{\sigma_v}$, as $\sigma \rightarrow 0^+$. However, I cannot distinguish between case (i) and case (ii). $\square$

B.7. Proof of Proposition 2

As $\sigma \rightarrow +\infty$, $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e^S - \underline{\gamma}_e^S = 1$ and $\lim_{\sigma \rightarrow +\infty} \alpha_e^S = 1 - G(\hat{k})$. Consider the two cases in Lemma 6: (i) $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e - \underline{\gamma}_e = 1$ and $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(k_1)$. Thus $\lim_{\sigma \rightarrow +\infty} \frac{\bar{\gamma}_e^S - \underline{\gamma}_e^S}{\alpha_e^S} \leq \lim_{\sigma \rightarrow +\infty} \frac{\bar{\gamma}_e - \underline{\gamma}_e}{\alpha_e}$ because $\hat{k} < k_1$. (ii) $\lim_{\sigma \rightarrow +\infty} \bar{\gamma}_e - \underline{\gamma}_e = 1 - k_3$ and $\lim_{\sigma \rightarrow +\infty} \alpha_e = 1 - G(1)$. From $\frac{1}{1+[1-G(\hat{k})]\frac{\mu_z}{\mu}} < \frac{1-k_3}{1-k_3+[1-G(1)]\frac{\mu_z}{\mu}}$, $\frac{1}{1-G(\hat{k})} < \frac{1-k_3}{1-G(1)}$, i.e., thus $\lim_{\sigma \rightarrow +\infty} \frac{\bar{\gamma}_e^S - \underline{\gamma}_e^S}{\alpha_e^S} \leq \lim_{\sigma \rightarrow +\infty} \frac{\bar{\gamma}_e - \underline{\gamma}_e}{\alpha_e}$.

As $\sigma \rightarrow 0^+$, by Lemma 4, $\hat{s} < s_1$, $\forall \sigma > 0$. Since $\frac{\bar{\gamma}_e^S - \underline{\gamma}_e^S}{\alpha_e^S} = \frac{\Phi(\hat{s}+\sigma) - \Phi(\hat{s}-\sigma)}{1-G(2B(\hat{s})-1)}$ and $\frac{\bar{\gamma}_e - \underline{\gamma}_e}{\alpha_e} = \frac{\Phi(s_1+\sigma) - \Phi(s_1-\sigma)}{1-G(2B(s_1)-1)}$, to show that $\frac{\bar{\gamma}_e^S - \underline{\gamma}_e^S}{\alpha_e^S} > \frac{\bar{\gamma}_e - \underline{\gamma}_e}{\alpha_e}$ for small σ , it is sufficient to show that there exists $\bar{\sigma} > 0$, s.t. $\forall \sigma \in (0, \bar{\sigma})$, $\frac{\Phi(s+\sigma) - \Phi(s-\sigma)}{1-G(2B(s)-1)}$ decreases in s . If $\frac{\mu_z}{\mu} < +\infty$, Lemma 9 gives us $\lim_{\sigma \rightarrow 0^+} s_1 \geq \lim_{\sigma \rightarrow 0^+} \hat{s} > 0$. Also, recall that $\lim_{\sigma \rightarrow 0^+} s_1 \sigma = \lim_{\sigma \rightarrow 0^+} \hat{s} \sigma = 0$.

Now consider the derivative of $\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)}$ with respect to s .

$$\begin{aligned} \frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} &= \frac{(\Phi(s+\sigma)-\Phi(s-\sigma)) G'(2B(s)-1) \frac{e^{-2s\sigma}}{(1+e^{-2s\sigma})^2} 4\sigma}{(1-G(2B(s)-1))^2} \\ &\quad + \frac{(\phi(s+\sigma)-\phi(s-\sigma))(1-G(2B(s)-1))}{(1-G(2B(s)-1))^2}. \end{aligned} \quad (\text{B.27})$$

Let $M = \max_{s \in [0, +\infty)} \left[\frac{4G'(2B(s)-1)}{1-G(2B(s)-1)} + 1 \right] < +\infty$. If $\lim_{\sigma \rightarrow 0^+} s > 0$ and $\lim_{\sigma \rightarrow 0^+} s\sigma = 0$, there exists $\bar{\sigma} > 0$, such that $\forall \sigma \in (0, \bar{\sigma})$, $s > Me^2\sigma$ and $s\sigma < 1$. Therefore by the mean value theorem,

$$\begin{aligned} \frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} &< \frac{2\sigma [\phi(s-\sigma)4G'(2B(s)-1)\sigma + \phi(s+\sigma)(1-G(2B(s)-1))(-(s-\sigma))]}{(1-G(2B(s)-1))^2} \\ &< \frac{2\sigma \left\{ \phi(s-\sigma) \left[\frac{4G'(2B(s)-1)}{1-G(2B(s)-1)} + 1 \right] \sigma + \phi(s+\sigma)(-s) \right\}}{1-G(2B(s)-1)} \\ &= \frac{2\sigma \{ \phi(s+\sigma)Me^{2s\sigma}\sigma - \phi(s+\sigma)s \}}{1-G(2B(s)-1)} \\ &< 0. \end{aligned}$$

Thus $\exists \bar{\sigma} > 0$, such that $\forall \sigma \in (0, \bar{\sigma})$, $s \in [\hat{s}, s_1]$ and $\frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} < 0$. Since $\hat{s} < s_1$, $\frac{\Phi(\hat{s}+\sigma)-\Phi(\hat{s}-\sigma)}{1-G(2B(\hat{s})-1)} > \frac{\Phi(s_1+\sigma)-\Phi(s_1-\sigma)}{1-G(2B(s_1)-1)}$. the proposition is then proved. $\square$

B.8. Proof of Proposition 3

The following Lemma is needed to proceed the proof.

Lemma 7. $\forall \sigma > 0, \exists \bar{C}(\sigma) > 0$, such that $\forall 0 \leq s \leq \bar{C}(\sigma)$, $\frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} > 0$.

Proof. Consider (B.27), $\forall \sigma > 0$, $\left. \frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} \right|_{s=0} > 0$. Therefore, $\exists \bar{C}(\sigma) > 0$ such that $\forall 0 \leq s \leq \bar{C}(\sigma)$,

$$\frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} > 0. \quad \square$$

First I show that $\bar{\sigma}_v = \sup_{x>0} \left\{ x | \forall \sigma_v \in (0, x), \frac{\bar{\gamma}_e^{\bar{S}} - \bar{\gamma}_e^{\bar{S}}}{\alpha_e^{\bar{S}}} > \frac{\bar{\gamma}_e - \gamma_e}{\alpha_e} \right\}$ is increasing in σ_e . Because $\frac{\mu_z}{\mu} < +\infty$ and $\frac{\mu_z}{\mu}$ sufficiently large, according to Proposition 2, there must exist a $\hat{\sigma}$ such that $\bar{\sigma} = \sup_{x>0} \left\{ x | \forall \sigma \in (0, x), \frac{\bar{\gamma}_e^{\bar{S}} - \bar{\gamma}_e^{\bar{S}}}{\alpha_e^{\bar{S}}} > \frac{\bar{\gamma}_e - \gamma_e}{\alpha_e} \right\}$. By definition $\bar{\sigma} = \frac{\bar{\sigma}_v}{\sigma_e}$, i.e., $\bar{\sigma}_v = \bar{\sigma}\sigma_e$, where $\bar{\sigma}$ is a constant. $\bar{\sigma}_v$ is increasing in σ_e . As $\sigma_e \rightarrow 0^+$, $\bar{\sigma}_v \rightarrow 0$ and as $\sigma_e \rightarrow +\infty$, $\bar{\sigma}_v \rightarrow +\infty$.

Next I prove that if $\frac{\mu_z}{\mu}$ is large enough, there exists a subsequence $\{(\frac{\mu_z}{\mu})_i\}$ such that $\bar{\sigma}_v$ decreases as $(\frac{\mu_z}{\mu})_i$ increases.

Let $\bar{C}(\sigma)$ be defined as $\sup_x \left\{ x | \forall s \in (0, x), \frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} > 0 \right\}$. By Lemma 7, such $\bar{C}(\sigma)$ exists for all $\sigma > 0$. Note that if $\frac{\mu_z}{\mu} \rightarrow +\infty$, $\hat{s}, s_1 \rightarrow 0$. Therefore, as $\frac{\mu_z}{\mu}$ becomes sufficiently large, there exists $\sigma(\frac{\mu_z}{\mu})$, such that:

$$\hat{s}, s_1 < C(\sigma(\frac{\mu_z}{\mu})).$$

Thus, $\left. \frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} \right|_{s=\hat{s}} > 0$, $\left. \frac{d \left(\frac{\Phi(s+\sigma)-\Phi(s-\sigma)}{1-G(2B(s)-1)} \right)}{ds} \right|_{s=s_1} > 0$. And since $\hat{s} < s_1$, $\frac{\bar{\gamma}_e^{\bar{S}} - \bar{\gamma}_e^{\bar{S}}}{\alpha_e^{\bar{S}}} < \frac{\bar{\gamma}_e - \gamma_e}{\alpha_e}$

This is to say, when $\frac{\mu_z}{\mu}$ is sufficiently large, a upper bound of $\bar{\sigma}$ is found, i.e., $\bar{\sigma} < \sigma(\frac{\mu_z}{\mu})$.

Therefore, there exists a subsequence $\{(\frac{\mu_z}{\mu})_i\}$ such that, as $(\frac{\mu_z}{\mu})_i$ increases, $\sigma(\frac{\mu_z}{\mu})$ decreases, and $\bar{\sigma}$ decreases, and as $(\frac{\mu_z}{\mu})_i \rightarrow +\infty$, $\sigma(\frac{\mu_z}{\mu}) \rightarrow 0$, $\bar{\sigma} \rightarrow 0$. $\square$

B.9. Proofs of Propositions 4 and 5

To prove Propositions 4 and 5, the following two lemmas need to be established.

Lemma 8. Suppose $s(\sigma)$ is continuously differentiable over $(0, +\infty)$, and $\lim_{\sigma \rightarrow 0^+} s(\sigma)\sigma = 0$, then:

$$\lim_{\sigma \rightarrow 0^+} (\phi(s(\sigma) + \sigma) - \phi(s(\sigma) - \sigma)) s'(\sigma) = 0,$$

$$\lim_{\sigma \rightarrow 0^+} (\Phi(s(\sigma) + \sigma) - \Phi(s(\sigma) - \sigma)) s'(\sigma) = 0.$$

In addition,

- (i) If $\lim_{\sigma \rightarrow 0^+} s(\sigma) = \pm\infty$, $|\sigma s'(\sigma)| \leq s(\sigma)$ for sufficiently small σ .
- (ii) If $-\infty < \lim_{\sigma \rightarrow 0^+} s(\sigma) < +\infty$, $\lim_{\sigma \rightarrow 0^+} \sigma s'(\sigma) = 0$.

Proof. (i) Suppose that $\lim_{\sigma \rightarrow 0^+} s(\sigma) = +\infty$. There exists $\epsilon > 0$ such that $\forall \sigma \in (0, \epsilon)$, $s(\sigma)\sigma > 0$ and $\frac{d(s(\sigma)\sigma)}{d\sigma} > 0$. Thus $\frac{d(s(\sigma)\sigma)}{d\sigma} = \sigma s'(\sigma) + s(\sigma) \geq 0$, and

$$|\sigma s'(\sigma)| \leq |s(\sigma)|,$$

for $\sigma \in (0, \epsilon)$. Similarly, if $\lim_{\sigma \rightarrow 0^+} s(\sigma) = -\infty$,

$$|\sigma s'(\sigma)| \leq |s(\sigma)|,$$

for sufficiently small σ .

Therefore, by the mean value theorem:

$$\begin{aligned} \lim_{\sigma \rightarrow 0^+} (\phi(s(\sigma) + \sigma) - \phi(s(\sigma) - \sigma)) s'(\sigma) &= \lim_{\sigma \rightarrow 0^+} \int_{s(\sigma)-\sigma}^{s(\sigma)+\sigma} -xe^{-\frac{x^2}{2}} dx s'(\sigma) \\ &= \lim_{\sigma \rightarrow 0^+} -2\sigma s(\sigma) e^{-\frac{s(\sigma)^2}{2}} s'(\sigma). \end{aligned}$$

Because $|\sigma s'(\sigma)| \leq |s(\sigma)|$ and $\lim_{\sigma \rightarrow 0^+} \left| -2s(\sigma)^2 e^{-\frac{s(\sigma)^2}{2}} \right| = 0$,

$$\lim_{\sigma \rightarrow 0^+} (\phi(s(\sigma) + \sigma) - \phi(s(\sigma) - \sigma)) s'(\sigma) = 0.$$

Similarly,

$$\begin{aligned} \lim_{\sigma \rightarrow 0^+} (\Phi(s(\sigma) + \sigma) - \Phi(s(\sigma) - \sigma)) s'(\sigma) &= \lim_{\sigma \rightarrow 0^+} \int_{s(\sigma)-\sigma}^{s(\sigma)+\sigma} e^{-\frac{x^2}{2}} dx s'(\sigma) \\ &= \lim_{\sigma \rightarrow 0^+} 2\sigma e^{-\frac{s(\sigma)^2}{2}} s'(\sigma). \end{aligned}$$

Additionally, $\lim_{\sigma \rightarrow 0^+} \left| -2s(\sigma) e^{-\frac{s(\sigma)^2}{2}} \right| = 0$ gives

$$\lim_{\sigma \rightarrow 0^+} (\Phi(s(\sigma) + \sigma) - \Phi(s(\sigma) - \sigma)) s'(\sigma) = 0.$$

(ii) Suppose that $\lim_{\sigma \rightarrow 0^+} s < +\infty$. On the one hand, $\lim_{\sigma \rightarrow 0^+} \frac{d(s(\sigma)\sigma)}{d\sigma} = \lim_{\sigma \rightarrow 0^+} \sigma s'(\sigma) + \lim_{\sigma \rightarrow 0^+} s(\sigma) = \lim_{\sigma \rightarrow 0^+} \sigma s'(\sigma) + s(0)$. On the other hand,

$$\left. \frac{d(s(\sigma)\sigma)}{d\sigma} \right|_{\sigma=0} = \lim_{\sigma \rightarrow 0^+} \frac{s(\sigma)\sigma - 0}{\sigma - 0} = s(0). \quad (\text{B.28})$$

Thus,

$$\lim_{\sigma \rightarrow 0^+} \sigma s'(\sigma) = 0,$$

and

$$\begin{aligned} \lim_{\sigma \rightarrow 0^+} (\phi(s(\sigma) + \sigma) - \phi(s(\sigma) - \sigma)) s'(\sigma) &= \lim_{\sigma \rightarrow 0^+} \int_{s(\sigma)-\sigma}^{s(\sigma)+\sigma} -xe^{-\frac{x^2}{2}} dx s'(\sigma) \\ &= \lim_{\sigma \rightarrow 0^+} (-2\sigma s'(\sigma)) \cdot \lim_{\sigma \rightarrow 0^+} s(\sigma) e^{-\frac{s(\sigma)^2}{2}} \\ &= 0, \\ \lim_{\sigma \rightarrow 0^+} (\Phi(s(\sigma) + \sigma) - \Phi(s(\sigma) - \sigma)) s'(\sigma) &= \lim_{\sigma \rightarrow 0^+} \int_{s(\sigma)-\sigma}^{s(\sigma)+\sigma} -xe^{-\frac{x^2}{2}} dx s'(\sigma) \\ &= \lim_{\sigma \rightarrow 0^+} (\sigma s'(\sigma)) \cdot \lim_{\sigma \rightarrow 0^+} e^{-\frac{s(\sigma)^2}{2}} \\ &= 0. \quad \square \end{aligned}$$

Lemma 9. $\lim_{\sigma \rightarrow 0^+} \hat{s} = s^*$, where $s^* \in (0, +\infty)$ is determined by the following:

$$s = \frac{2\phi(s)}{2 - 2\Phi(s) + \frac{\mu_-}{\mu}}.$$

Proof. Because $G(\cdot)$, $\Phi(\cdot) \in C^2$. The implicit function theorem and the uniqueness of $\hat{s}$ show that $\hat{s}(\sigma)$ is a continuously differentiable function over $(0, +\infty)$.

When $\sigma = 0$, $\bar{\gamma}_e^S - \underline{\gamma}_e^S = 0$ and $\frac{A^S}{\sigma_v} = 0$. Eq. (B.7) gives that $B(\hat{s}) = \frac{1}{2}$ and $\hat{s}(\sigma)\sigma = 0$.

Recall that:

$$\frac{A^S}{\sigma_v} = \frac{\Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma)}{2 - \Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma) + (1 - G(2B(\hat{s}) - 1)) \frac{\mu_z}{\mu}}, \quad (\text{B.29})$$

$G(\cdot)$, $\Phi(\cdot) \in C^2$, and $\frac{A^S}{\sigma_v}$ is differentiable of σ over $(0, +\infty)$.

Taking the derivative, one can get

$$\begin{aligned} \frac{d\left(\frac{A^S}{\sigma_v}\right)}{d\sigma} &= \frac{(\phi(\hat{s} + \sigma) - \phi(\hat{s} - \sigma)) \frac{d\hat{s}}{d\sigma} + (\phi(\hat{s} + \sigma) + \phi(\hat{s} - \sigma))}{\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu}} \\ &\quad + \frac{[\Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma)] \left[(\phi(\hat{s} + \sigma) + \phi(\hat{s} - \sigma)) \frac{d\hat{s}}{d\sigma} + (\phi(\hat{s} + \sigma) - \phi(\hat{s} - \sigma)) \right]}{\left[\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu} \right]^2} \\ &\quad + \frac{2G'(2B(\hat{s}) - 1) \frac{\mu_z}{\mu} [\Phi(\hat{s} + \sigma) - \Phi(\hat{s} - \sigma)] \left(\frac{\partial B(\hat{s})}{\partial \hat{s}} \frac{d\hat{s}}{d\sigma} + \frac{\partial B(\hat{s})}{\partial \sigma} \right)}{\left[\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu} \right]^2}. \end{aligned}$$

Lemma 8 gives:

$$\lim_{\sigma \rightarrow 0^+} \frac{d\left(\frac{A^S}{\sigma_v}\right)}{d\sigma} = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(\hat{s})}{2 - 2\Phi(\hat{s}) + \frac{\mu_z}{\mu}}. \quad (\text{B.30})$$

On the other hand, from Eq. (B.7),

$$\frac{A^S}{\sigma_v} = 2B(\hat{s}) - 1.$$

Taking derivative with respect to σ gives:

$$\frac{d\left(\frac{A^S}{\sigma_v}\right)}{d\sigma} = 2B(\hat{s}) [1 - B(\hat{s})] \left(2\sigma \frac{d\hat{s}}{d\sigma} + 2\hat{s} \right).$$

Using Lemma 8 and $\lim_{\sigma \rightarrow 0^+} \hat{s}(\sigma)\sigma = 0$, one can obtain:

$$\lim_{\sigma \rightarrow 0^+} \frac{d\left(\frac{A^S}{\sigma_v}\right)}{d\sigma} = \lim_{\sigma \rightarrow 0^+} \left(\sigma \frac{d\hat{s}}{d\sigma} + \hat{s} \right). \quad (\text{B.31})$$

Combining Eqs. (B.30) and (B.31) gives:

$$\lim_{\sigma \rightarrow 0^+} \left(\sigma \frac{d\hat{s}}{d\sigma} + \hat{s} \right) = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(\hat{s})}{2 - 2\Phi(\hat{s}) + \frac{\mu_z}{\mu}}.$$

Suppose that $\lim_{\sigma \rightarrow 0^+} \hat{s} = +\infty$, then similar as the proof of Lemma 8, $\lim_{\sigma \rightarrow 0^+} \sigma \frac{d\hat{s}}{d\sigma} + \hat{s} > 0$, which contradicts with $\lim_{\sigma \rightarrow 0^+} \frac{2\phi(\hat{s})}{2 - 2\Phi(\hat{s}) + \frac{\mu_z}{\mu}} = 0$.

Then I need to show that the limit cannot be zero. Because the limit cannot be infinity, $\lim_{\sigma \rightarrow 0^+} \sigma s'(\sigma) = 0$ from Lemma 8. Let $f(s) = \frac{2\phi(s)}{2 - 2\Phi(s) + \frac{\mu_z}{\mu}} - s$. One can check that there is a unique $s^* \in (0, +\infty)$ such that $f(s^*) = 0$. Therefore,

$$\lim_{\sigma \rightarrow 0^+} \hat{s} = s^* \in (0, +\infty). \quad \square$$

I then proceed to prove the propositions.

Case I: Without a crossing network

By Lemmas 8 and 9, $\frac{A^S}{\sigma_v}$, $\hat{\alpha}_e$, $\bar{\gamma}_e^S$, and $\underline{\gamma}_e^S$ are differentiable functions of σ , and

$$\lim_{\sigma \rightarrow 0^+} \frac{d\left(\frac{A^S}{\sigma_v}\right)}{d\sigma} = s^* \in (0, +\infty).$$

Also, taking derivative of $B(\hat{s})$ with respect to σ gives:

$$\frac{dB(\hat{s})}{d\sigma} = \frac{\partial B(\hat{s})}{\partial \hat{s}} \frac{d\hat{s}}{d\sigma} + \frac{\partial B(\hat{s})}{\partial \sigma} = B(\hat{s}) (1 - B(\hat{s})) \left(2\sigma \frac{d\hat{s}}{d\sigma} + 2\hat{s} \right), \quad (\text{B.32})$$

and the derivative of $\hat{\alpha}_e$ is

$$\frac{d\hat{\alpha}_e}{d\sigma} = -G'(2B(\hat{s}) - 1)B(\hat{s})(1 - B(\hat{s})) \left(2\sigma \frac{d\hat{s}}{d\sigma} + 2\hat{s} \right).$$

When σ is sufficiently small,

$$\lim_{\sigma \rightarrow 0^+} \frac{d\hat{\alpha}_e}{d\sigma} = -\frac{G'(0)s^*}{2} \in (-\infty, 0).$$

Similarly, take derivative of $\bar{\gamma}_e^{\mathbb{S}} - \underline{\gamma}_e^{\mathbb{S}}$ with respect to σ gives:

$$\frac{d(\bar{\gamma}_e^{\mathbb{S}} - \underline{\gamma}_e^{\mathbb{S}})}{d\sigma} = [\phi(\hat{s} + \sigma) - \phi(\hat{s} - \sigma)] \frac{d\hat{s}}{d\sigma} + [\phi(\hat{s} + \sigma) + \phi(\hat{s} - \sigma)],$$

and letting $\sigma \rightarrow 0^+$, then:

$$\lim_{\sigma \rightarrow 0^+} \frac{d(\bar{\gamma}_e^{\mathbb{S}} - \underline{\gamma}_e^{\mathbb{S}})}{d\sigma} = 2\phi(\hat{s}) \in (0, +\infty).$$

Note that since $\sigma = \frac{\sigma_v}{\sigma_e}$, the following conclusions can be made:

Given σ sufficiently small, as σ_v increases (or σ_e decreases),

- (i) $\frac{A}{\sigma_e}$ strictly increases.
- (ii) $\bar{\gamma}_e^{\mathbb{S}} - \underline{\gamma}_e^{\mathbb{S}}$ strictly increases, and $\alpha_e^{\mathbb{S}}$ strictly decreases.

Case II, With a crossing network

Note that when $\sigma = 0$, $\bar{\gamma}_e = \underline{\gamma}_e$ and $\bar{\gamma}_d = \underline{\gamma}_d$. Therefore $\frac{A}{\sigma_v} = 0$ and $\bar{R} = \underline{R}$. Eqs. (B.9) and (B.10) show that $B(s_0) = \frac{1}{2}$ and $B(s_1) = \frac{1}{2}$. If $0 < \sigma < +\infty$, by Theorem 2, that $0 < s_0 < s_1 < \infty$. Therefore, $\bar{\gamma}_e > \underline{\gamma}_e$, $\bar{\gamma}_d > \underline{\gamma}_d$, $\frac{A}{\sigma_v} > 0$, $\bar{R} > \underline{R}$, and $\frac{1}{2} < B(s_0) < B(s_1) < 1$. Then it is sufficient to conclude the following:

Given σ sufficiently small, as σ_v increases (or as σ_e decreases),

- (i) $\frac{A}{\sigma_v}$ increases, and $\bar{R} - \underline{R}$ increases.
- (ii) $\bar{\gamma}_e - \underline{\gamma}_e$, $\bar{\gamma}_d - \underline{\gamma}_d$ increases, α_e decreases, and α_d increases.

Let (s_0, s_1) be any equilibrium. Since $G(\cdot)$, and $\Phi(\cdot)$ are twice differentiable, by the implicit function theorem, there exist continuously differentiable functions $s_0(\sigma)$, $s_1(\sigma)$ defined on $(0, +\infty)$.

When $\sigma \in (0, +\infty)$. By Eq. (B.9), $B(s_0) = \frac{\bar{R}}{\bar{R} + \underline{R}} \in (0, 1)$. Alternatively,

$$\frac{\bar{R}}{\underline{R}} = \frac{1}{\frac{1}{B(s_0)} - 1},$$

and the derivative can be found as:

$$\begin{aligned} \frac{d\left(\frac{\bar{R}}{\underline{R}}\right)}{d\sigma} &= \frac{1}{\underline{R}^2} \left[\frac{d\bar{R}}{d\sigma} \underline{R} - \frac{d\underline{R}}{d\sigma} \bar{R} \right] \\ &= \frac{1}{[1 - B(s_0)]^2} \left(\frac{\partial B(s_0)}{\partial s_0} \frac{ds_0}{d\sigma} + \frac{\partial B(s_0)}{\partial \sigma} \right) \\ &= \frac{B(s_0)}{1 - B(s_0)} \left(2\sigma \frac{ds_0}{d\sigma} + 2s_0 \right). \end{aligned}$$

Also, knowing $\lim_{\sigma \rightarrow 0^+} B(s_0) = \frac{1}{2}$ and $\lim_{\sigma \rightarrow 0^+} \bar{R} = \lim_{\sigma \rightarrow 0^+} \underline{R} = 1$, one has:

$$\lim_{\sigma \rightarrow 0^+} \frac{d\bar{R}}{d\sigma} - \lim_{\sigma \rightarrow 0^+} \frac{d\underline{R}}{d\sigma} = \lim_{\sigma \rightarrow 0^+} \left(2\sigma \frac{ds_0}{d\sigma} + 2s_0 \right).$$

Eq. (B.10) shows that:

$$2B(s_1) - 1 - [B(s_1)\underline{R} - (1 - B(s_1))\bar{R}] = \frac{A}{\sigma_v}.$$

Taking derivatives on both sides gives:

$$\begin{aligned} \frac{d\left(\frac{A}{\sigma_v}\right)}{d\sigma} &= (2 - \underline{R} - \bar{R})B(s_0)[1 - B(s_0)] \left(\sigma \frac{ds_1}{d\sigma} + s_1 \right) \\ &\quad + [1 - B(s_0)] \frac{d\bar{R}}{d\sigma} - B(s_0) \frac{d\underline{R}}{d\sigma}, \end{aligned}$$

and because $\frac{A}{\sigma_v} = \frac{\Phi(s_1 + \sigma) - \Phi(s_1 - \sigma)}{2 - \Phi(s_1 + \sigma) - \Phi(s_1 - \sigma) + (1 - G(2B(s_1) - 1)) \frac{\mu_z}{\mu}}$,

$$\begin{aligned} & \frac{d\left(\frac{A}{\sigma_v}\right)}{d\sigma} \\ &= \frac{(\phi(s_1 + \sigma) - \phi(s_1 - \sigma)) \frac{ds_1}{d\sigma} + (\phi(s_1 + \sigma) + \phi(s_1 - \sigma))}{\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu}} \\ &+ \frac{[\Phi(s_1 + \sigma) - \Phi(s_1 - \sigma)] \left[(\phi(s_1 + \sigma) + \phi(s_1 - \sigma)) \frac{ds_1}{d\sigma} + (\phi(s_1 + \sigma) - \phi(s_1 - \sigma)) \right]}{\left[\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu} \right]^2} \\ &+ \frac{2G'(2B(s_1) - 1) \frac{\mu_z}{\mu} [\Phi(s_1 + \sigma) - \Phi(s_1 - \sigma)] \left(\frac{\partial B(s_1)}{\partial s_1} \frac{ds_1}{d\sigma} + \frac{\partial B(s_1)}{\partial \sigma} \right)}{\left[\bar{\gamma}_e + \underline{\gamma}_e + \alpha_e \frac{\mu_z}{\mu} \right]^2}. \end{aligned}$$

Similarly to the procedures in the proof of Lemma 9, one obtains:

$$\lim_{\sigma \rightarrow 0^+} \frac{d\left(\frac{A}{\sigma_v}\right)}{d\sigma} = \frac{1}{2} \left(\lim_{\sigma \rightarrow 0^+} \frac{d\bar{R}}{d\sigma} - \lim_{\sigma \rightarrow 0^+} \frac{dR}{d\sigma} \right) = \lim_{\sigma_n \rightarrow 0^+} \left(\sigma \frac{ds_0}{d\sigma} + s_0 \right), \quad (\text{B.33})$$

and

$$\lim_{\sigma \rightarrow 0^+} \frac{d\left(\frac{A}{\sigma_v}\right)}{d\sigma} = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(s_1)}{2 - 2\Phi(s_1) + \frac{\mu_z}{\mu}}. \quad (\text{B.34})$$

Combining Eqs. (B.33) and (B.34) gives:

$$\lim_{\sigma \rightarrow 0^+} \left(\sigma \frac{ds_0}{d\sigma} + s_0 \right) = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(s_1)}{2 - 2\Phi(s_1) + \frac{\mu_z}{\mu}}$$

Suppose $\lim_{\sigma \rightarrow 0^+} s_0 = +\infty$. Using the similar argument as in the proof of Lemma 9, one obtains $\lim_{\sigma \rightarrow 0^+} \left(\sigma \frac{ds_0}{d\sigma} + s_0 \right) > 0$. However, as $s_0 \rightarrow +\infty$, $ws_1 \rightarrow +\infty$ and $\frac{2\phi(s_1)}{2 - 2\Phi(s_1) + \frac{\mu_z}{\mu}} \rightarrow 0$. This is a contradiction. Therefore, it must be that $\lim_{\sigma \rightarrow 0^+} s_0 < +\infty$.

By Lemma 8, $\lim_{\sigma \rightarrow 0^+} \sigma \frac{ds_0}{d\sigma} = 0$. Therefore:

$$\lim_{\sigma \rightarrow 0^+} s_0 = \lim_{\sigma \rightarrow 0^+} \frac{2\phi(s_1)}{2 - 2\Phi(s_1) + \frac{\mu_z}{\mu}}.$$

Define $\lim_{\sigma \rightarrow 0^+} s_0 \triangleq s_0(0^+)$, $\lim_{\sigma \rightarrow 0^+} s_1 \triangleq s_1(0^+)$, one has:

$$\begin{aligned} \lim_{\sigma \rightarrow 0^+} \frac{d\left(\frac{A}{\sigma_v}\right)}{d\sigma} &= \lim_{\sigma \rightarrow 0^+} s_0 = s_0(0^+) \geq 0, \\ \lim_{\sigma \rightarrow 0^+} \frac{d(\bar{\gamma}_e - \underline{\gamma}_e)}{d\sigma} &= 2\phi(s_1(0^+)) \geq 0, \\ \lim_{\sigma \rightarrow 0^+} \frac{d\alpha_e}{d\sigma} &= -\frac{G'(0)s_1(0^+)}{2} \leq 0, \\ \lim_{\sigma \rightarrow 0^+} \frac{d\alpha_d}{d\sigma} &= \frac{G'(0)(s_1(0^+) - s_0(0^+))}{2} \geq 0, \\ \lim_{\sigma \rightarrow 0^+} \frac{d(\alpha_e + \alpha_d)}{d\sigma} &= -\frac{G'(0)s_0(0^+)}{2} \leq 0, \end{aligned}$$

which concludes the proof. $\square$

Appendix C. A model with endogenous information acquisition

The following defines the equilibrium of a model with endogenous information acquisition.

Definition 3 (A Model with Endogenous Information Acquisition). A symmetric endogenous equilibrium is an information precision level, σ_e^* , chosen by the informed traders, a strategy for the informed speculators, $h_I^*(s)$, a strategy for the uninformed liquidity traders, $h_U^*(d)$, an exchange spread, A^* , two expected execution rate in the crossing network $\bar{R}^*, \underline{R}^*$, and a set of participation fractions $\bar{\gamma}_e^*, \underline{\gamma}_e^*, \bar{\gamma}_d^*, \underline{\gamma}_d^*, \alpha_e^*$, and α_d^* , s.t.

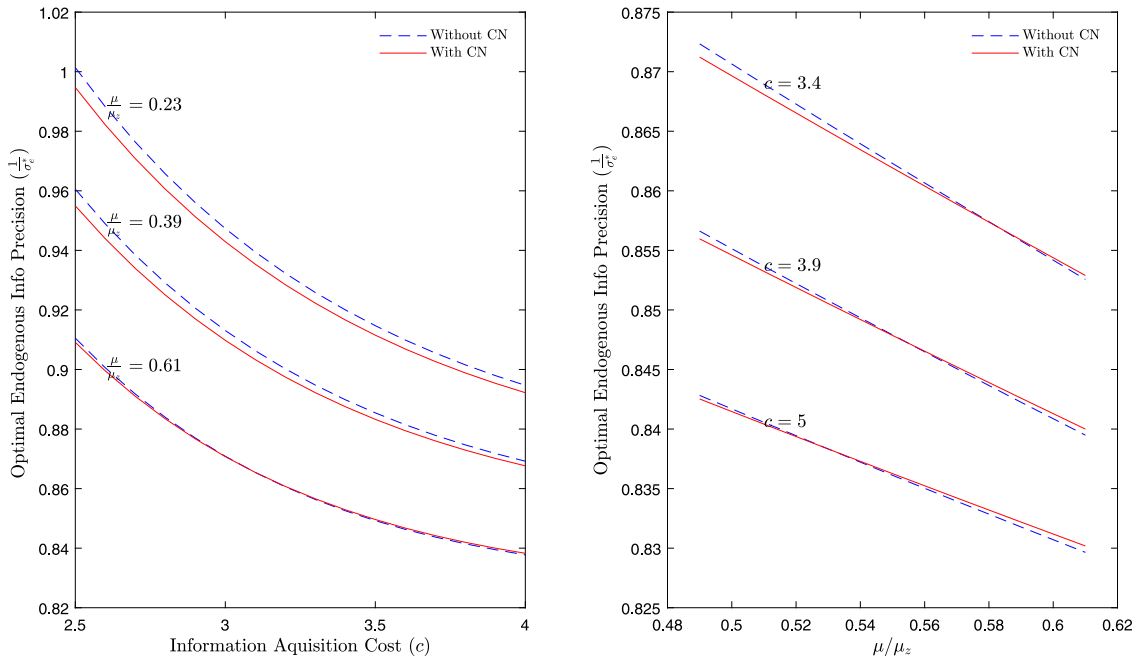


Fig. 17. The optimal endogenous information precision level, σ_e^* . The left figure shows σ_e^* as a function of the information acquisition difficulty level c , selecting three levels of the relative size of informed traders $\frac{\mu}{\mu_z}$. The right figure shows σ_e^* as a function of the relative size of informed traders $\frac{\mu}{\mu_z}$, choosing three levels of information acquisition difficulty measure, c .

- (i) σ_e^* is optimal for informed speculators, giving other informed traders choosing σ_e^* ;
- (ii) $h_I^*(s)$ is optimal for informed speculators with signal s , whereas $h_{U,i}^*(d)$ is optimal for uninformed liquidity traders with (per unit) delay cost d , given A^* , $\bar{R}^*$, and $\underline{R}^*$.
- (iii) the exchange spread A^* makes a market maker in the exchange break-even on average, given $\bar{\gamma}_e^*$, $\underline{\gamma}_e^*$, $\bar{\gamma}_d^*$, $\underline{\gamma}_d^*$, α_e^* , and α_d^* ;
- (iv) the crossing network operates using a mid-pricing and a rationing execution mechanism. $\underline{R}^*$ and $\bar{R}^*$ are the respective expected execution probability for orders that are in the right and in the wrong directions;
- (v) $\bar{\gamma}_e^*$ and $\underline{\gamma}_e^*$ measure the respective fractions of informed traders in the exchange who trade in the right and wrong directions. $\bar{\gamma}_d^*$ and $\underline{\gamma}_d^*$ measure the respective fractions of informed traders in the crossing network who trade in the right and wrong directions. α_e^* and α_d^* measure the respective fraction of uninformed traders who trade in the exchange and in the crossing network in period 1.

Fig. 17 shows a numerical example of the endogenous model. In this example, the cost function of information acquisition is assumed to be $C(\sigma_e) = .1(\frac{\sigma_e}{\sigma_0})^{-c}$. The coefficient c can be regarded as the difficulty of acquiring information.

(i) **The choice of the optimal level of information precision.** Fig. 17 shows that the informed traders' optimal level of the endogenous symmetric information precision, measured by $\frac{1}{\sigma_e^*}$, decrease in the information acquisition cost coefficient, c , and the relative size of the informed speculators, $\frac{\mu}{\mu_z}$, respectively, with or without a crossing network. Furthermore, as shown in Fig. 17, in a less costly information acquisition environment (low $\frac{\mu}{\mu_z}$, low c), adding a crossing network discourages information production, inducing a lower level of information precision in a market with a crossing network, compared to when there is not a crossing network.

By contrast, in a more costly information acquisition environment (high $\frac{\mu}{\mu_z}$, high c), adding a crossing network induces the informed to choose a higher level of information precision. The endogenous equilibrium results may help provide explain the empirical discovery of Brogaard and Pan (2018), who find that dark trading leads to greater firm-specific fundamentals in stock prices.

(ii) **The resulting crossing network impact on price discovery.** Fig. 11 shows that the amplification effect of crossing networks found in an exogenous model preserves in the endogenous model. In an economy where the price discovery level is already sufficiently low (for example, when $\frac{\mu}{\mu_z}$ is high, or c is high and hence a low information precision in equilibrium), introducing a crossing network to the market lowers the price discovery furthermore. By contrast, when the price discovery level is sufficiently high (for example, when $\frac{\mu}{\mu_z}$ is low, or c is low and hence a high information precision in equilibrium), adding a crossing network improves the price discovery. More intuition was discussed in Section 5.2.

(iii) **The resulting crossing network impact on transaction costs.** Figs. 18 and 19 plot the transaction costs and market participation in equilibrium of the endogenous model, respectively. Consistent with the argument in Proposition 1, as Fig. 18 shows,

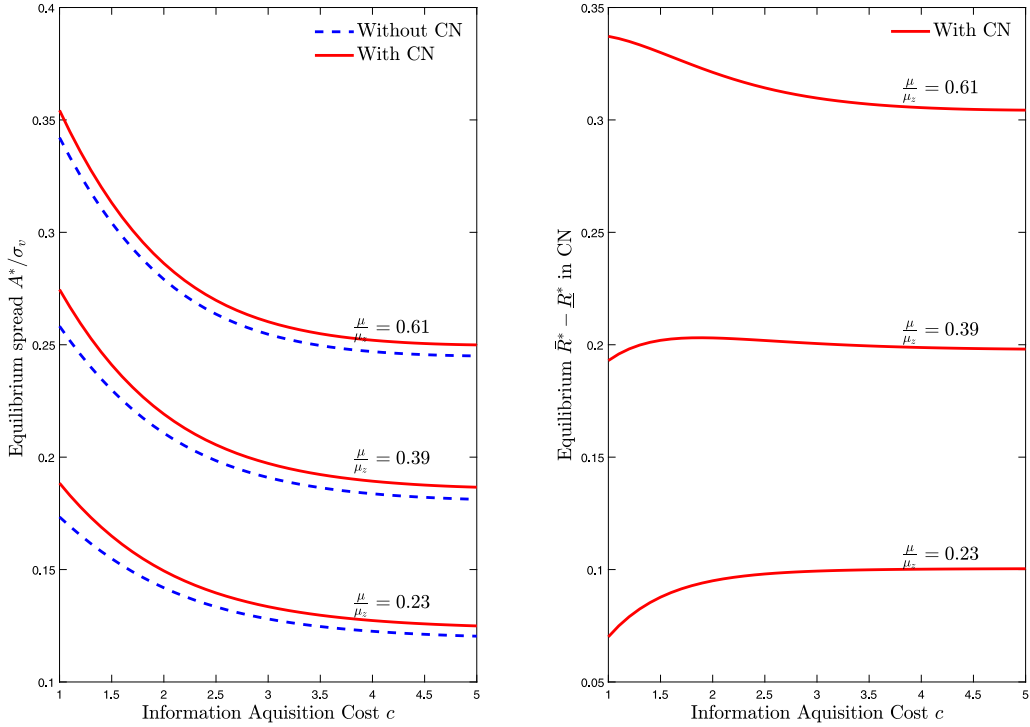


Fig. 18. The transaction costs in equilibrium of the endogenous model. The left figure shows the equilibrium exchange spreads $\frac{A^*}{\sigma_v}$, with and without the crossing network, as a function of the information acquisition cost coefficient c , selecting three different levels of $\frac{\mu}{\mu_z}$. The right figure shows the adverse selection cost $\bar{R}^* - \underline{R}^*$ in the crossing network in the equilibrium of the endogenous model.

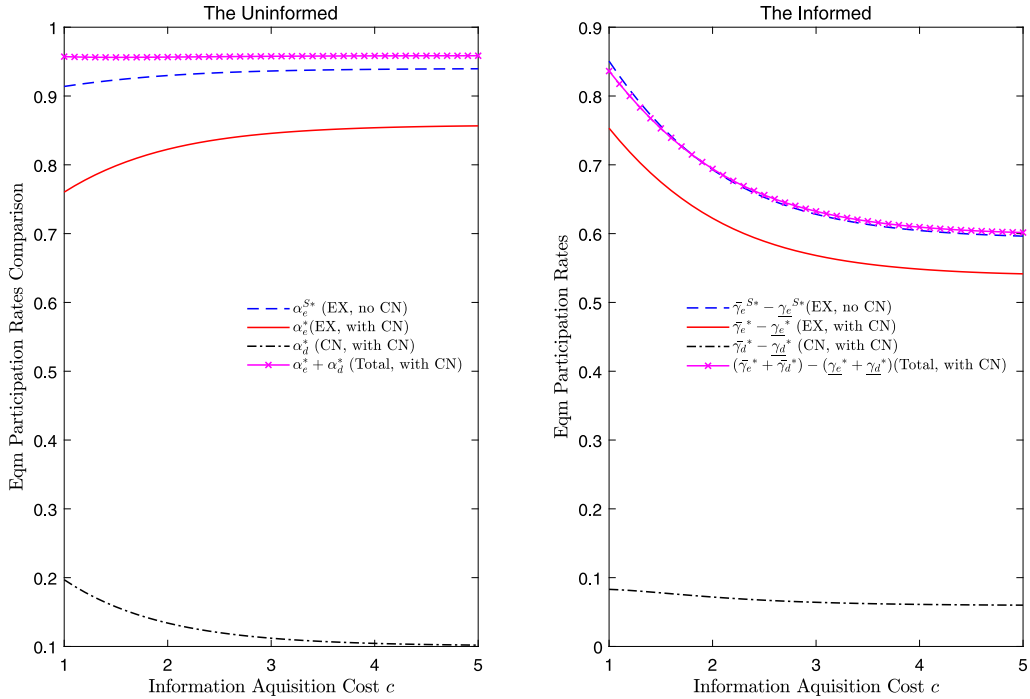


Fig. 19. The participation rate in equilibrium of the endogenous model. The left figure shows the uninformed traders' exchange participation rate without a crossing network, α_e^{S*} , and the uninformed's exchange, crossing network, and total participation rate, α_e^* , α_d^* , $\alpha_e^* + \alpha_d^*$, with a crossing network, as a function of c . The right figure shows the informed traders right-wrong participation in the exchange without a crossing network, $\bar{\gamma}_e^{S*} - \underline{\gamma}_e^{S*}$, and their corresponding participation in the exchange, $\bar{\gamma}_e^* - \underline{\gamma}_e^*$, in the crossing network, $\bar{\gamma}_d^* - \underline{\gamma}_d^*$ and the total $(\bar{\gamma}_e^* + \bar{\gamma}_d^*) - (\underline{\gamma}_e^* + \underline{\gamma}_d^*)$, when there is a crossing network. In these figures, $\frac{\mu}{\mu_z} = 0.39$.

when it is fixed the relative size of the informed traders, $\frac{\mu}{\mu_z}$, the normalized exchange spread ($\frac{\Delta^*}{\sigma_v}$) decreases in the information acquisition cost coefficient, c , because the optimal precision level decreases in c . Also, when c is fixed, a larger relative size of informed traders (higher $\frac{\mu}{\mu_z}$) could drive the transaction cost higher in both the exchange and the crossing network ($\frac{\Delta^*}{\sigma_v}$ and $\bar{R}^* - \bar{R}^*$). Fig. 19 shows the traders' participation rate before and after the entry of a crossing network in the endogenous model. Similar to the exogenous model's result, the crossing network diverges part of the traders from the exchange to the crossing network. Also, as information costs c increases and optimal precision level $\frac{1}{\sigma_e}$ decreases, the participation of the uninformed traders increases on the exchange and decreases in the crossing network. By contrast, as c increases and $\frac{1}{\sigma_e}$ decreases, the meaningful informed trading on the exchange decreases. This again echos the argument that, when information precision is low (because c is high), a crossing network attracts more informed traders from the exchange than it does for the uninformed, and therefore the price discovery in the exchange decreases.

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